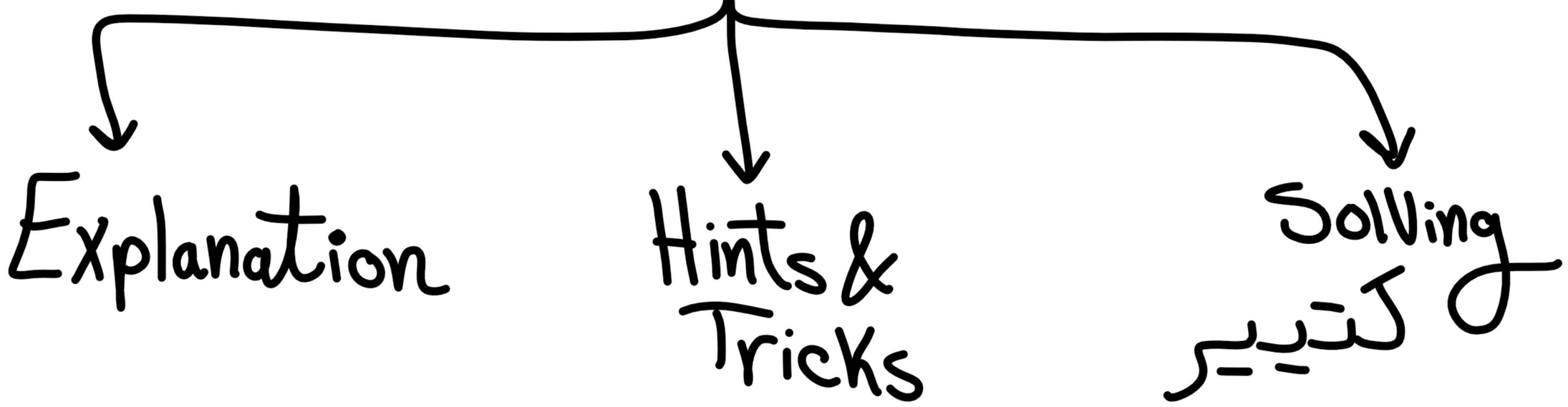
Chapter ①

ReVision



#Questions Resources

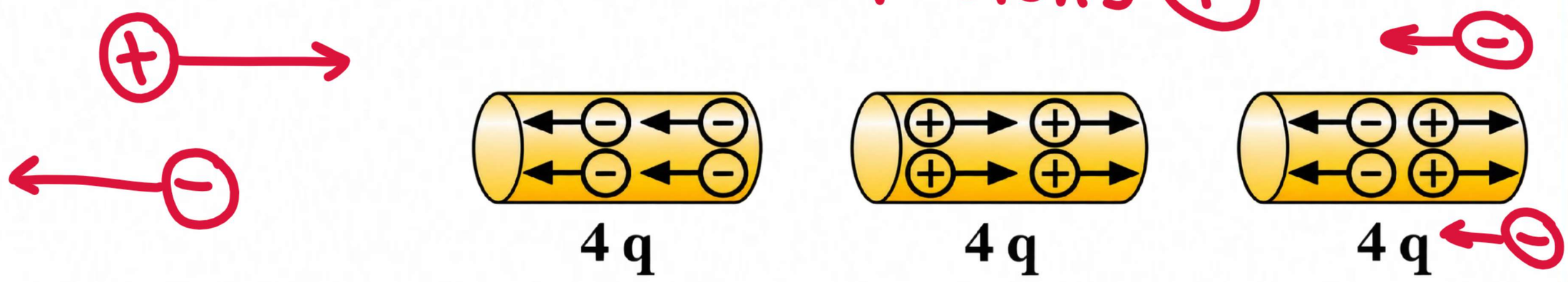
1. Previous years Exams [2021 ⟶ 2026]
2. Moasser
3. Tafouk
4. Azhar

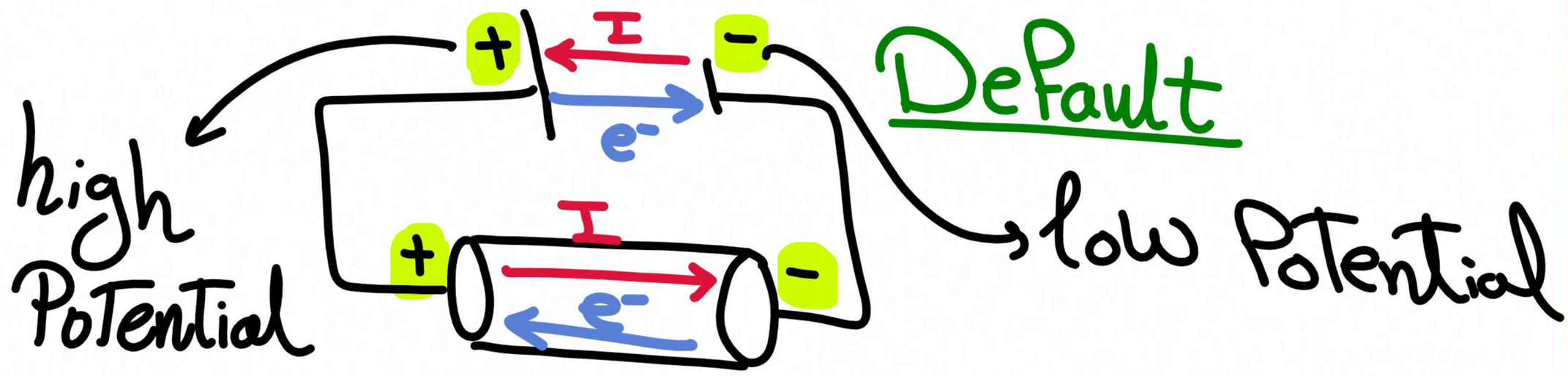
Mr.Abdelwahab

Electric Charge

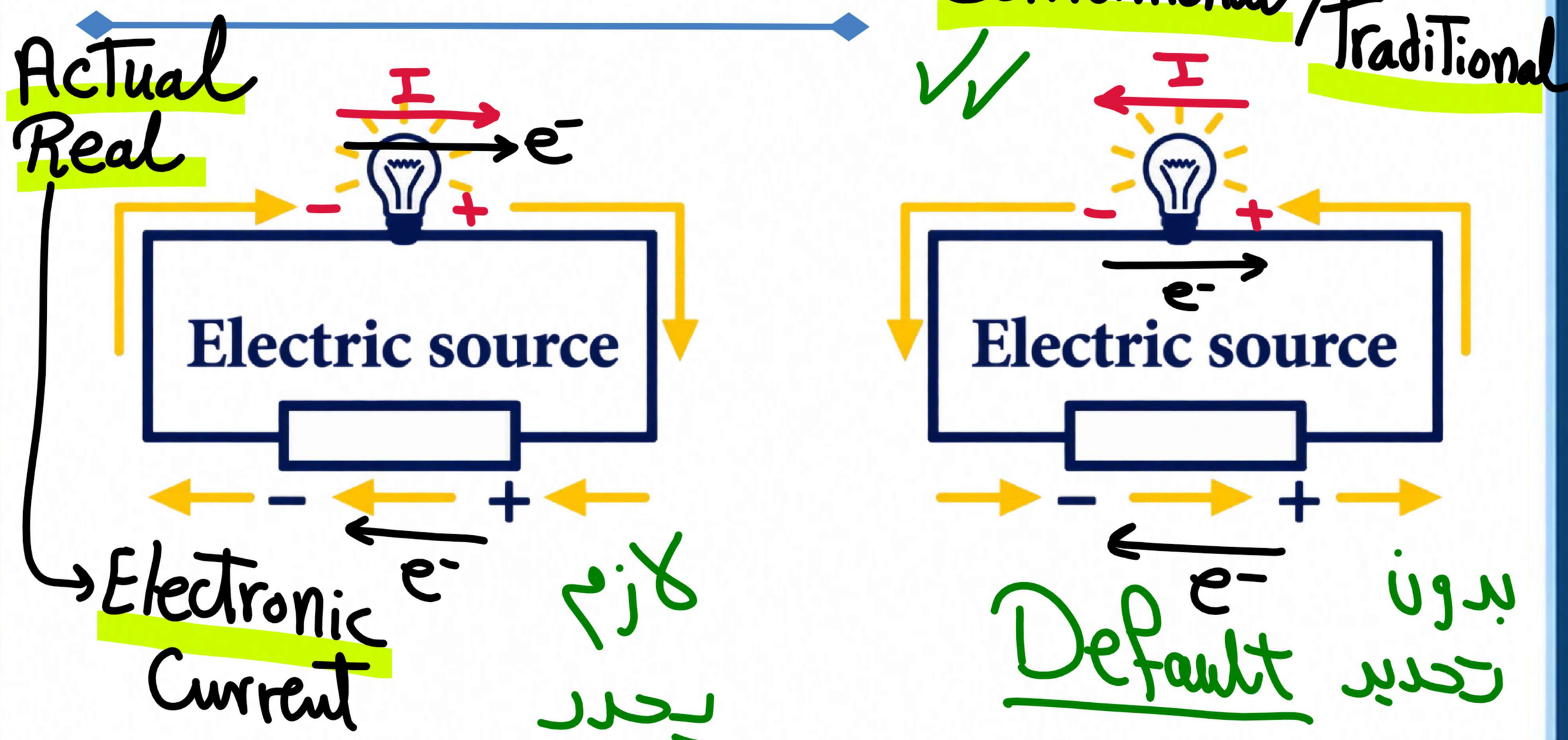
free electrons ⊖

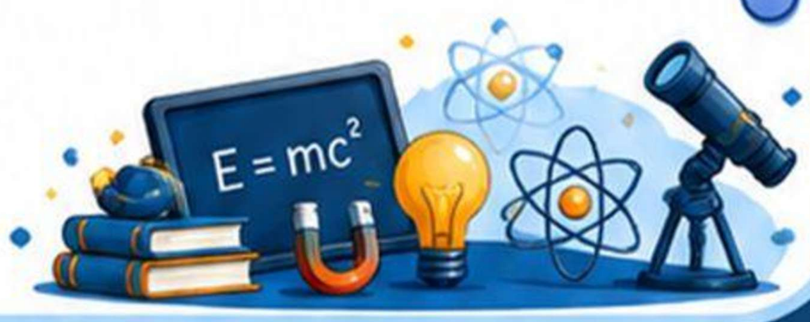
Protons ⊕





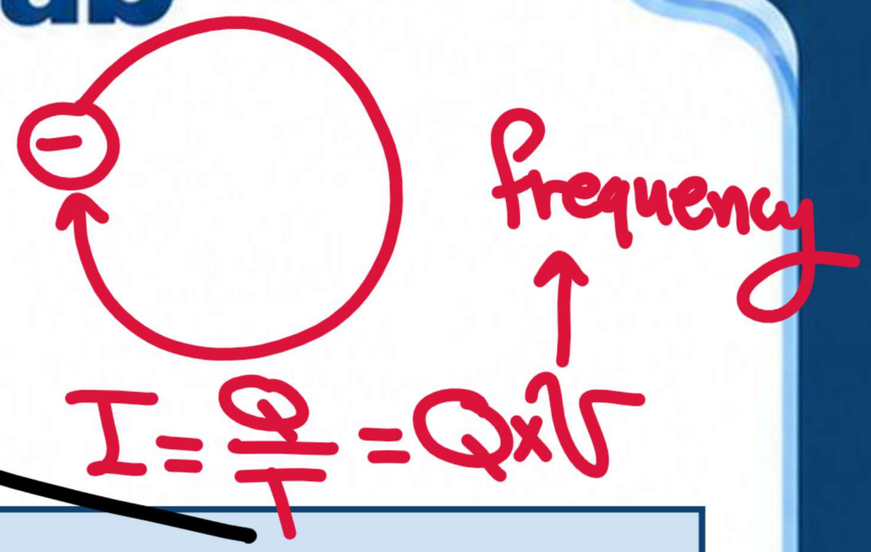
Direction of electric current:





Electric current intensity (I):

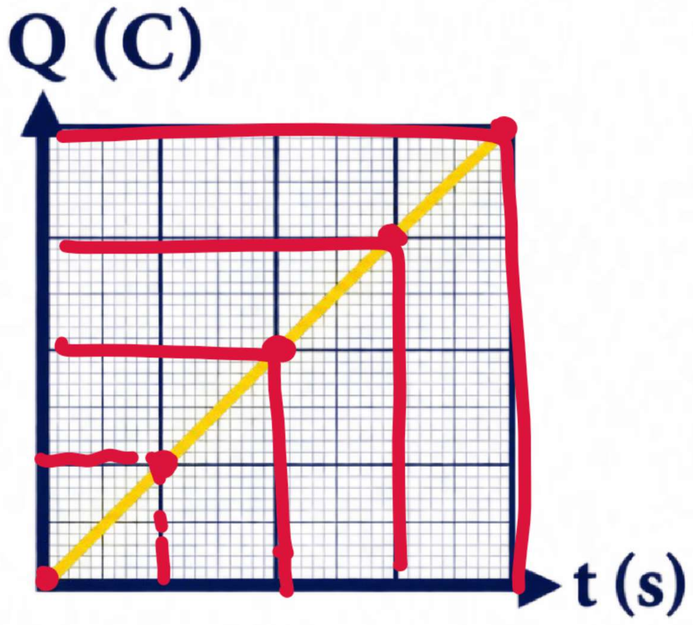
Ampere [A]



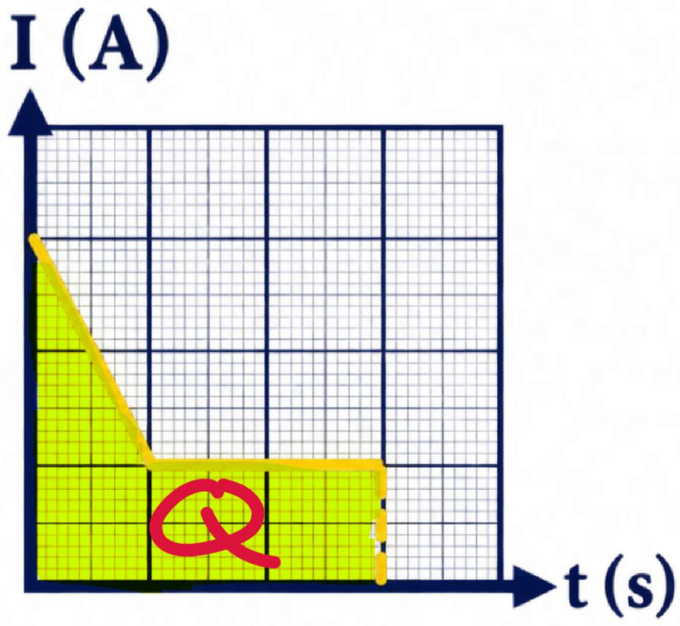
In a closed electric circuit containing an electric source [battery]	When an electric charge (q) moves with a linear speed (ϑ) along the circumference of a circular path of radius (r)
$I = \frac{Q}{t} = \frac{N \times e}{t}$	$I = \frac{Q \times \vartheta}{2\pi r}$ → Linear Speed

Slop:

Area = y-axis × X-axis

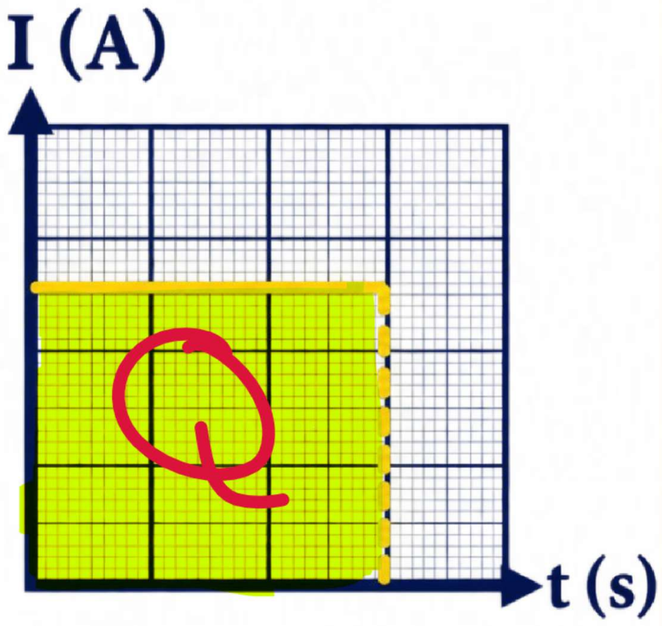


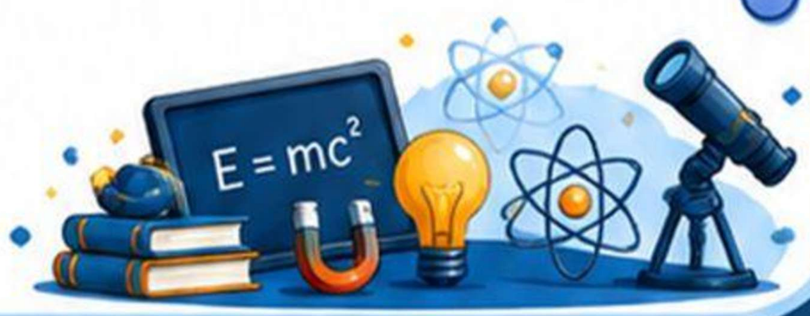
Slope $= \frac{Q}{t} = I$



$Q = I \times t$

= Area under the graph

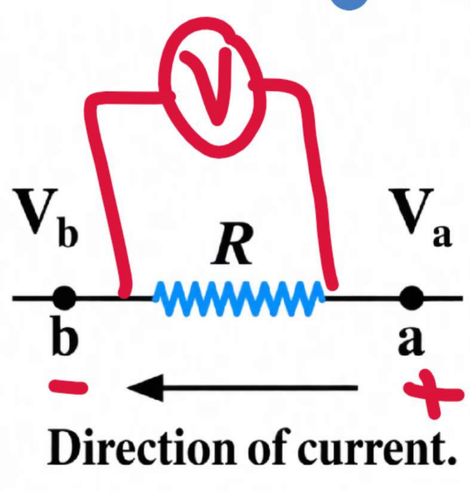




Potential difference (V):

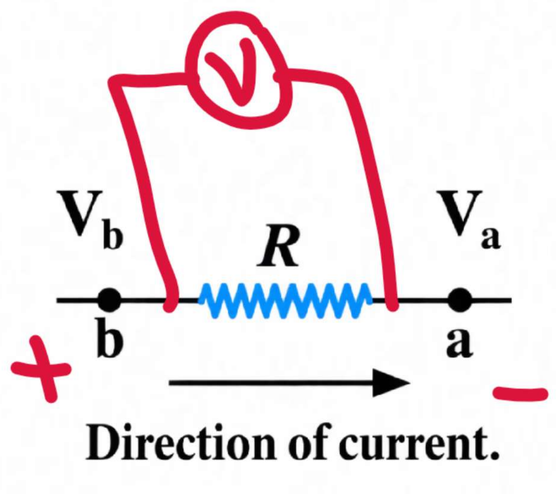
$V = V_{high} - V_{low}$

(+) الكبير — الصغير



$V_b < V_a$

$V = V_a - V_b$



$V_b > V_a$

$V = V_b - V_a$

$V = 10 - 3$

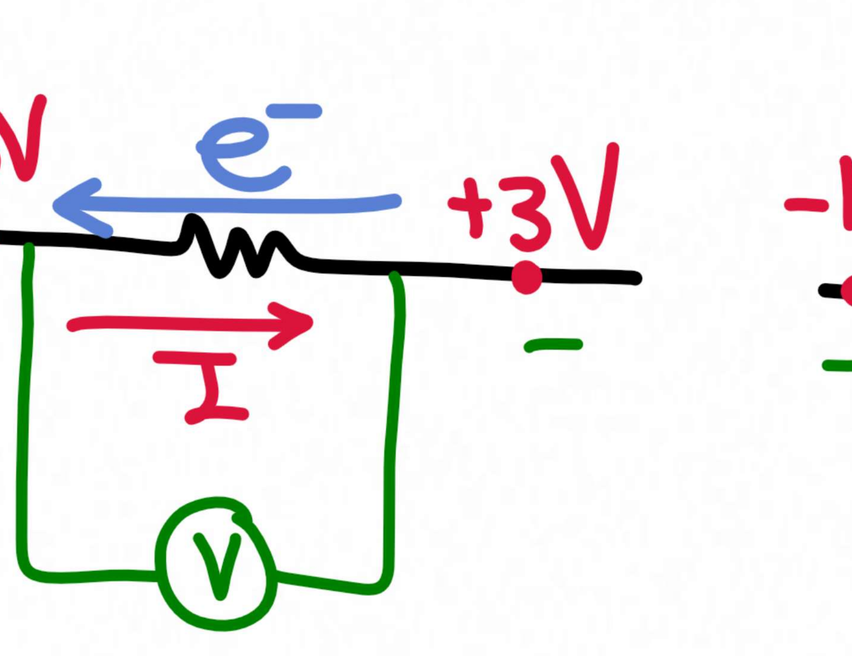
$= +7V$

$V = -3 - (-10)$

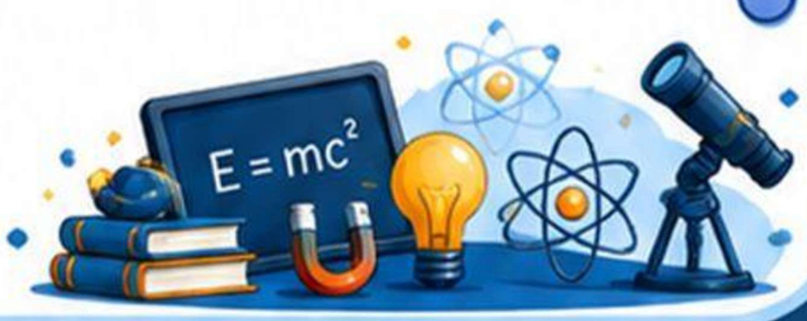
$= +7V$

$V = +3 - (-10)$

$= +13V$



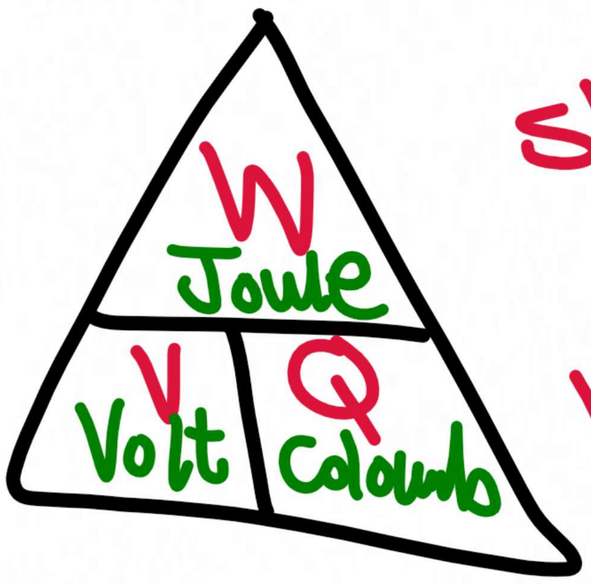
$V = 5 - 0 = 5V$



اضغط هنا

YouTube Channel Physics Universe Mr.Abdelwahab
Facebook Page physics universe
WHATSAPP 01113577752
Website https://www.etqanway.com

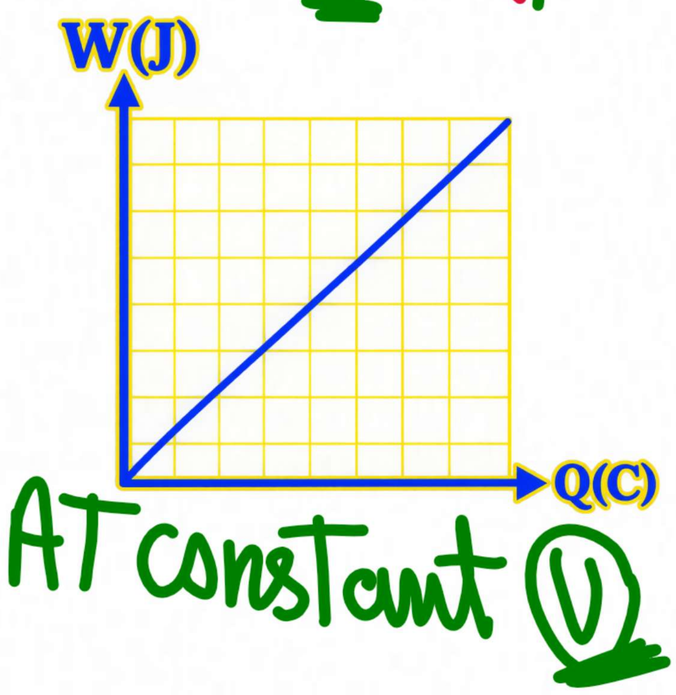
Electric potential difference between two points (V)



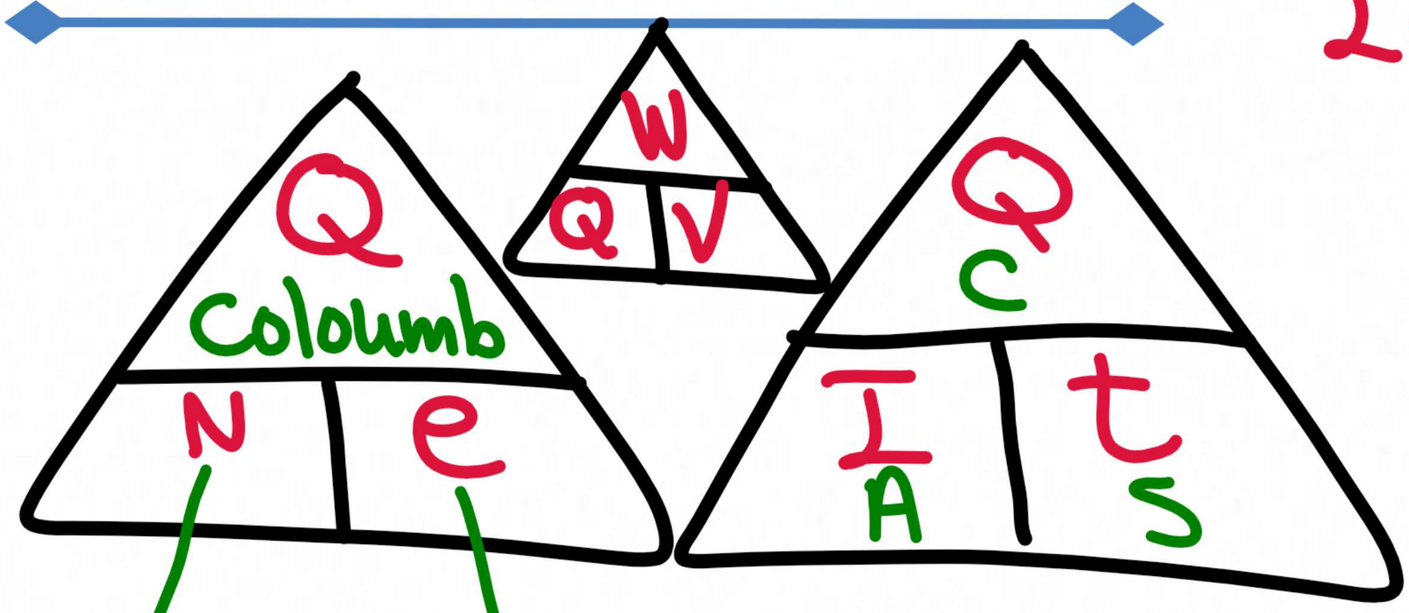
$$slope = \frac{W}{Q} = V$$

$$V = \frac{W}{Q} \rightarrow \frac{V_1}{V_2} = \frac{W_1}{W_2} \times \frac{Q_2}{Q_1}$$

$$W = V \times Q$$



Quantity of Electric Charges (Q)



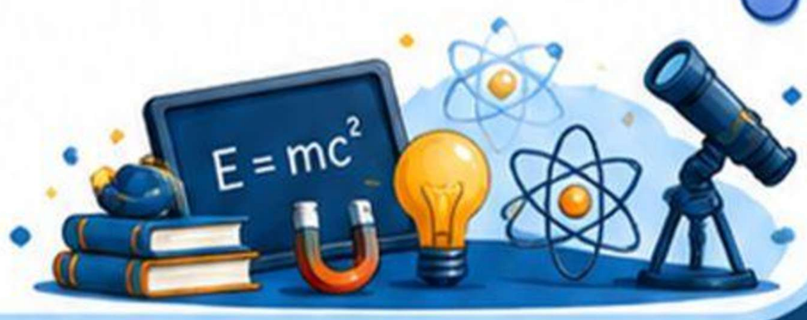
No. of Charges

1.6×10^{-19} C / Constant

$$I = \frac{Q}{t} = \frac{N \times e}{t}$$

Rate of electrons

A/C or elec./s



اضغط هنا

YouTube Channel
Physics Universe Mr.Abdelwahab

Facebook Page
physics universe

WHATSAPP
01113577752

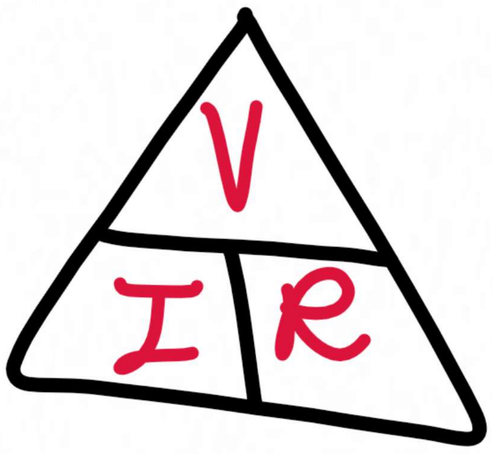
Website
https://www.etqanway.com

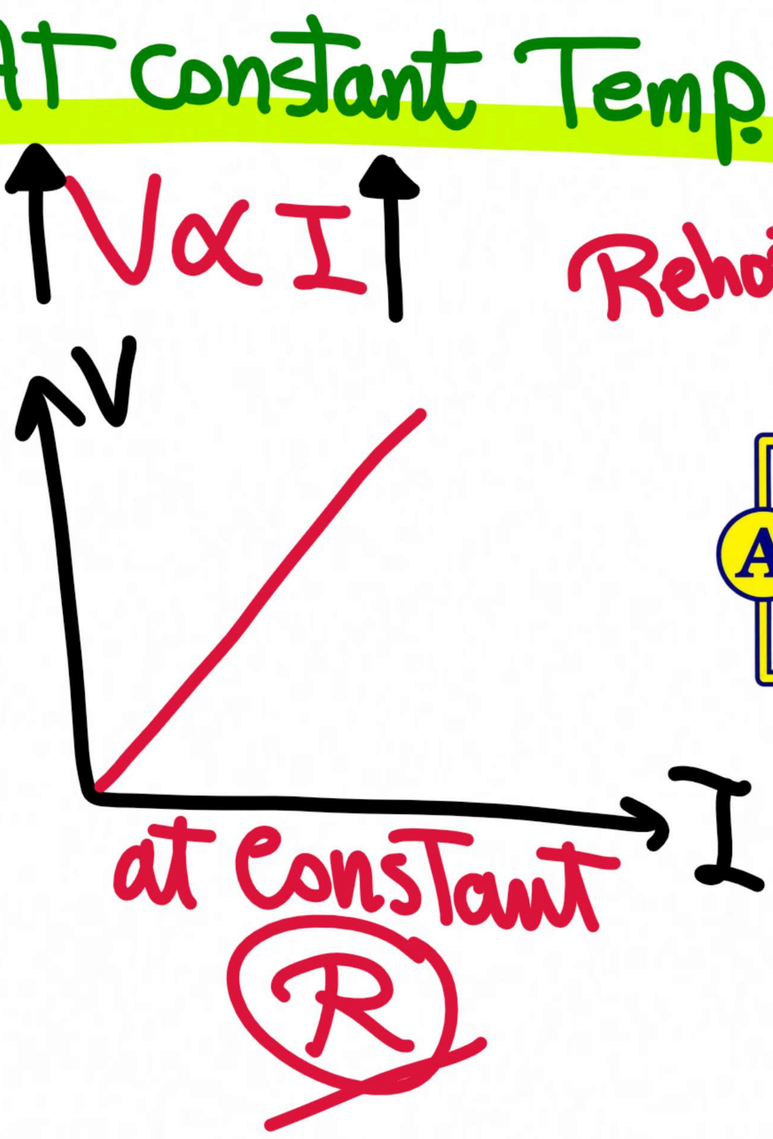
Mr.Abdelwahab

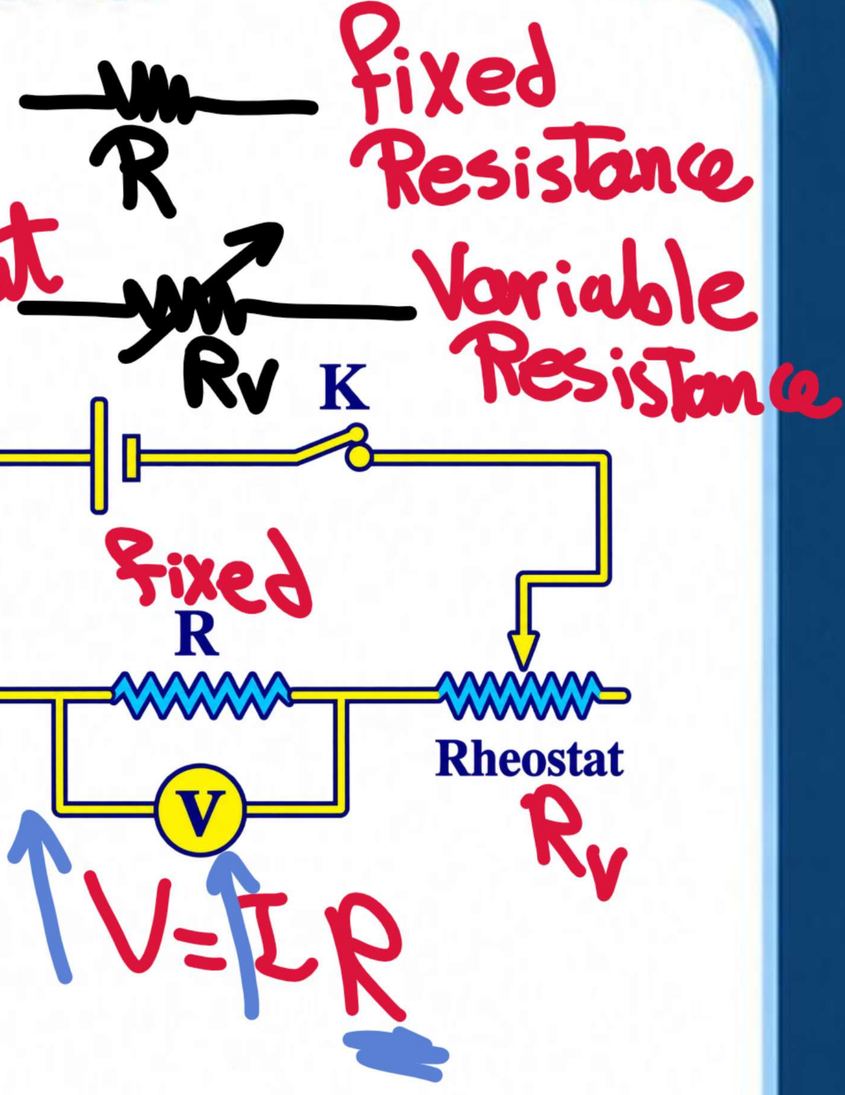
Ohm's law:

At constant Temp.

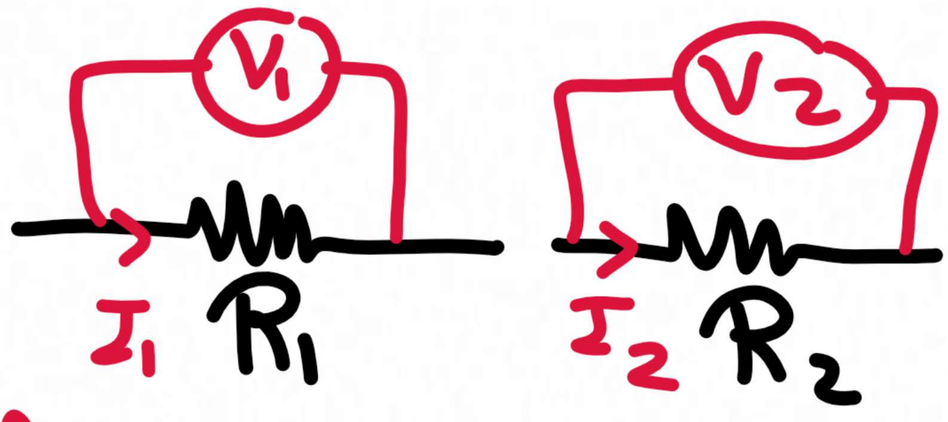
$V \propto I$







When the potential difference across each of the two resistors



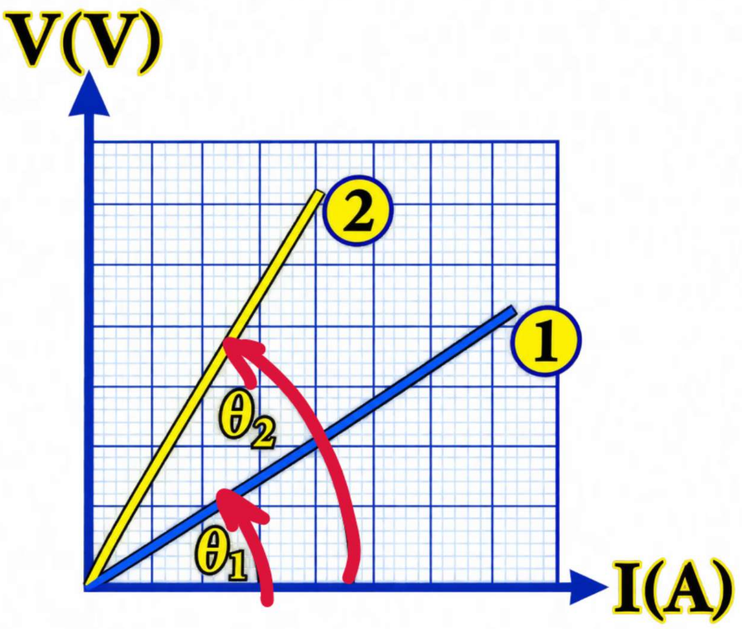
$$Slope = \frac{V}{I} = R = \tan\theta$$

$$\theta_2 > \theta_1$$

$$\tan\theta_2 > \tan\theta_1$$

$$Slope_2 > Slope_1$$

$$R_2 > R_1$$

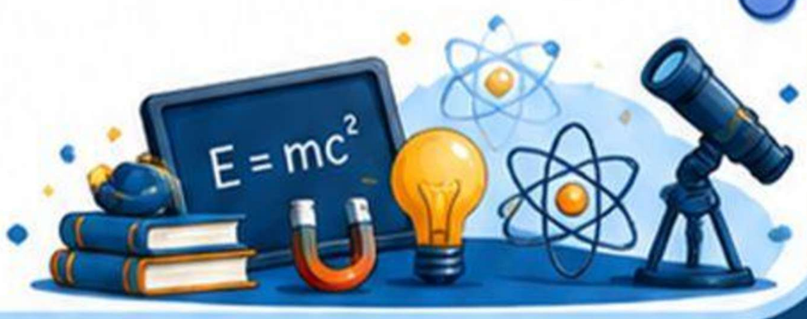


#take care

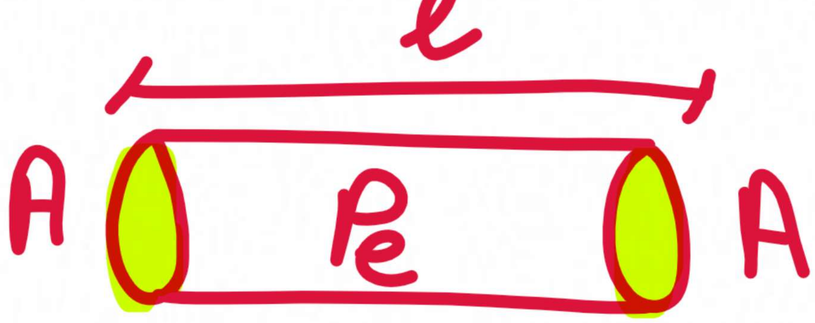
$\theta_2 > \theta_1$

$R_2 < R_1$

$$Slope = \frac{I}{V} = \frac{1}{R} = \tan\theta$$



Electrical resistance (R):



$R \propto \rho_e \uparrow$

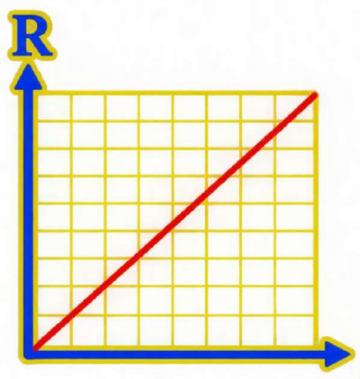
$R \propto l \uparrow$

$R \propto \frac{1}{A} \downarrow$

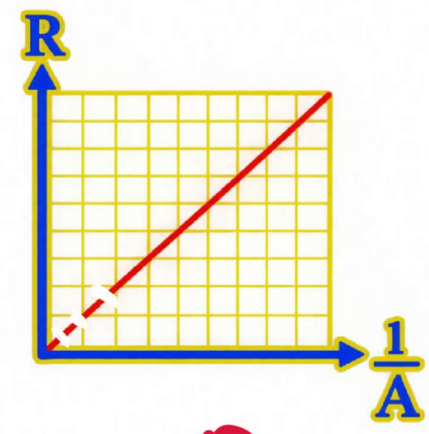
$$R = \frac{\rho_e \times l}{A}$$

Very Important 2027

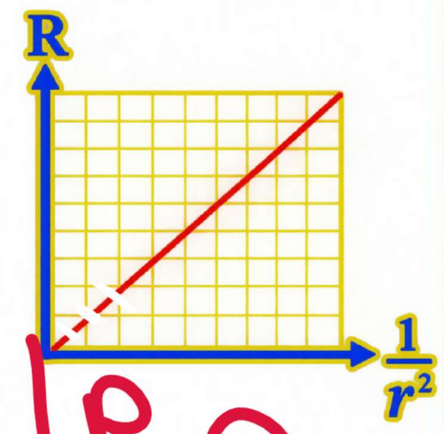
$A = \pi r^2$



$$Slope = \frac{R}{l} = \frac{\rho_e}{A}$$

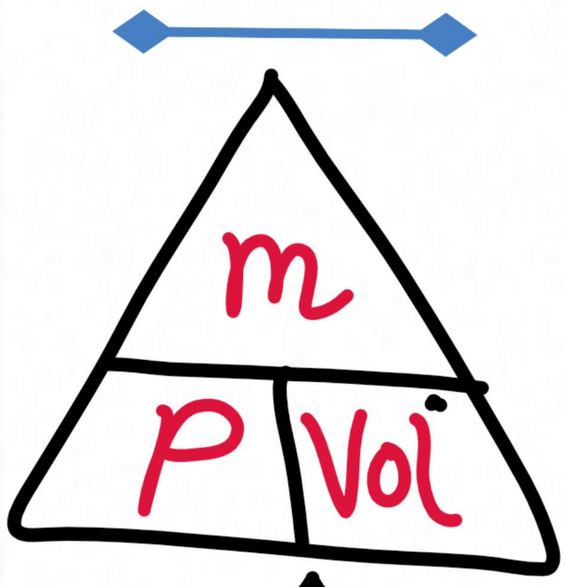


$$Slope = \frac{R}{1/A} = R A = \rho_e \times L$$



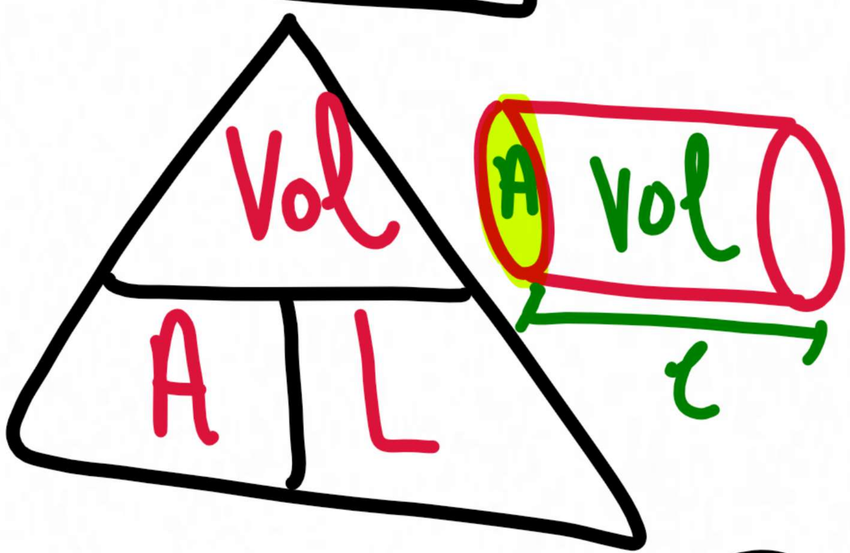
$$\frac{R}{1/r^2} = R \times r^2 = \frac{\rho_e l}{\pi}$$

Density



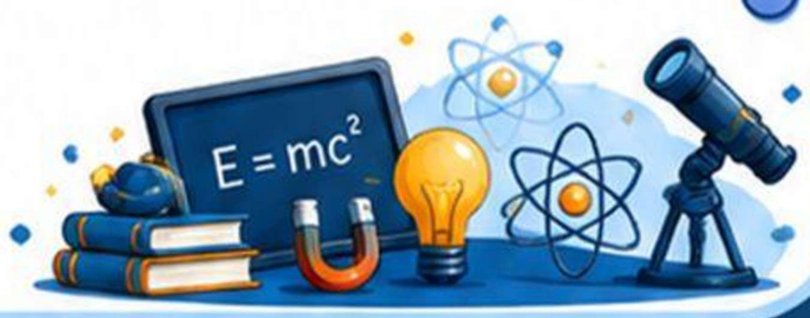
$$R = \frac{\rho_e . L}{A} = \frac{\rho_e . \frac{Vol}{A}}{A} = \frac{\rho_e \times Vol}{A^2} = \frac{\rho_e \times \frac{m}{\rho}}{A^2}$$

$$R = \frac{\rho_e \times m}{\rho \times A^2}$$



$$R = \frac{\rho_e \times L}{A} = \frac{\rho_e \times L}{\frac{Vol}{L}} = \frac{\rho_e \times L^2}{Vol} = \frac{\rho_e \times L^2}{\frac{m}{\rho}}$$

$$R = \frac{\rho_e \rho \times L^2}{m}$$



$R = \frac{\rho e \times l^2}{Vol}$

$R = \frac{V}{I}$

$R = \frac{\rho e \times Vol}{A^2}$

$R = \frac{\rho e \times L}{A}$

$R = \frac{\rho e \rho l^2}{m}$

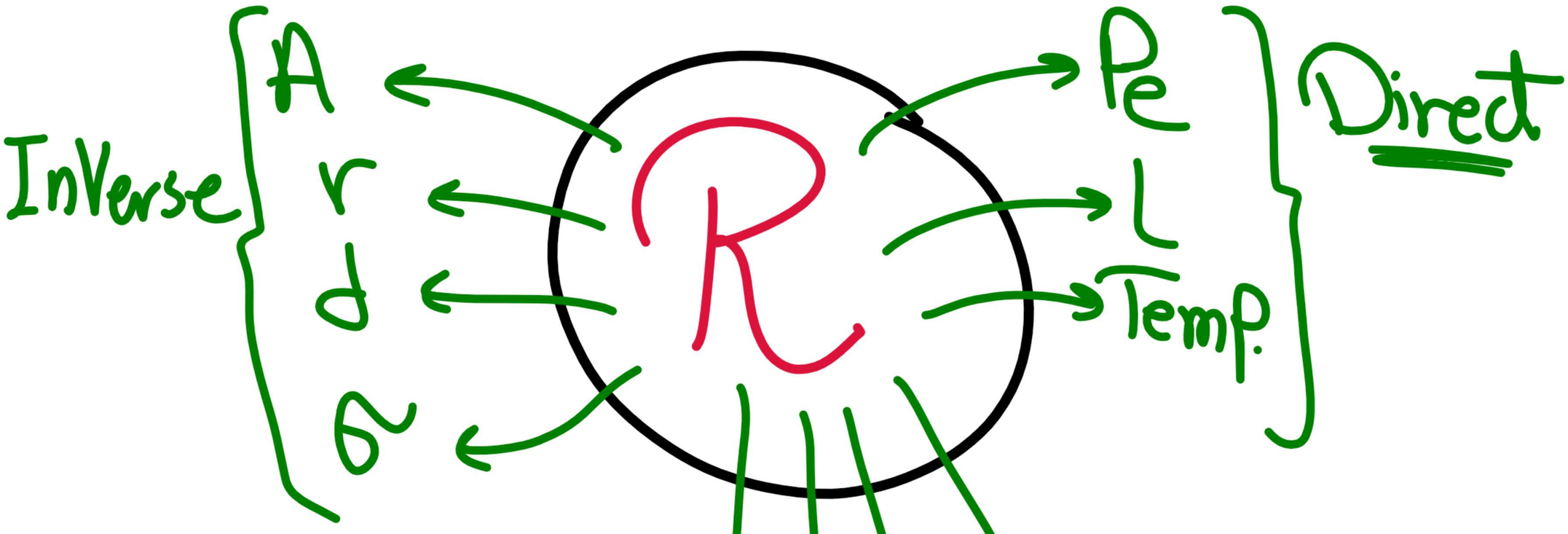
$R = \frac{L}{\sigma A}$

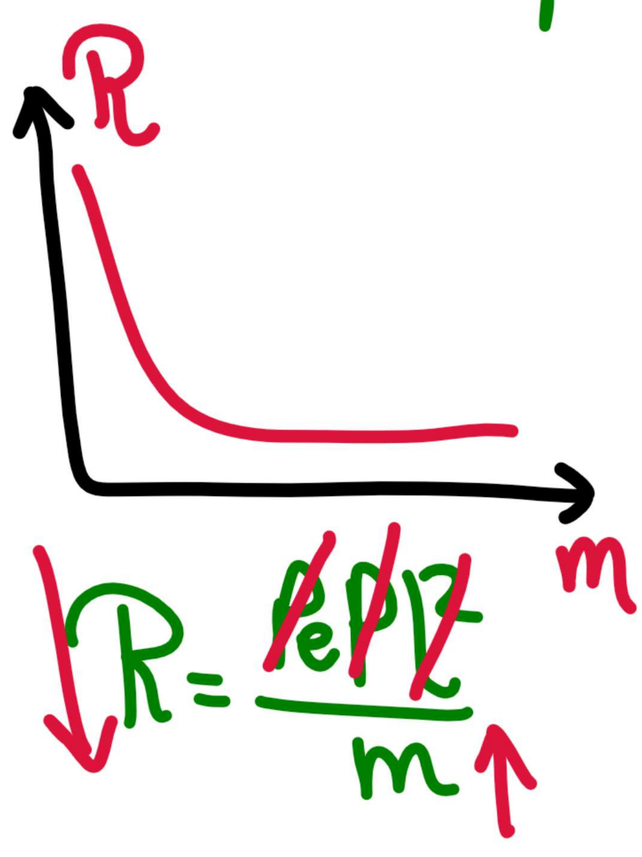
Ω

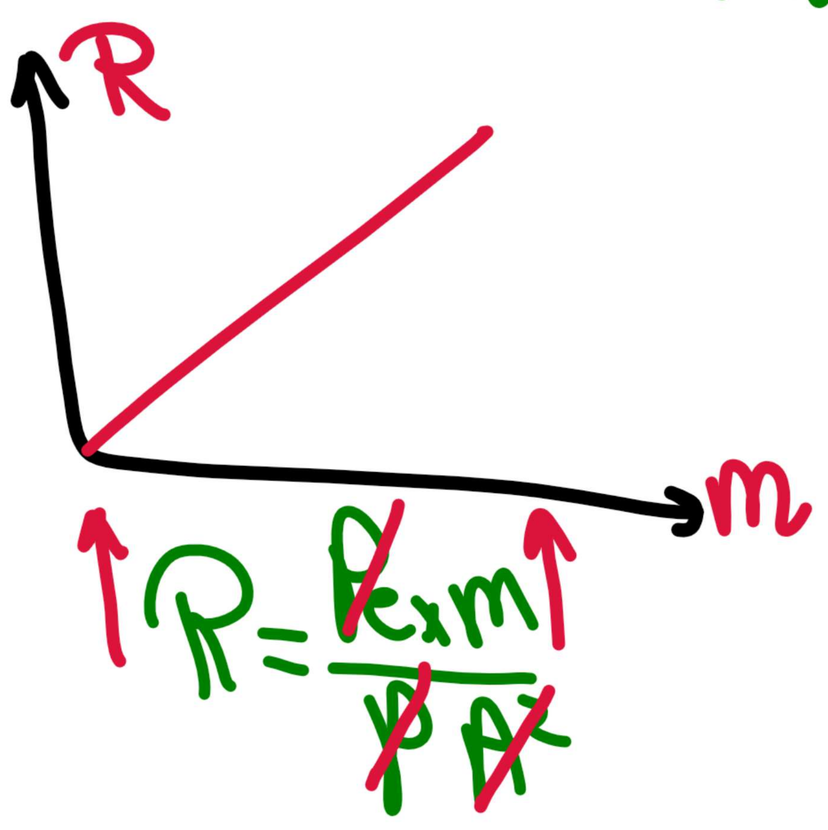
ohm unit

$R = \frac{\rho e \times m}{\rho \times A^2}$

factors affecting Resistance



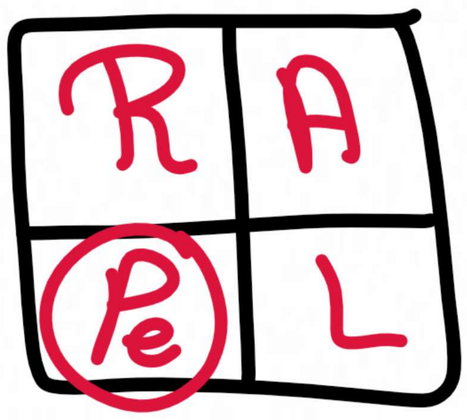


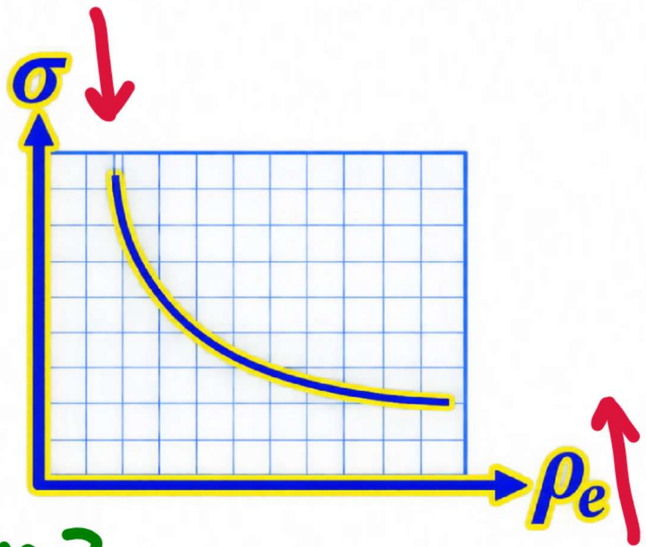


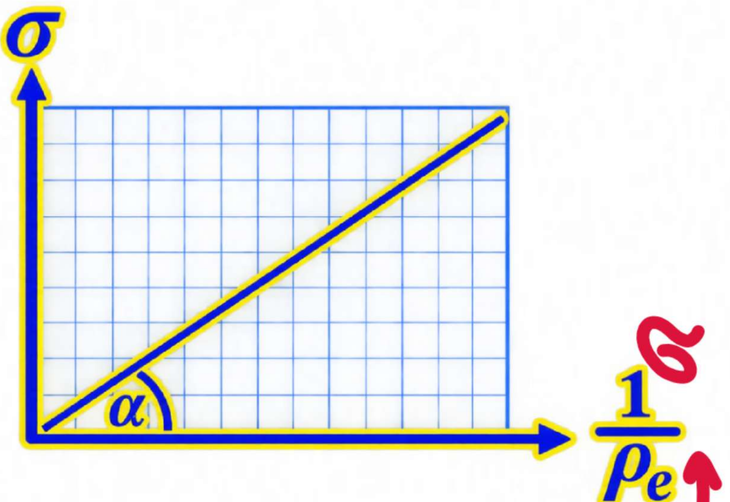
Resistivity & conductivity

ρ_e σ

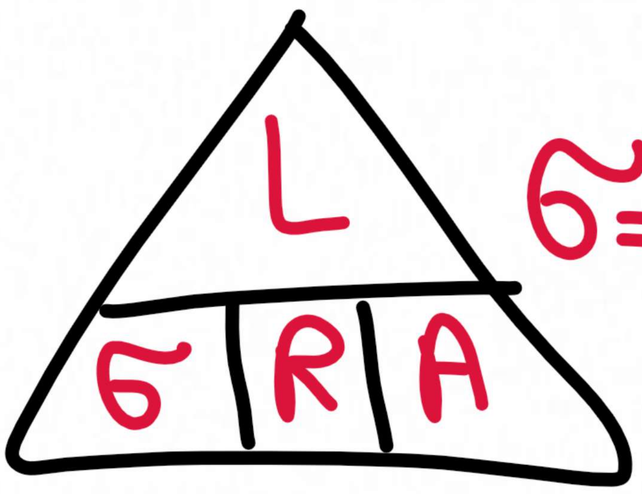
$\Omega.m$ $\Omega^{-1}.m^{-1}$







$$\rho_e = \frac{R \times A}{L} = \frac{\Omega.m^2}{m} = \Omega.m$$



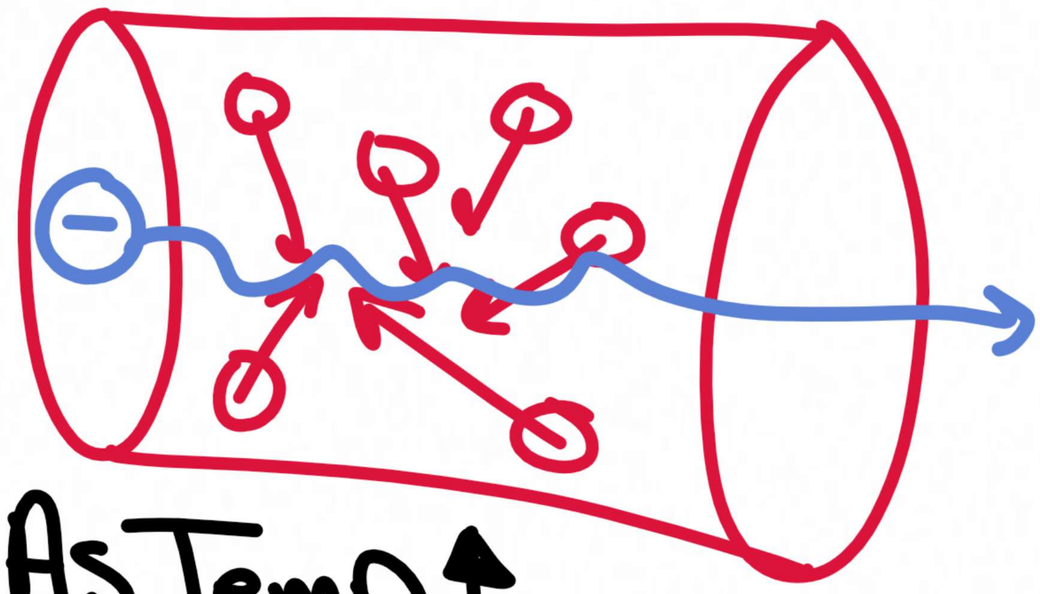
$$\sigma = \frac{L}{RA} = \frac{m}{\Omega.m^2} = \frac{1}{\Omega.m} = \Omega^{-1}.m^{-1}$$

ρ_e ↑

- material
  - $\rho_{e\,wood} > \rho_{e\,cu}$
- Temp. ↑

σ ↓

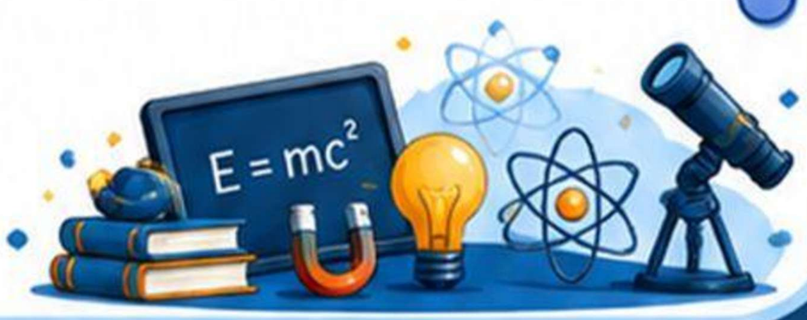
- material
  - $\sigma_{wood} < \sigma_{cu}$
- Temp. ↑



As Temp. ↑

K.E. molecules ↑ σ ↓

Collisions ↑ ρ_e ↑ R ↑



Mr.Abdelwahab

To compare the resistances of two conductors:

$$R = \frac{\rho_e \times l}{A}$$

Take Care

$$\frac{A_2}{A_1} = \left[\frac{r_2}{r_1}\right]^2 = \left[\frac{d_2}{d_1}\right]^2$$

Two different conductors	When reshaping [withdrawing] the conductor
$\frac{R_1}{R_2} = \frac{\rho_{e_1}}{\rho_{e_2}} \times \frac{L_1}{L_2} \times \frac{A_2}{A_1}$	Reshape/Strech/drawn Pulled/Pressed/Melted $\frac{R_1}{R_2} = \left[\frac{l_1}{l_2}\right]^2 = \left[\frac{A_2}{A_1}\right]^2 = \left[\frac{r_2}{r_1}\right]^4 = \left[\frac{d_2}{d_1}\right]^4$

Two different materials

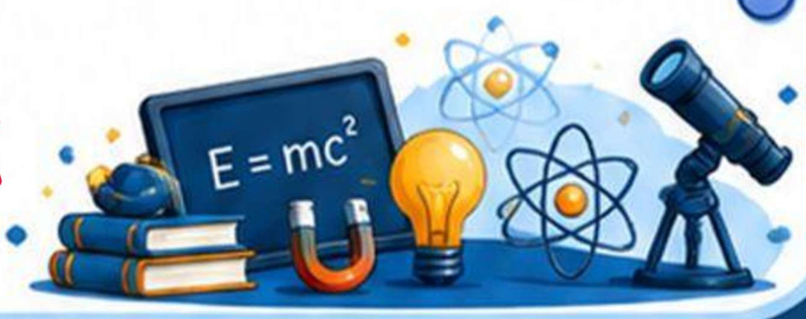
The material resistivities of two conductors	The material conductivities of two conductors.
$\rho_e = \frac{RA}{L}$ $\frac{\rho_{e_1}}{\rho_{e_2}} = \frac{R_1}{R_2} \times \frac{A_1}{A_2} \times \frac{L_2}{L_1}$	$\sigma = \frac{L}{RA} = \frac{L}{R\pi r^2}$ $\frac{\sigma_x}{\sigma_y} = \frac{L_x}{L_y} \times \frac{R_y}{R_x} \times \left(\frac{r_y}{r_x}\right)^2$

#Take Care

1. Same material
2. Same Temp.

$\Rightarrow \rho_{e_1} = \rho_{e_2}$, $\sigma_1 = \sigma_2$

★ Same Conductor $\rightarrow R = \frac{\rho_e l}{A}$ all const.



Mr.Abdelwahab

When a wire or conductor of length (L), radius (r), and resistance (R1) is reshaped, then: Strech/drawn

$$R \alpha l^2 \alpha \frac{1}{A^2} \alpha \frac{1}{r^4} \alpha \frac{1}{d^4}$$

Changing the length

$$R_2 = n^2 R_1$$

* When the length increases from (ℓ) to ($n\ell$), then:

$$\frac{R_1}{R_2} = \left[\frac{\ell}{n\ell}\right]^2 = \frac{1}{n^2}$$

* When the length decreases from (ℓ) to ℓ/n, then:

$$\frac{R_1}{R_2} = \left[\frac{\ell}{\frac{\ell}{n}}\right]^2 = \frac{n^2}{1} \rightarrow R_2 = \frac{1}{n^2} R_1$$

Changing the radius

* When the radius increases from (r) to (nr), then:

$$\frac{R_1}{R_2} = \left[\frac{nr}{r}\right]^4 = \frac{n^4}{1} \rightarrow R_2 = \frac{1}{n^4} \times R_1$$

* When the radius decreases from (r) to r/n, then:

$$\frac{R_1}{R_2} = \left[\frac{\frac{r}{n}}{r}\right]^4 = \frac{1}{n4} \rightarrow R_2 = n^4 R_1$$

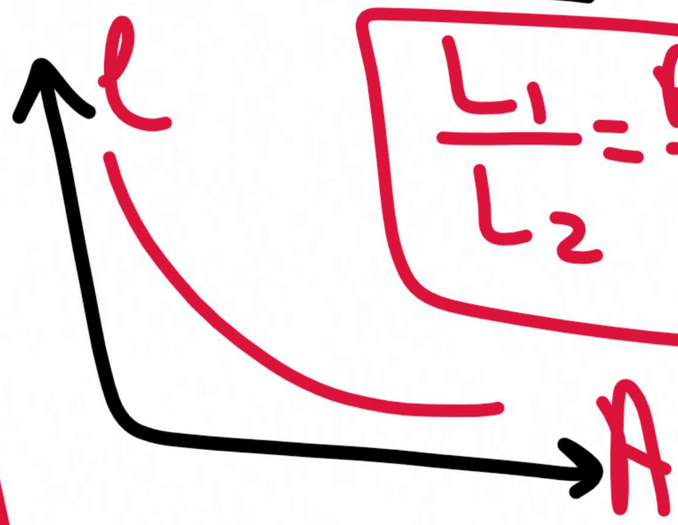
Note

$$\frac{R_1}{R_2} = \left[\frac{L_1}{L_2}\right]^2$$

$$\frac{R_1}{R_2} = \left[\frac{r_2}{r_1}\right]^4$$

Reshape
Strech
drawn
Pulled
Pressed

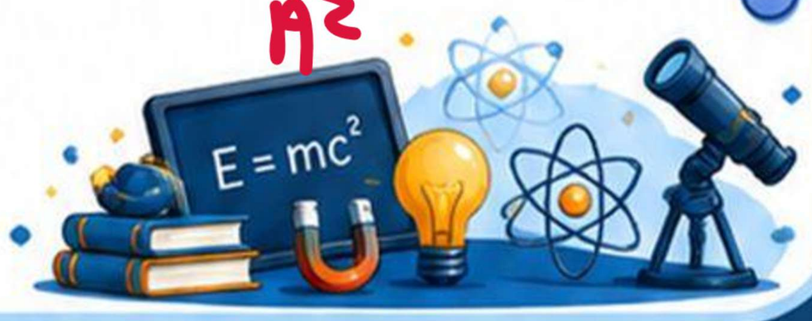
Const. Volume



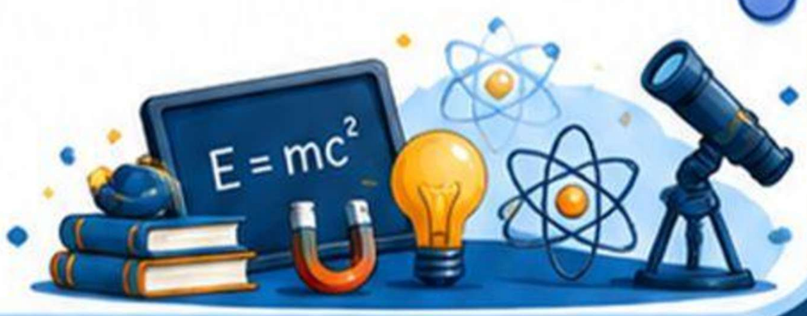
$$\frac{L_1}{L_2} = \frac{A_2}{A_1}$$

$$R = \frac{\rho \times L^2}{Vol}$$

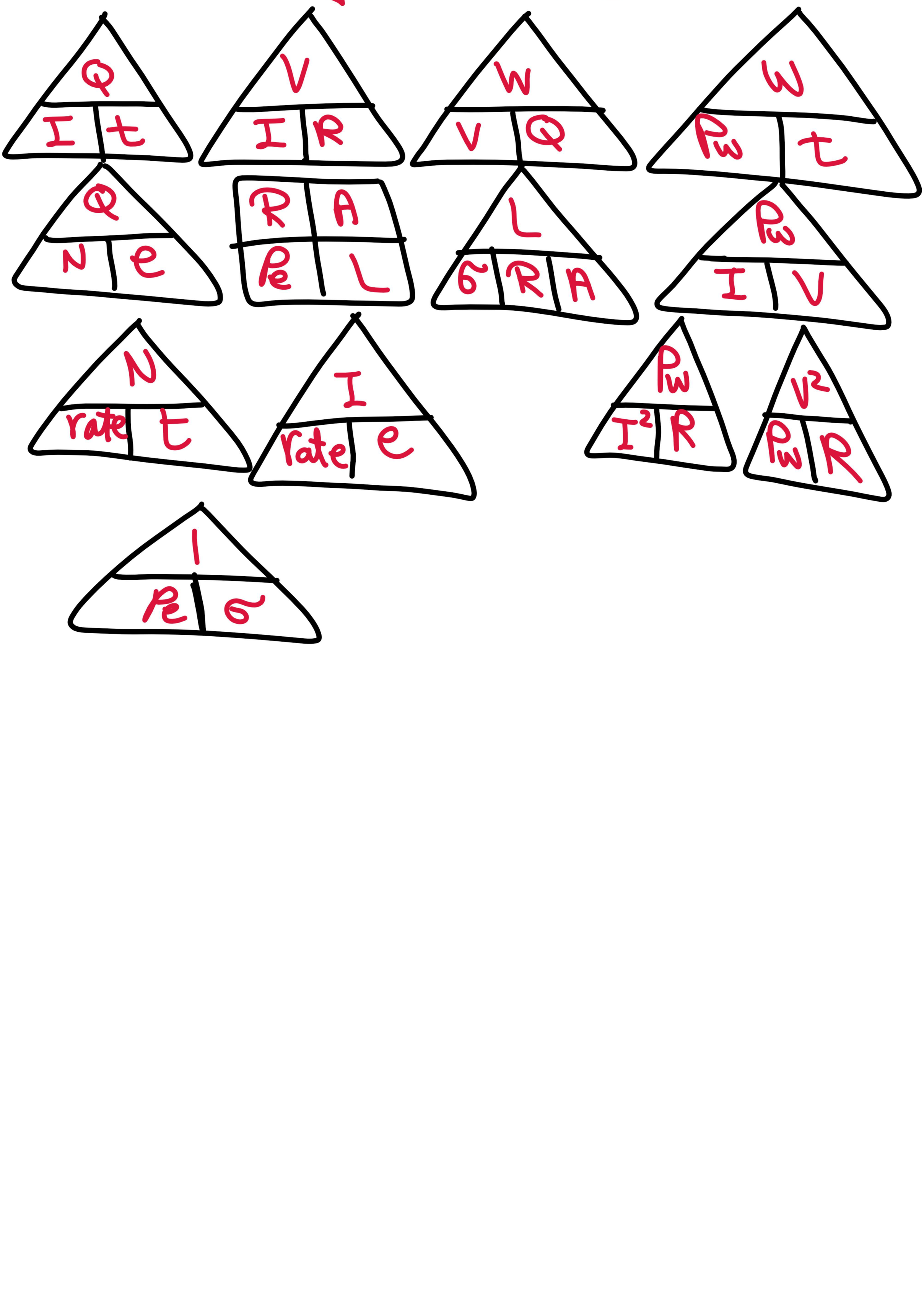
$$R = \frac{\rho \times Vol}{A^2}$$



يلا بينا...
نتأكد من فهمك


E = mc²

Rules Summary



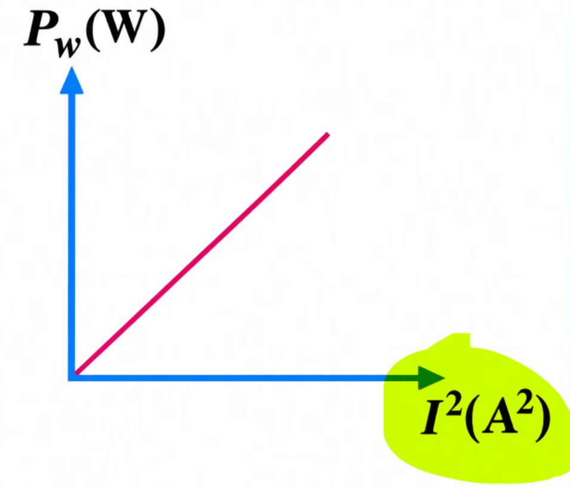
Questions

The opposite graph represents the relation between the power consumed (P_w) in a conductor and the square of the current intensity (I^2) through this conductor, so the slope of the line representing the relationship equals ..... .

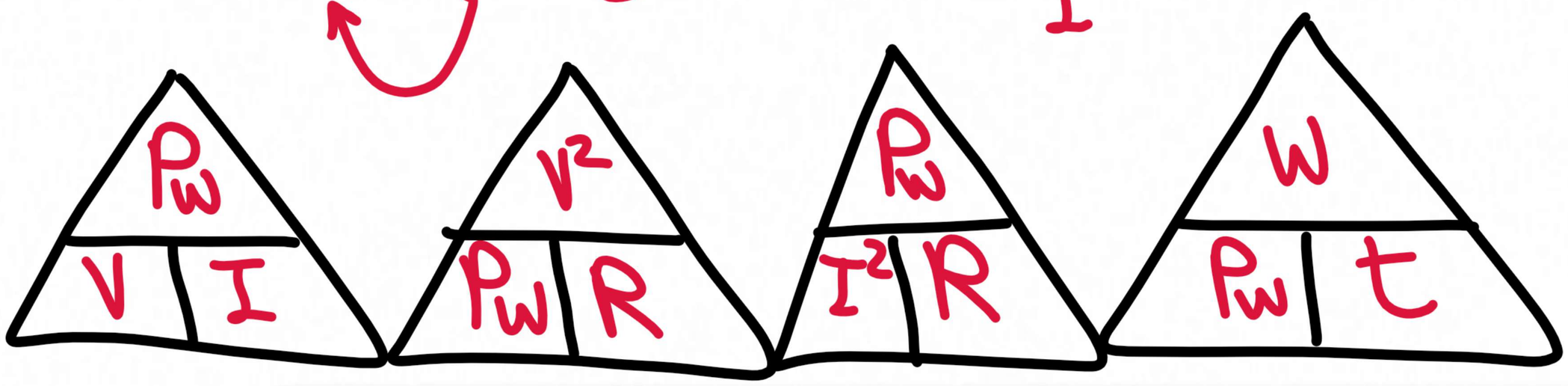
$P_w = \frac{W}{t} = VI = I^2R = \frac{V^2}{R}$

(Exp. 2025)

- the resistance of the conductor
- the reciprocal of the resistance of the conductor
- the potential difference across the conductor
- the square of the potential difference across the conductor



$P_w = I^2R$ $\quad$ $\text{slope} = \frac{P_w}{I^2} = R$



Questions

An electric lamp is labelled (100 V, 80 W) which means ..... .

$P_w = \frac{V^2}{R} \rightarrow 80 = \frac{100^2}{R} \rightarrow R = \frac{10000}{80}$

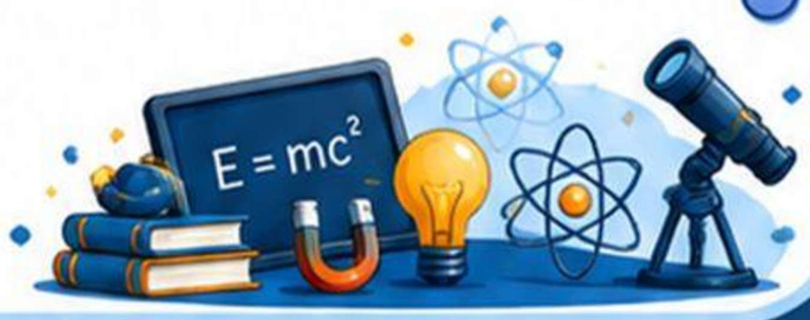
(Exp. 2025)

- the electrical resistance of the lamp is 0.8 Ω
- the electrical resistance of the lamp is 1.25 Ω
- when the potential difference across the lamp is 100 V, a current of 0.8 A flows through it
- when the potential difference across the lamp is 100 V, a current of 1.25 A flows through it

$R = 125\,\Omega$

$\frac{100}{1.25} = 80\,\Omega$

$R = \frac{V}{I} = \frac{100}{0.8} = \frac{1000}{8} = 125\,\Omega$



Mr.Abdelwahab

Questions

Two copper wires have the same length, if the cross-sectional area of the second wire is three times that of the first wire, the ratio between the resistance of the first wire and the resistance of the second wire equals ……. .

(Exp. 2021)

- ✗ 1/3
- ✓ 3/1
- ✗ 6/1
- ✗ 1/6

$A_2 = 3A_1$

$$\frac{R_1}{R_2} = \frac{A_2}{A_1} = \frac{3A_1}{A_1} = \frac{3}{1}$$

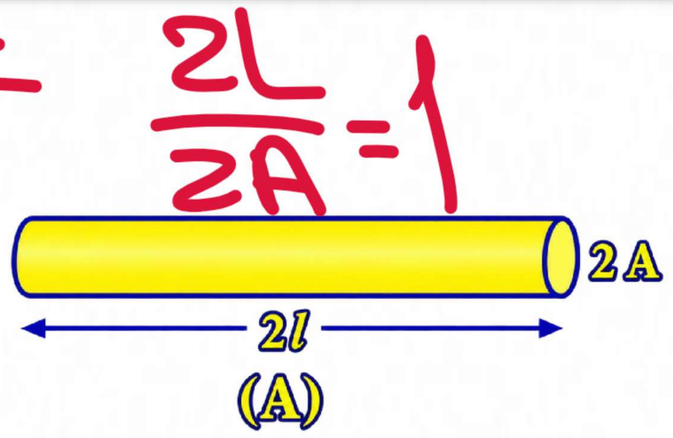
$$R = \frac{\rho_e \times l}{A}$$

Questions

The opposite figure shows four conductors of uniform cross-sections made of the same materials but with different dimensions, so their correct arrangement in an ascending order according to their electrical resistances is …… .

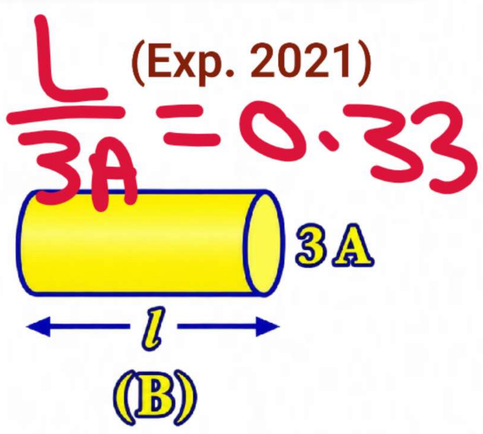
- ✗ B → C → A → D
- ✗ D → A → C → B
- ✗ C → A → B → D
- ✓ D → B → A → C

$$R = \frac{\rho_e \, l}{A}$$

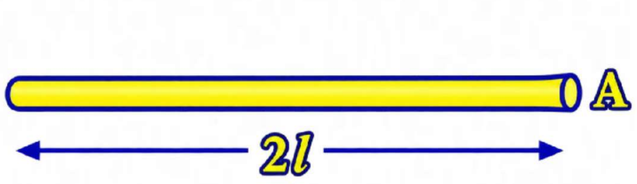


$\frac{2L}{2A} = 1$

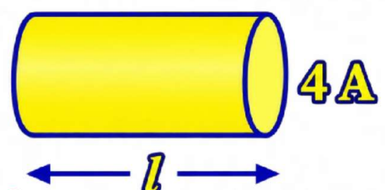
(Exp. 2021)



$\frac{L}{3A} = 0.33$



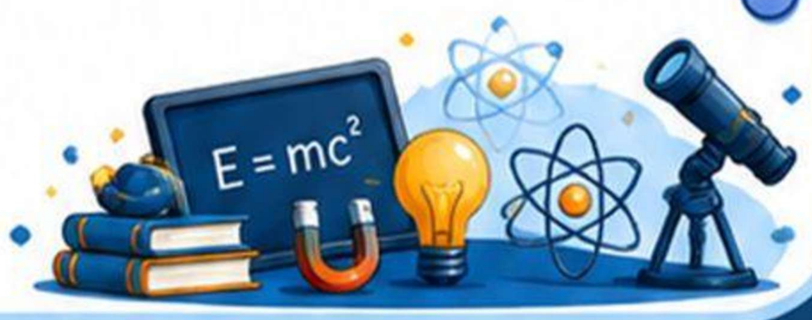
$\frac{2L}{A} = 2$



$\frac{L}{4A} = 0.25$

D → B → A → C

0.25 0.33 1 2



Mr.Abdelwahab

Questions

A conductor of length ℓ and cross-sectional area 3 A is connected between a potential difference V so that a current of intensity I passes in it, when the same potential difference is used with a different conductor of the same material as the first one, it was found that a current of intensity 3 I passes in it, so the length and the cross-sectional area of the second conductor are …….. .

	Length	Cross-sectional area
a	2ℓ	18 A
b	3ℓ	3 A
c	8ℓ	2 A
d	$\frac{1}{3}\ell$	$\frac{1}{3}$ A

$$\frac{1}{I} \propto \frac{\ell}{A} \rightarrow I \propto \frac{A}{\ell}$$ (Exp. 2021)

$$\frac{I_1}{I_2} = \frac{A_1}{A_2} \times \frac{L_2}{L_1}$$

$$\frac{I}{3I} = \frac{3A}{18A} \times \frac{L_2}{\ell} \quad \frac{L_2}{\ell} = \frac{1}{3}$$

Questions

Two wires are made of the same material, the diameter of the first wire is 3 times the diameter of the second wire and the resistance of the second wire is 4 times the resistance of the first wire, so the length of the second wire is ……. the length of the first wire.

- 3/4
- 72/2
- 4/9
- 36/3

$r_1 = 3 \times 2r_2$

$d_1 = 3d_2$

$r_1 = 3r_2$

$R_2 = 4R_1$

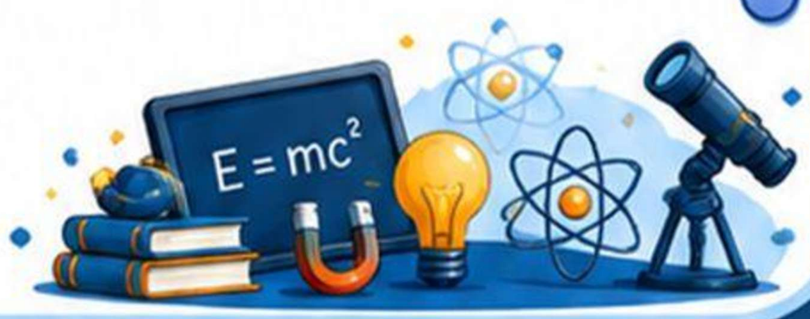
$L_2 = \ldots L_1$

$\frac{L_2}{L_1} = \ldots$

$$R = \frac{\rho_e L}{A} \rightarrow RA = \rho_e L$$ (1^{st} -session 21)

$$\frac{L_2}{L_1} = \frac{R_2}{R_1} \times \frac{A_2}{A_1} = \frac{R_2}{R_1} \times \left[\frac{r_2}{r_1}\right]^2$$

$$\frac{L_2}{L_1} = \frac{4R_1}{R_1} \times \left[\frac{r_2}{3r_2}\right]^2 = \frac{4}{9} *$$



Mr.Abdelwahab

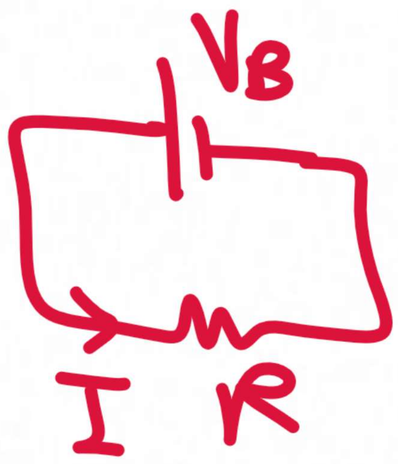
Questions

A current of I intensity passes through a conductor of ℓ length and cross-sectional area A, when the battery is changed with a different one, the current passing through the same conductor becomes 3 I, so the cross-sectional area of the conductor becomes …….. .

(2nd -session 21)

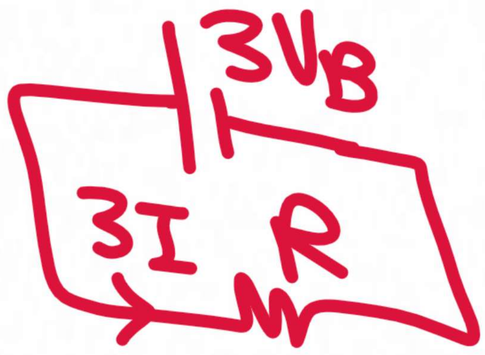
Same Conductor

- A
- 3 A
- ⅓ A
- 6 A



$$R = \frac{\rho_e \times \ell}{A}$$

all constant



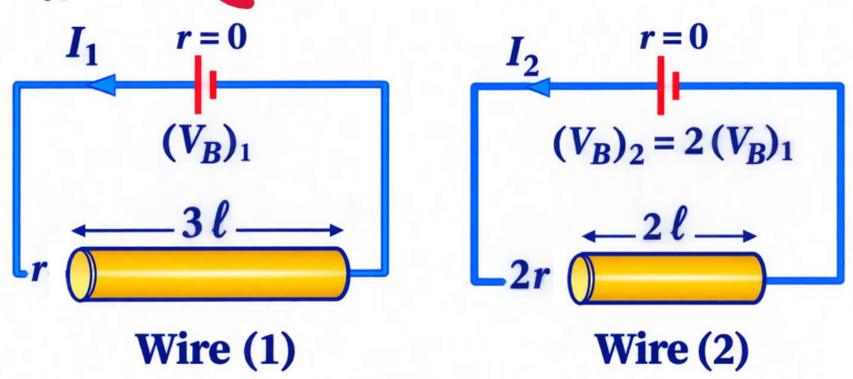
Questions

Two wires (1) and (2) are made of the same material, wire (ℓ) has length (3ℓ) and radius (r) while wire (2) has length (2ℓ) and radius (2 r) as shown in the figure. So, the ratio (I_1 / I_2) = ….. .

(1st -session 22)

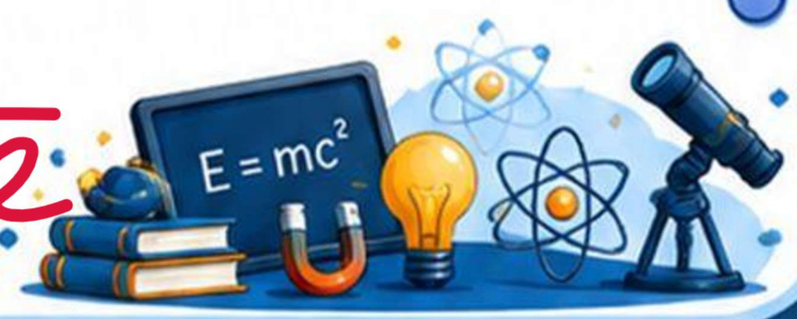
$$I = \frac{V_B}{R} = \frac{V_B}{\left[\frac{\rho_e \ell}{A}\right]} = \frac{V_B \times A}{\rho_e \times L} = \frac{V_B \times \pi \times r^2}{\rho_e \times L}$$

- $^{12}/_{1}$
- $^{1}/_{12}$
- $^{3}/_{2}$
- $^{1}/_{6}$



$$\frac{I_1}{I_2} = \frac{V_{B_1}}{V_{B_2}} \times \left[\frac{r}{r_2}\right]^2 \times \frac{L_2}{L_1}$$

$$= \frac{V}{2V} \times \left[\frac{r}{2r}\right]^2 \times \frac{2\ell}{3\ell} = \frac{1}{12}$$



Mr.Abdelwahab

Questions

The graph shows the relation between the electric resistance (R) and the reciprocal of the volume $^1/_{vol}$ for many wires of same length but of different materials. The correct arrangement of these wires concerning the conductivity (σ) of their materials is …….. .

(2nd -session 22)

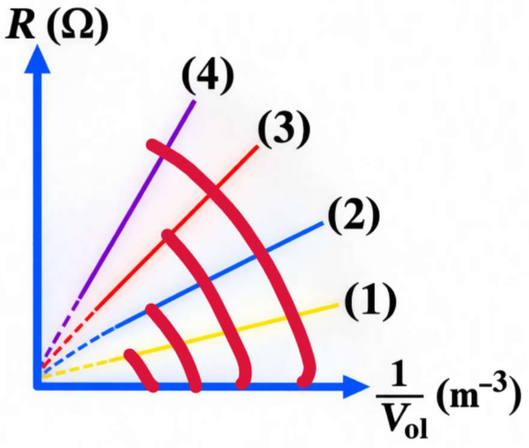
$\sigma = \frac{L}{RA} = \frac{L}{R \times Vol} = \frac{L^2}{R \times Vol}$

- $\sigma_4 > \sigma_1 > \sigma_3 > \sigma_2$
- $\sigma_1 > \sigma_3 > \sigma_2 > \sigma_4$
- ✓ $\sigma_1 > \sigma_2 > \sigma_3 > \sigma_4$
- $\sigma_4 > \sigma_3 > \sigma_2 > \sigma_1$

$Slope = \frac{R}{\frac{1}{Vol}} = R \times Vol = \frac{L^2}{\sigma}$

Same length

$\frac{L^2}{\sigma} = \tan\theta$



$\theta_4 > \theta_3 > \theta_2 > \theta_1$

$\sigma_4 < \sigma_3 < \sigma_2 < \sigma_1$

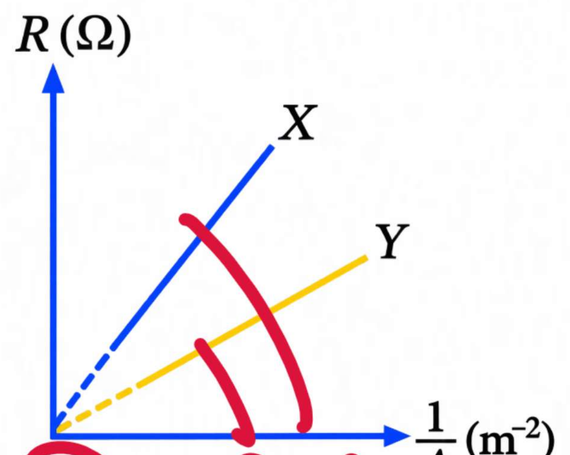
Questions

The two groups X and Y are wires of the same length and of different materials, the opposite figure represents the relation between the electric resistance (R) of the wire and the reciprocal of the cross-sectional area (1/A). which of the following is correct?

$Slope = \frac{R}{1/A} = RA = \rho_e \times l = \tan\theta$

(Exp. 23)

- ✓ Group X is made of a material with a higher resistivity.
- ✗ Group Y is made of a material with a higher resistivity.
- ✗ The electric conductivity of the two groups is equal.
- ✗ Group X is made of a material with a higher electric conductivity.



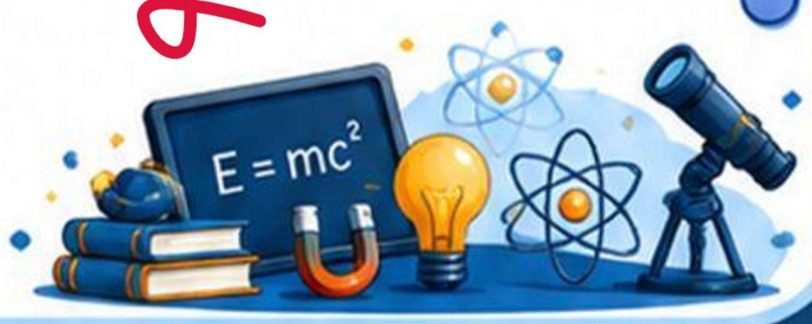
$\uparrow \rho_e \times l = \tan\theta \uparrow$

$\theta_x > \theta_y$

$\rho_{ex} > \rho_{ey}$

$\sigma_x < \sigma_y$

$R = \frac{\rho_e l}{A}$



Mr.Abdelwahab

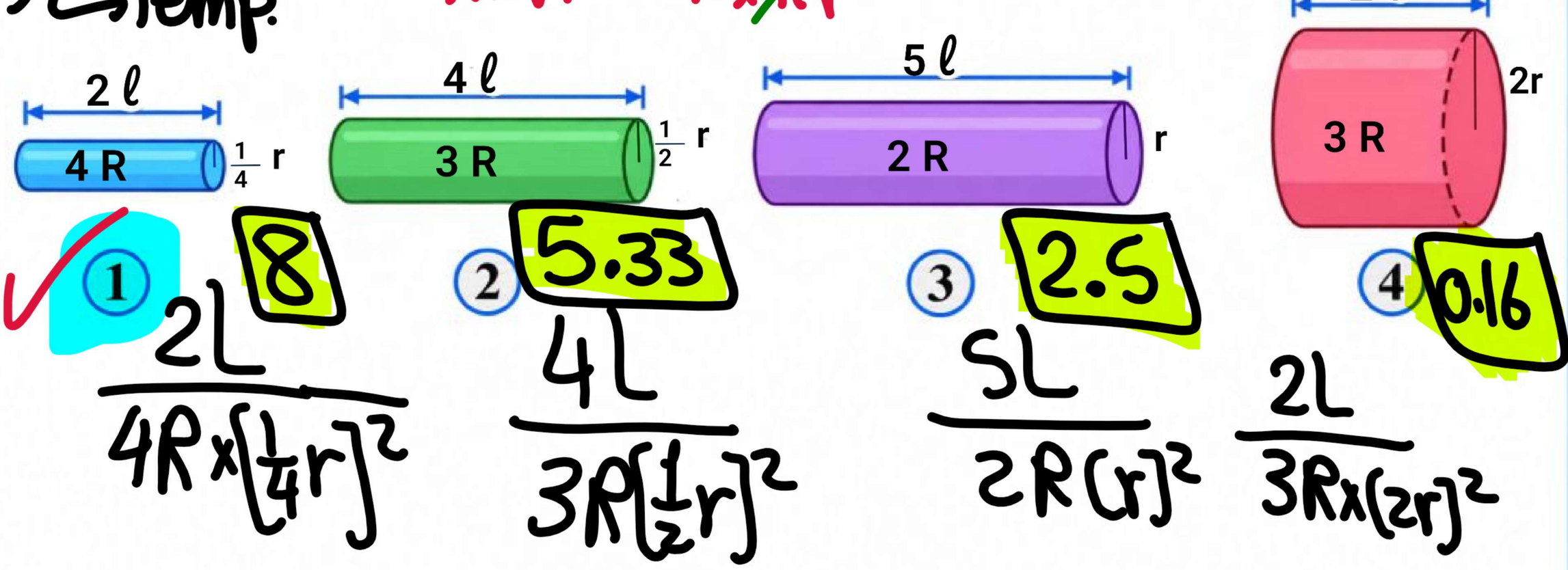
Questions

The following figures illustrate four wires of different materials, using the data on the figures, which wire material has the greatest conductivity at the same temperature?

(1st -session 23)

material
Temp.

$\sigma = \frac{L}{R \times A} = \frac{L}{R \times \pi r^2}$



① $\frac{2L}{4R \times [\frac{1}{4}r]^2}$ 8

② $\frac{4L}{3R[\frac{1}{2}r]^2}$ 5.33

③ $\frac{5L}{2R[r]^2}$ 2.5

④ $\frac{2L}{3R \times [2r]^2}$ 0.16

Questions

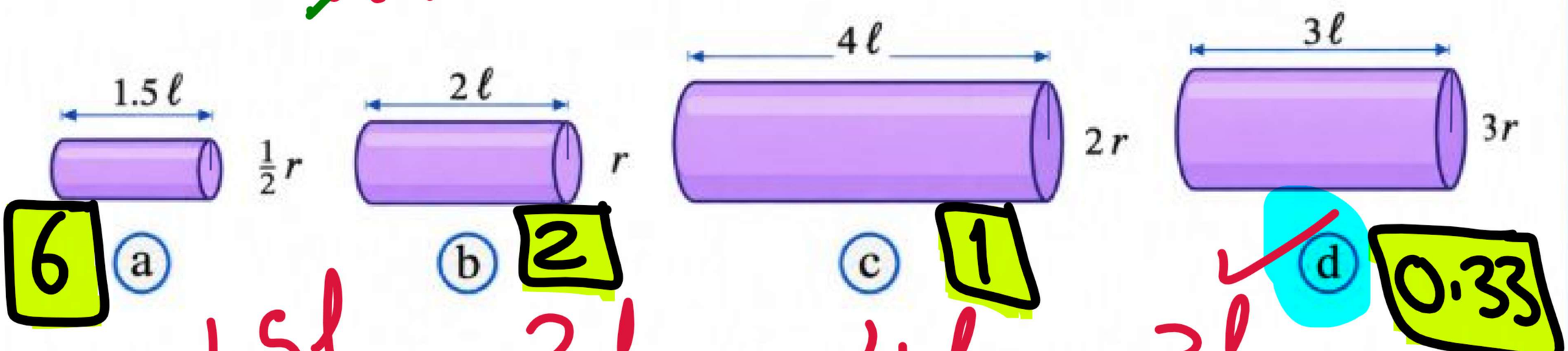
The figure represents four aluminum wires. Which wire has the least electric resistance?

Pe

$R = \frac{\rho_e l}{A}$

(2nd -session 23)

$R = \frac{\rho_e l}{\pi r^2}$

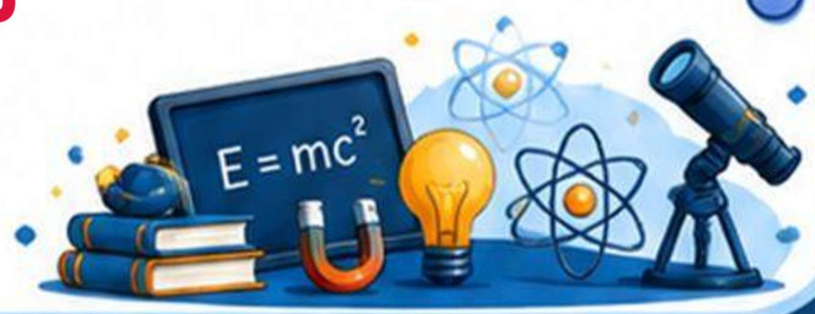


a 6 $\frac{1.5l}{[\frac{1}{2}r]^2}$

b 2 $\frac{2l}{[r]^2}$

c 1 $\frac{4l}{[2r]^2}$

d 0.33 $\frac{3l}{[3r]^2} =$



Questions

The figure represents a metallic conductor with different cross-sections. It is connected to the terminals of a battery in closed electric circuit. Knowing that the length of part (A) = The length of part (B). So, the ratio: (V_A / V_B) = …….. .

- ✓ R_A / R_B
- R_B / R_A
- $2R_A / R_B$
- $4R_A / R_B$

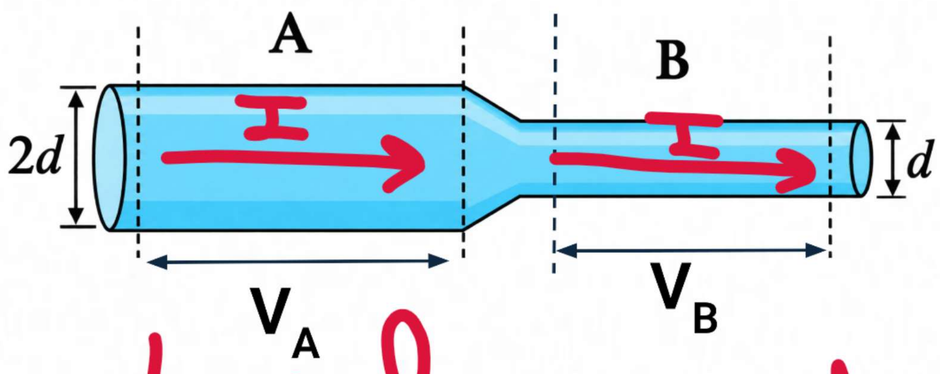
$N_A = N_B \times e$

$Q_A = Q_B \div t$

$I_A = I_B$

$V = IR \rightarrow \frac{V_A}{V_B} = \frac{R_A}{R_B}$

(1st -session 24)



Law of conservation of electric charges

Questions

The graph shows the relation between the change in the resistance of a number of wires made of three different materials X, Y and Z having the same cross-sectional area at the same temperature and the length. Which of the following choices is correct?

(1st session 24)

- $\sigma_z < \sigma_y < \sigma_x$
- $\sigma_z = \sigma_y = \sigma_x$
- ✓ $\sigma_z > \sigma_y > \sigma_x$
- $\sigma_z > \sigma_x > \sigma_y$

$\sigma = \frac{1}{\rho_e}$

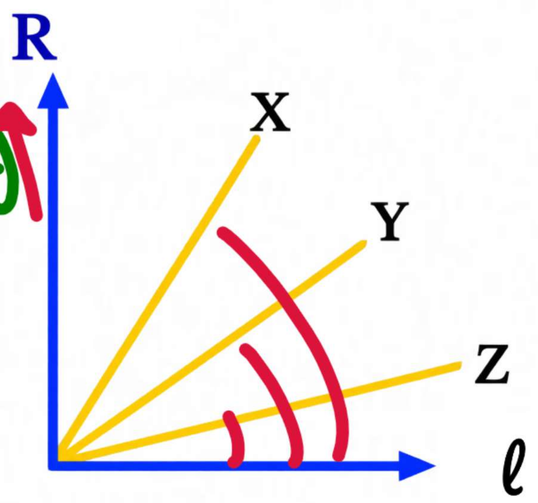
$R = \frac{\rho_e l}{A}$

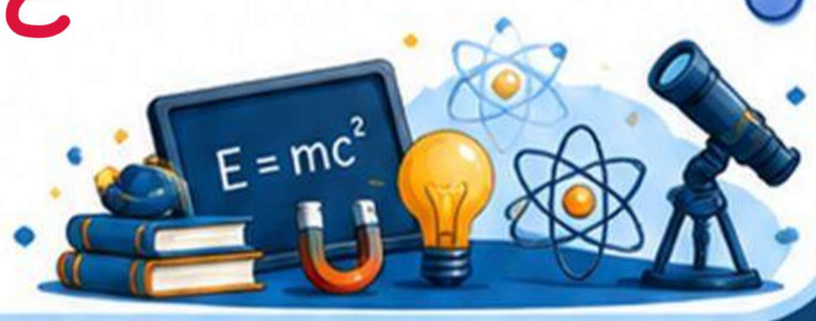
$\text{slope} = \frac{R}{l} = \frac{\rho_e l}{A} = \tan\theta$

$\theta_x > \theta_y > \theta_z$

$\rho_{ex} > \rho_{ey} > \rho_{ez}$

$\sigma_x < \sigma_y < \sigma_z$





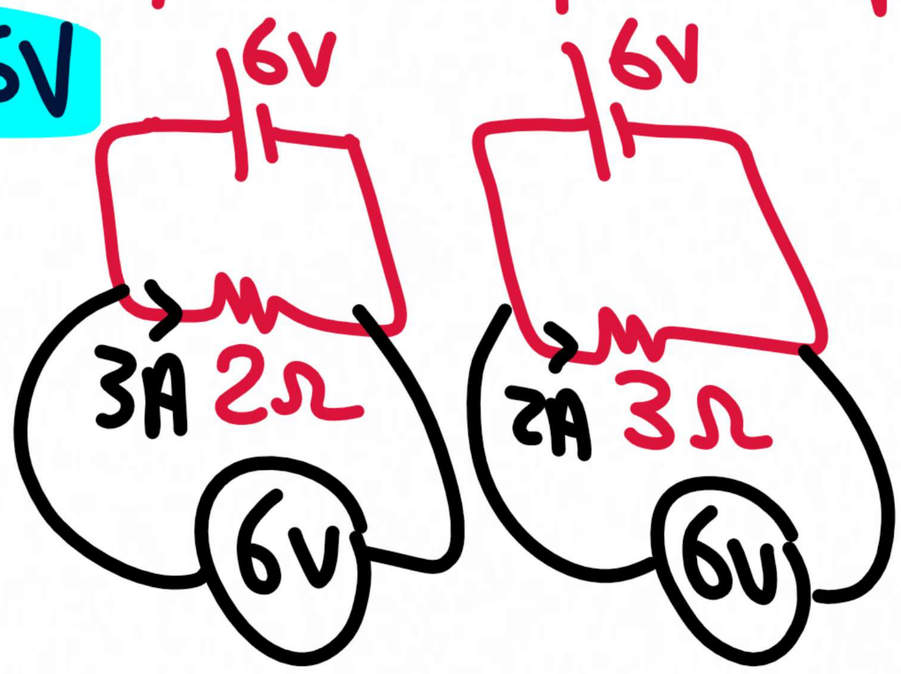
Mr.Abdelwahab

Figure (1) represents an electric circuit which contains a conductor AB at temperature T_1, so the voltmeter reading was V_1. When the temperature of the conductor AB is raised to T_2, the reading of the voltmeter becomes V_2 as in figure (2). Which of the following choices is correct?

(2nd -session 24)

$R \propto \rho_e \propto Temp.\uparrow$

- V_2 = Zero
- $V_1 = V_2 = 6V$
- $V_1 > V_2$
- $V_1 < V_2$



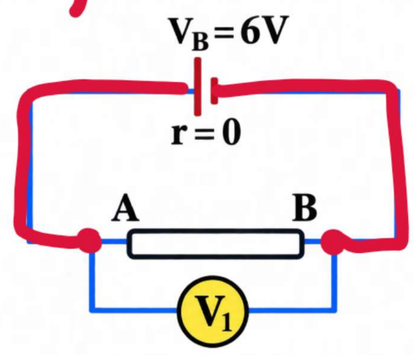


Figure (1)

$V_1 = 6V$

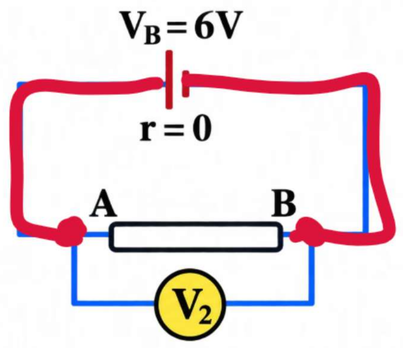


Figure (2)

$V_2 = 6V$

The opposite graph shows the relation between the resistances of two sets of wires A and B which are made of two different materials having the same cross-sectional area at constant temperature and the length, so which choice is correct?

$R = \frac{\rho_e l}{A}$

$Slope = \frac{R}{L} = \frac{\rho_e \uparrow}{A} = \tan\theta \uparrow$

$\theta_B > \theta_A$

$\rho_{eB} > \rho_{eA}$

$\sigma_B < \sigma_A$

(2nd -session 24)

$R \propto l$

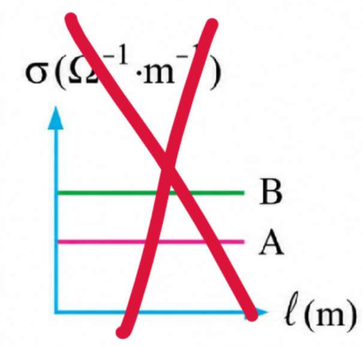


Figure (1)

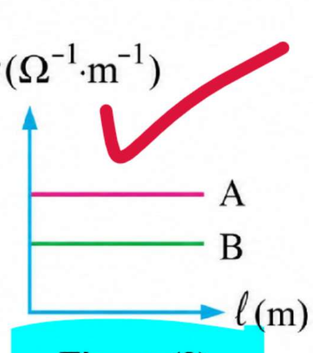


Figure (2)

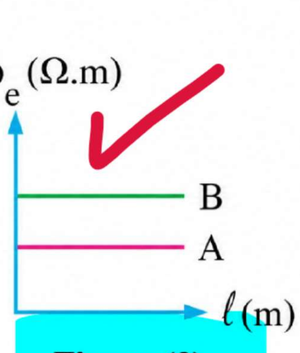


Figure (3)

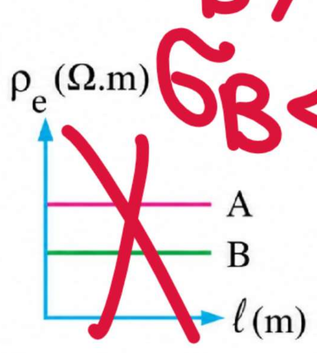
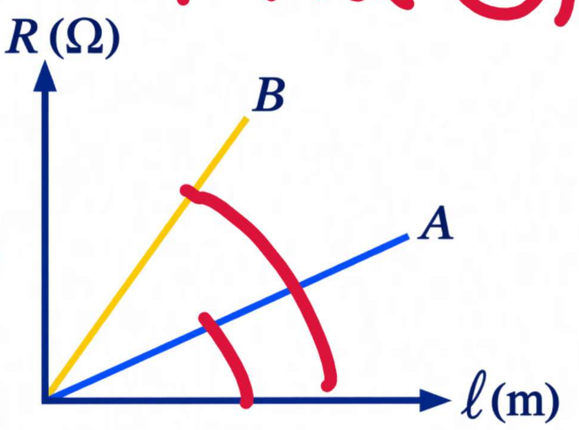
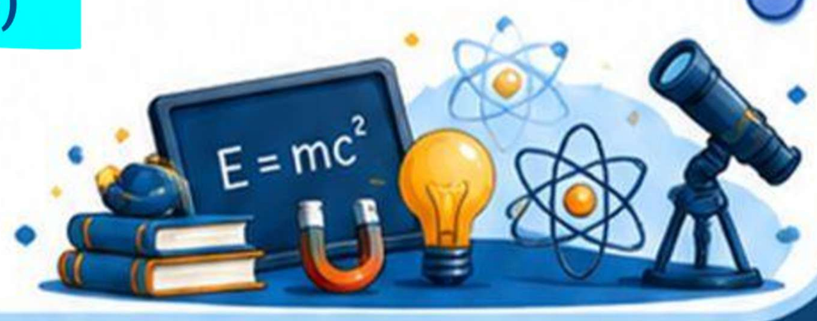


Figure (4)



$\sigma_B > \sigma_A$ $\sigma_B < \sigma_A$ $\rho_{eB} > \rho_{eA}$ $\rho_{eA} > \rho_{eB}$

- Figures (1) and (3)
- Figures (1) and (4)
- Figures (2) and (4)
- Figures (2) and (3)

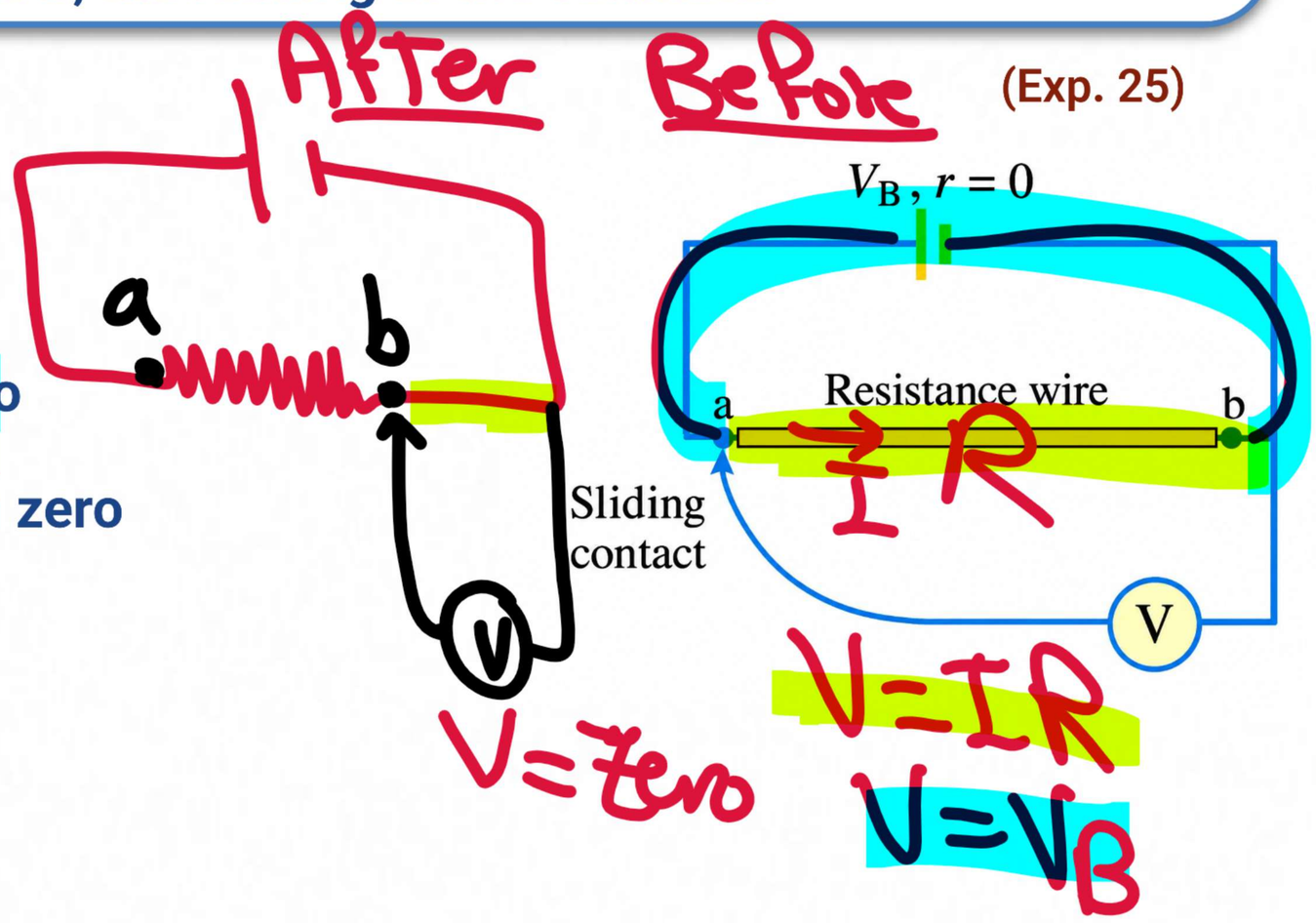


Questions

A battery of emf VB that has a negligible internal resistance is connected to a non-insulated resistance wire ab of uniform cross-section, a voltmeter and a sliding contact as shown in the opposite figure, so when moving the sliding contact from point a to b, the reading of the voltmeter

- increases
- remains unchanged
- decreases until it reaches zero
- decreases but it doesn't reach zero

(Exp. 25)



Questions

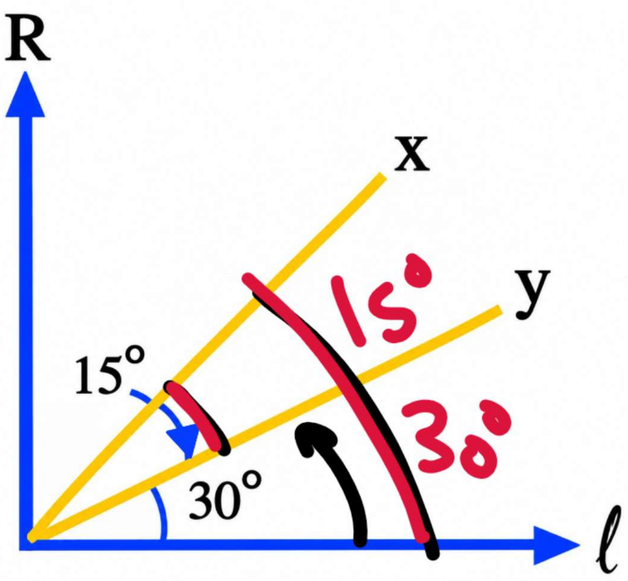
Two long copper wires x and y are different in thickness, where the opposite graph represents the relation between the resistance (R) and the length (ℓ) taken from each wire, so the ratio between the cross-sectional areas of the two wires (A_X / A_Y) is …… .

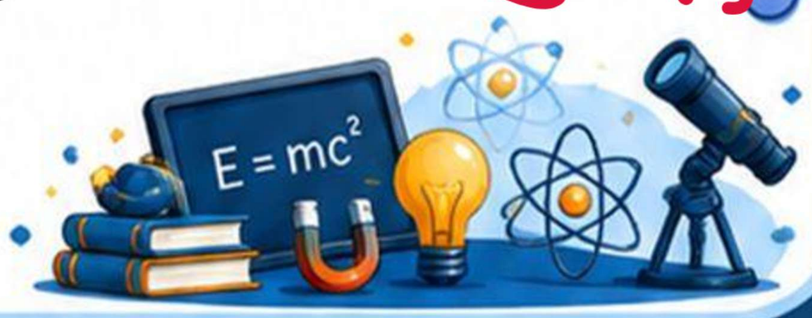
(Exp.25)

$$R = \frac{\rho_e \ell}{A}$$

$$\text{Slope} = \frac{R}{\ell} = \frac{\rho_e}{A} = \tan\theta \uparrow$$

$$\frac{A_x}{A_y} = \frac{\tan\theta_y}{\tan\theta_x} = \frac{\tan 30}{\tan 45} = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}$$





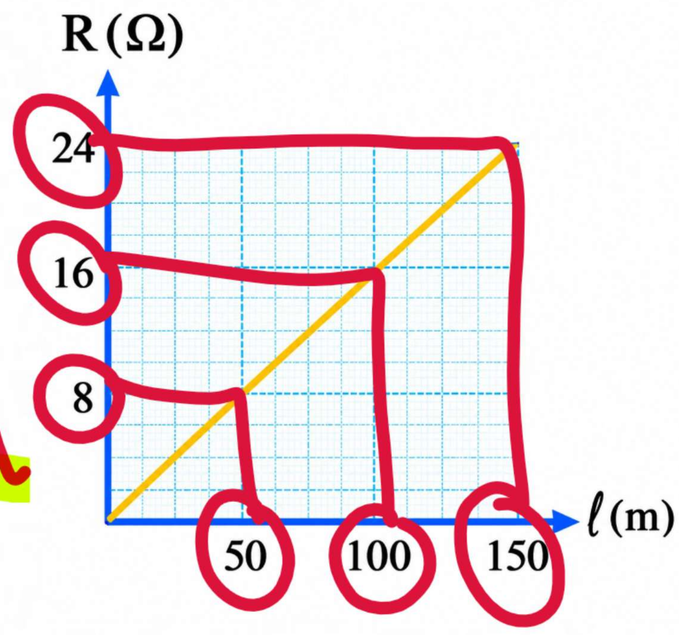
Questions

The opposite graph represents the relation between the electric resistance (R) of each wire of a group of wires made of the same material and having the same cross-sectional area that is $0.1\ cm^2$ and length (ℓ) for each of these wires, then the resistivity of the material of these wires (ρ_e) equals

(Exp. 25)

$$\rho_e = \frac{RA}{L} = \frac{24 \times 0.1 \times 10^{-4}}{150}$$

$$\rho_e = 1.6 \times 10^{-6}\ \Omega.m$$



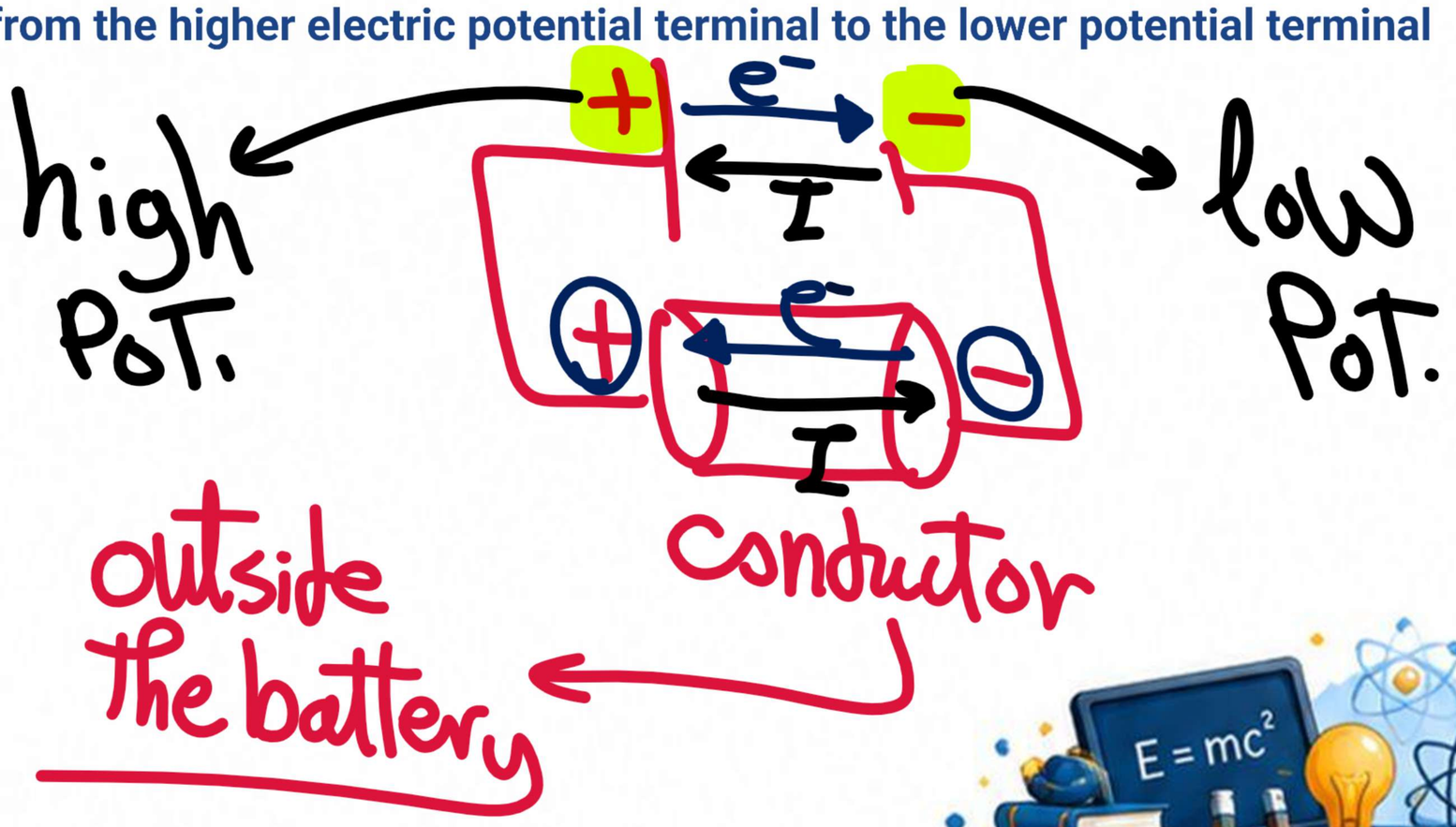
Questions

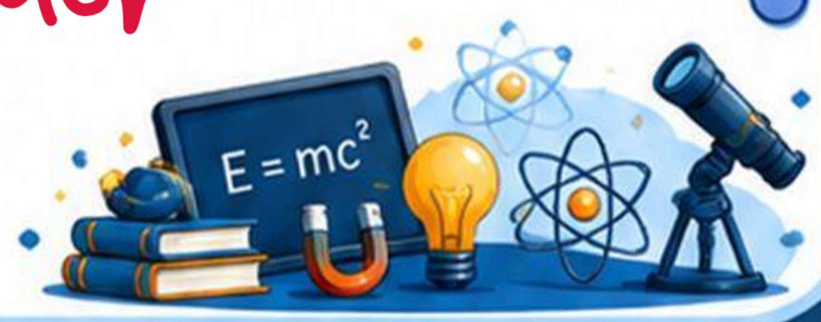
The electric current through a metallic conductor connected to the terminals of a voltage source is the flow of ..... .

Conductor (2nd session 25)

- Free electrons from the lower electric potential terminal to the higher potential terminal
- Free electrons from the higher electric potential terminal to the lower potential terminal
- Positive ions from the lower electric potential terminal to the higher potential terminal
- Positive ions from the higher electric potential terminal to the lower potential terminal

Battery





Questions

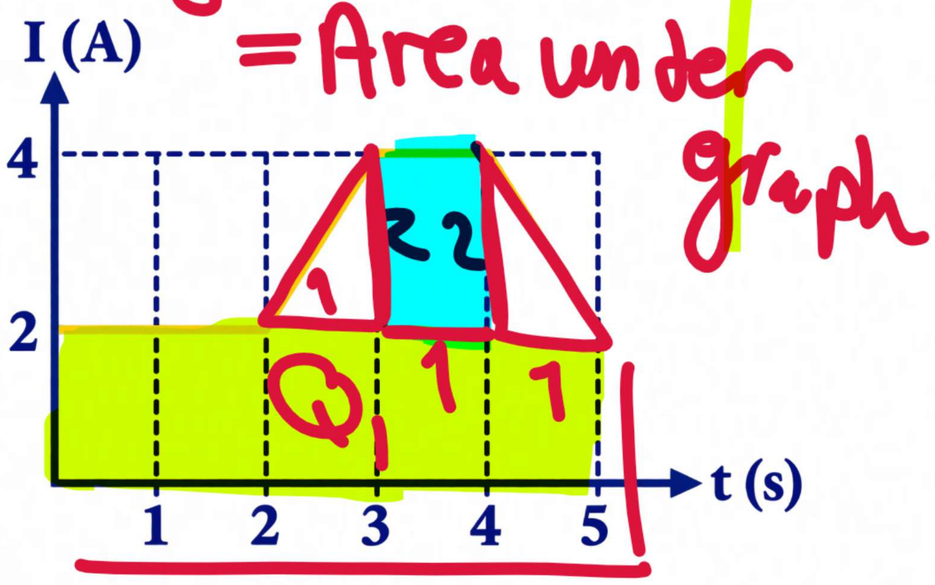
The figure shows the relation between the current flowing through a cross-section of a conductor and time. The quantity of electric charge passing through the cross-section during the whole 5 seconds is equal to ……. Coulombs.

(Exp. 25)

$Q = I \times t$

= y-axis × X-axis

= Area under graph



$Q_1 = 5 \times 2 = 10C$

$+ Q_2 = \frac{1}{2} \times 1 \times 2 = 1C$

$+ Q_3 = \frac{1}{2} \times 1 \times 2 = 1C$

$+ Q_4 = 2 \times 1 = 2C$

$10 + 1 + 1 + 2 = 14C$

2

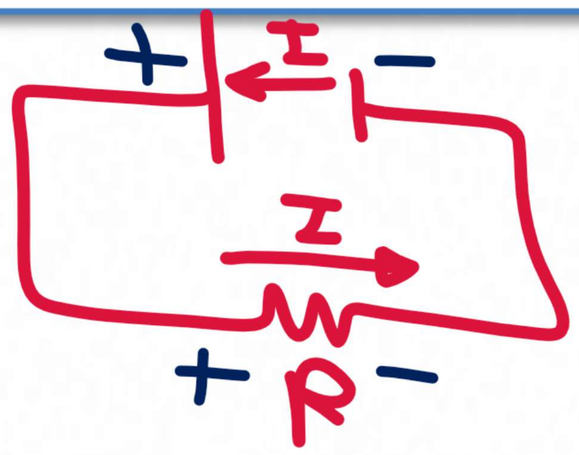
4

14

20

Questions

The figure shows an electric circuit containing two batteries (E_1) and (E_2), and a resistor (R), where ($E_1 > E_2$). Which of the following choices shows the correct order of the electric potentials (V_A), (V_B), and (V_c)?



$V_A > V_B > V_c$

$V_c > V_A > V_B$

$V_c > V_B > V_A$

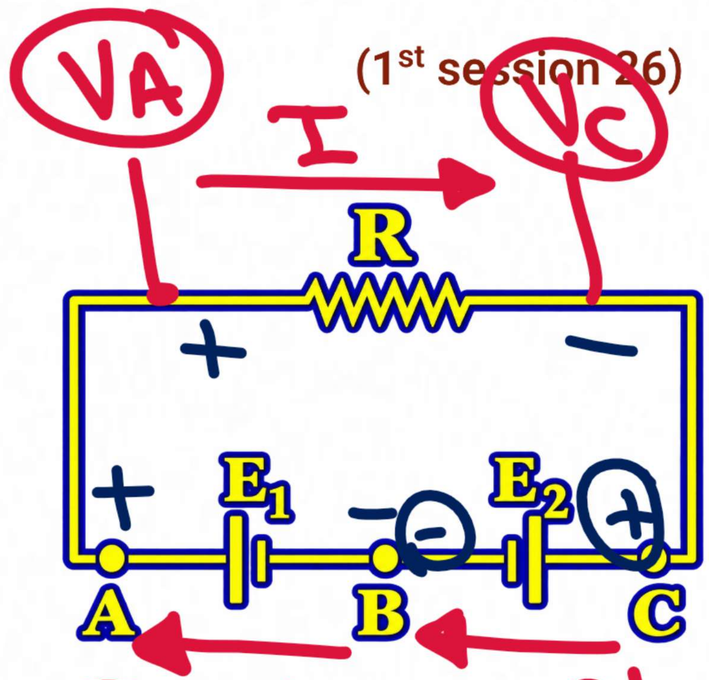
$V_A > V_c > V_B$

$V_A > V_B$

$V_c > V_B$

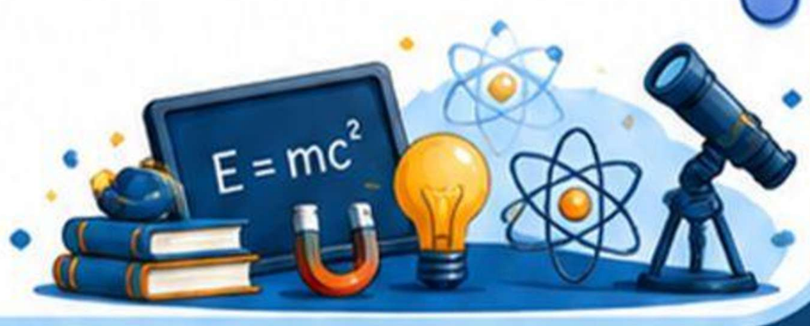
$V_A > V_c$

(1st session 26)



Discharging

Charging



Questions

conductor has length (ℓ) and cross-sectional radius (r), and another conductor is made of the same material and has the same length, but its radius is (r/3) Therefore, the resistance of the second conductor is …….

$R = \frac{\rho \ell}{\pi r^2}$

(Al-Azhar - Second session 2025)

$\frac{R_2}{R_1} = \left[\frac{r_1}{r_2}\right]^2 = \left[\frac{r}{r/3}\right]^2 = \frac{9}{1}$

- Equal to ⅓ of the first conductor's resistance
- Equal to 3 times the first conductor's resistance
- Equal to 6 times the first conductor's resistance
- Greater than the first conductor's resistance by 8 times its value

$R_2 = 9 \times R_1$

$R_1 = R \xrightarrow[\text{inc. by } 8]{\text{inc. To } 9} R_2 = 9R$

$\rho_e \downarrow \; R \downarrow \; \sigma \uparrow$

$R \downarrow$

Questions

Which change decreases the electrical resistance of a metallic conductor?

Temp. ↓

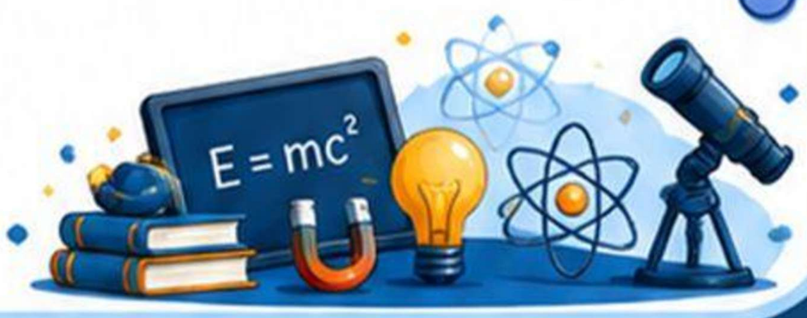
Collision ↓ KE ↓

- Connecting it to a battery of high emf
- Connecting it to a battery of low emf
- Decreasing the conductor's temperature
- Increasing the conductor's temperature

$\rho_e \propto$ Temp.

(1st session 26)

$R = \frac{\rho_e \times \ell}{A}$



Mr.Abdelwahab

Questions

Pe Reshape/drawn

A copper wire is stretched until its length becomes three times its original length. The ratio of its resistivity before stretching to its resistivity after stretching is equal to ……. .

(Al-Azhar – First session 2026)

- 1/1
- 1/3
- 1/9
- 3/1

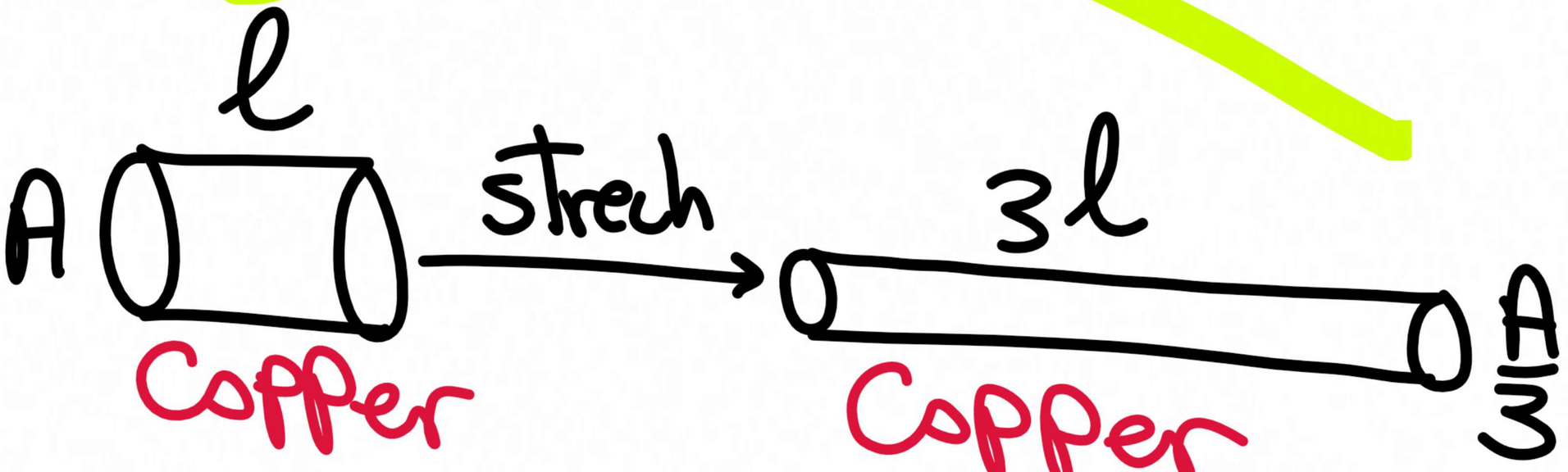
$l_1 = l \longrightarrow l_2 = 3l$

Pe → material X

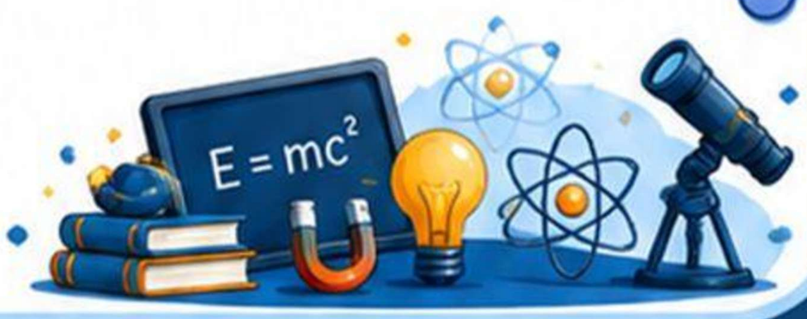
Const. Pe → Temp. X

$\frac{Pe_1}{Pe_2}$

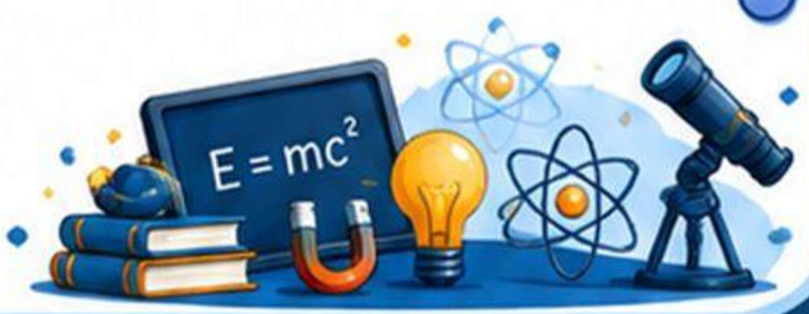
Resistance $\frac{R_1}{R_2} = \left[\frac{l_1}{l_2}\right]^2 = \left[\frac{L}{3L}\right]^2 = \frac{1}{9}$



Temprature constant

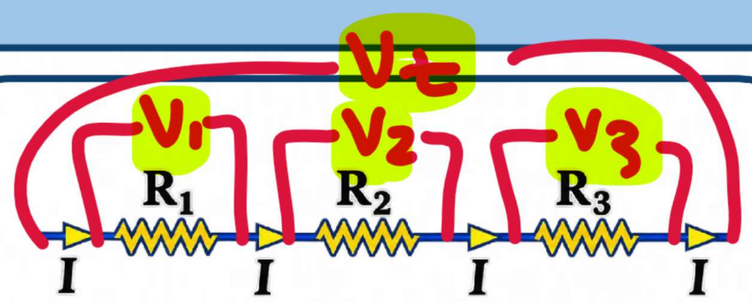


Lesson Two: Resistors Connection



Total resistance (R_{eq})

In Series

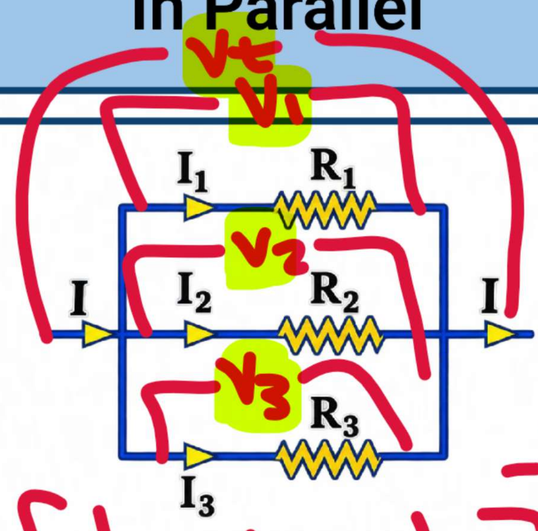


$$R_{eq} = R_1 + R_2 + R_3$$

$$I_1 = I_2 = I_3 = I_t$$

$$V_t = V_1 + V_2 + V_3$$

In Parallel



$$R_{eq} = \left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right]^{-1}$$

$$V_t = V_1 = V_2 = V_3$$

$$I_t = I_1 + I_2 + I_3$$

- If the resistors are identical (Equal), each having a value of (R), and their number is (N)

Parallel $\longrightarrow R_{eq} = \frac{R}{N}$

Series $\longrightarrow R_{eq} = R \times N$

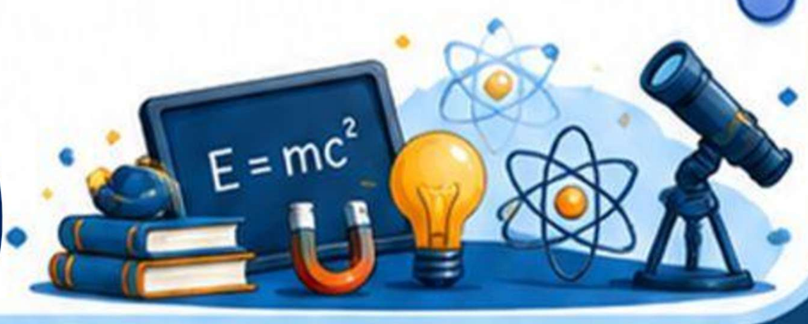
$R_{Series} = R \times N \longrightarrow R_s = N \times R_p \times N$

$R_{Parallel} = \frac{R}{N} \rightarrow R = N \times R_p$

$R_s = N^2 \times R_p$

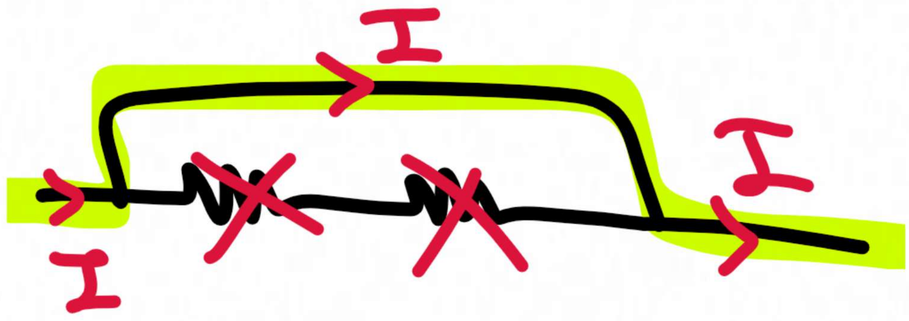
$N = \sqrt{\frac{R_s}{R_p}}$

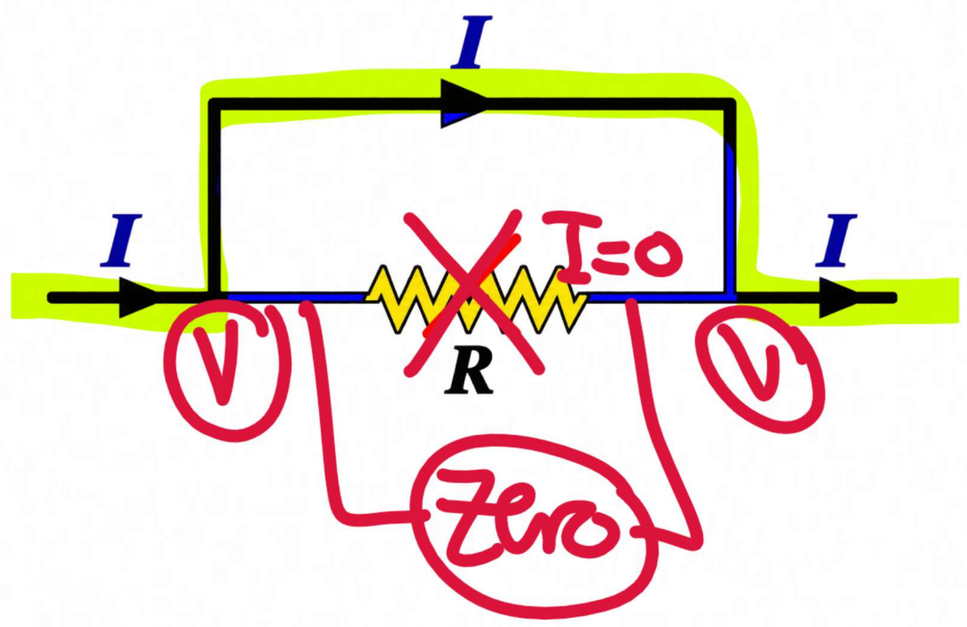
$R = \sqrt{R_s \times R_p}$



Cases of canceling an electrical resistance

Short Circuit

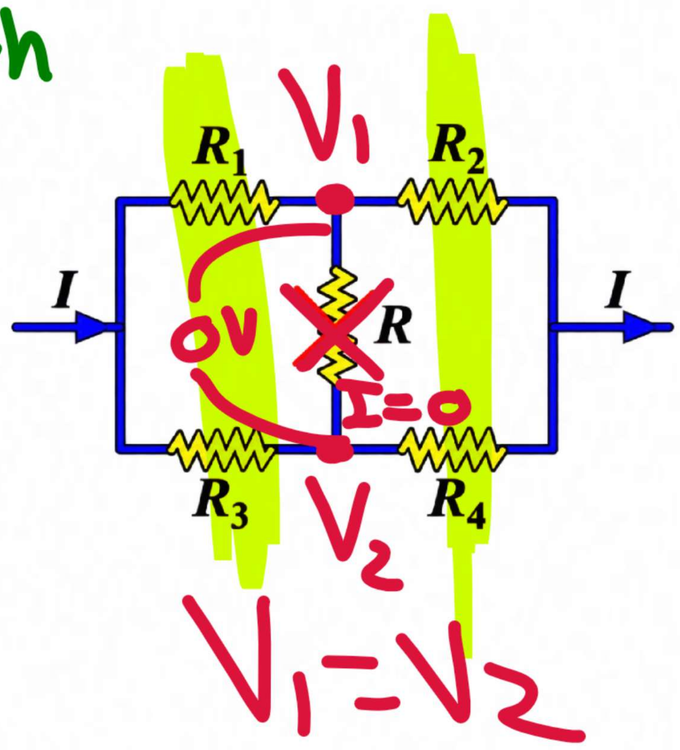


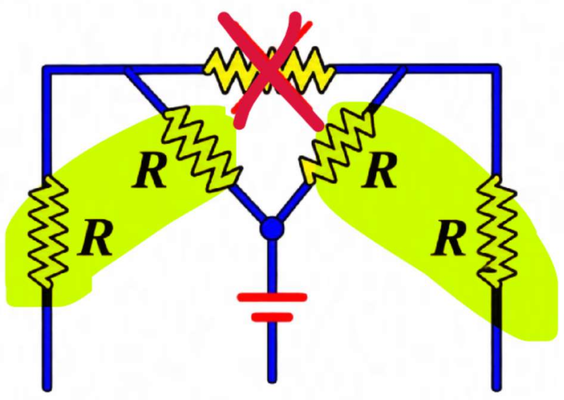


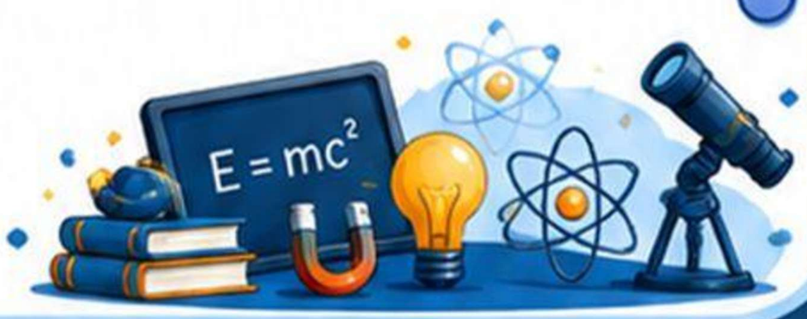
V=0 Winston Bridge

Zero branch

$$\frac{R_1}{R_2}=\frac{R_3}{R_4}$$

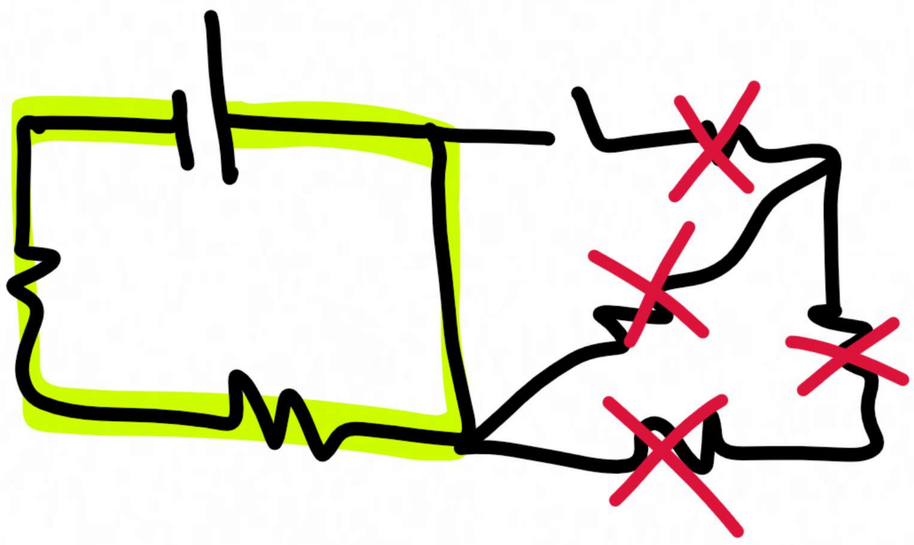


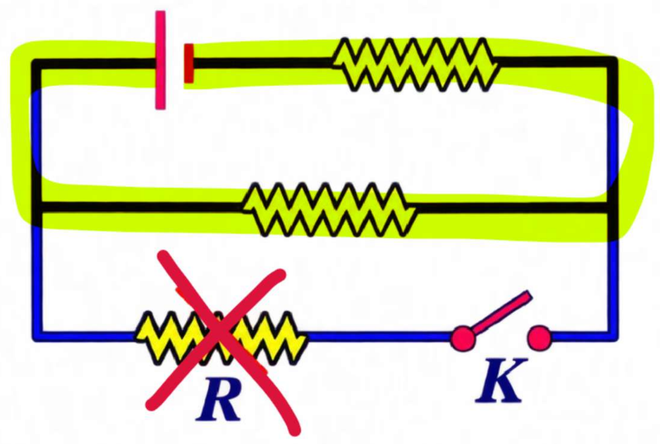




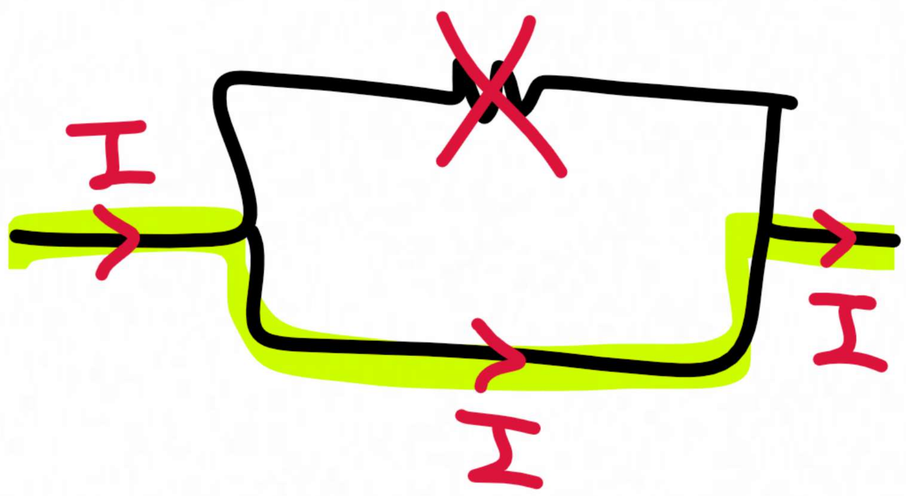
Cases of canceling an electrical resistance

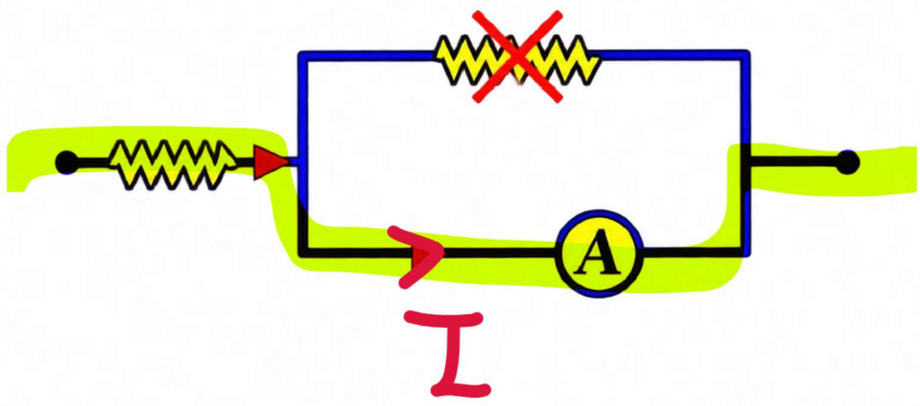
Open key



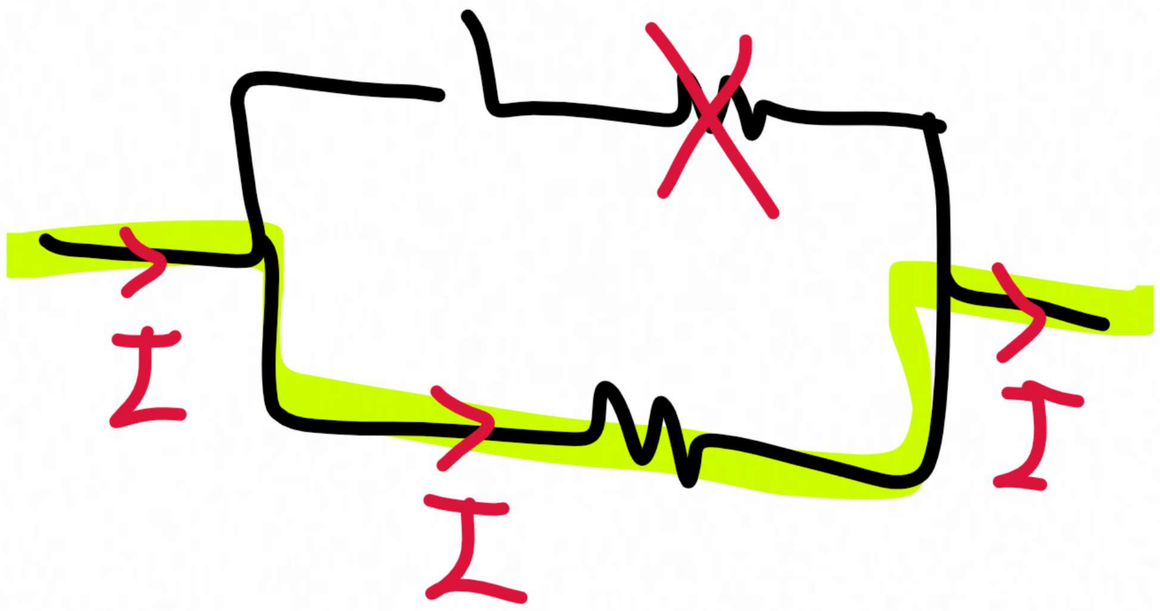


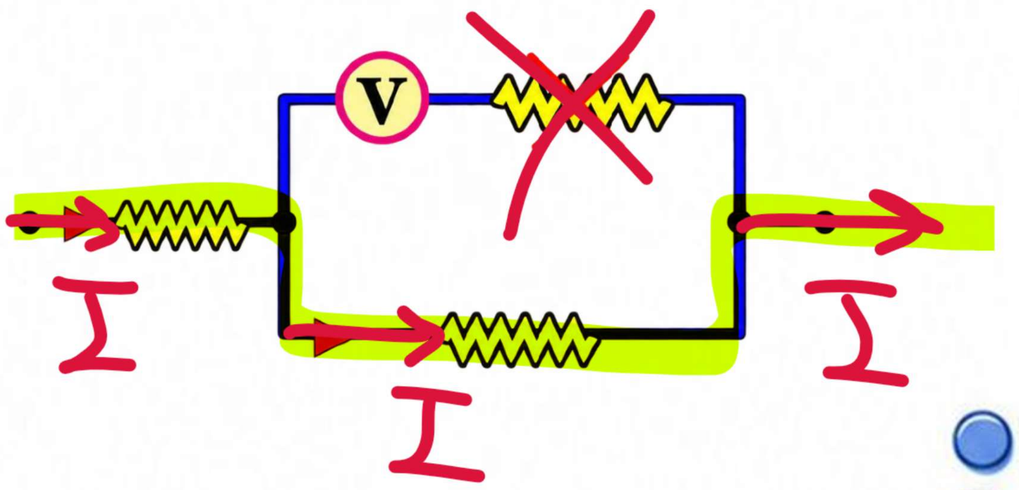
Ideal Ammeter

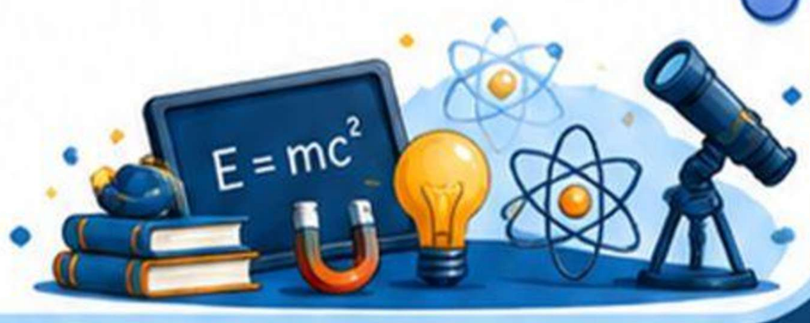




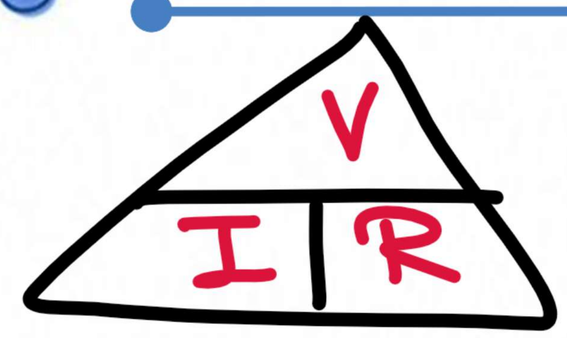
Ideal Ammeter



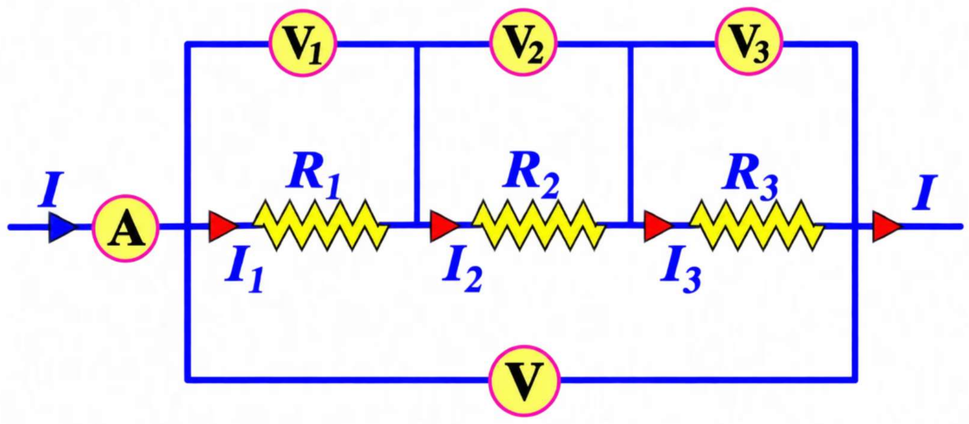




Distribution

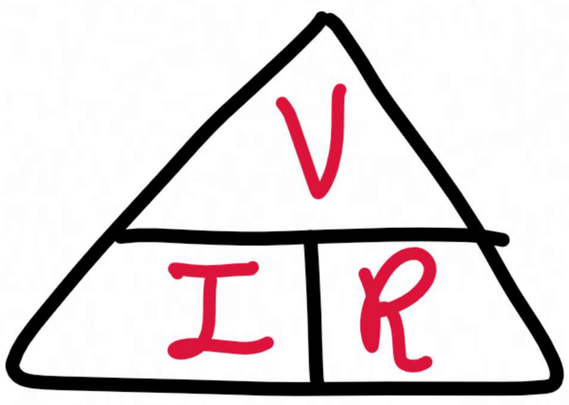


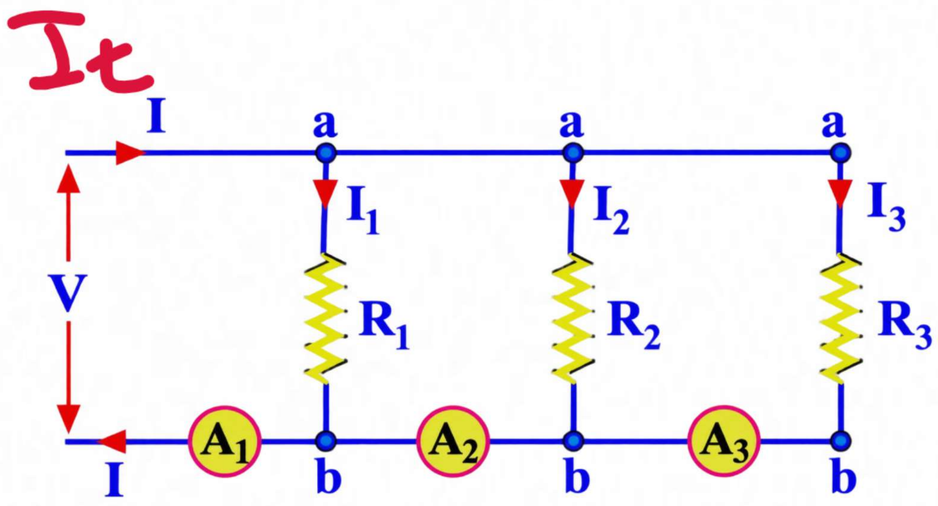
$$I_1 = I_2 = I_3 = I$$



$$I = \frac{V_t}{R_t} = \frac{V_1}{R_1} = \frac{V_2}{R_2} = \frac{V_3}{R_3}$$

$$V_1 = I R_1 \qquad V_2 = I R_2 \qquad V_3 = I R_3$$

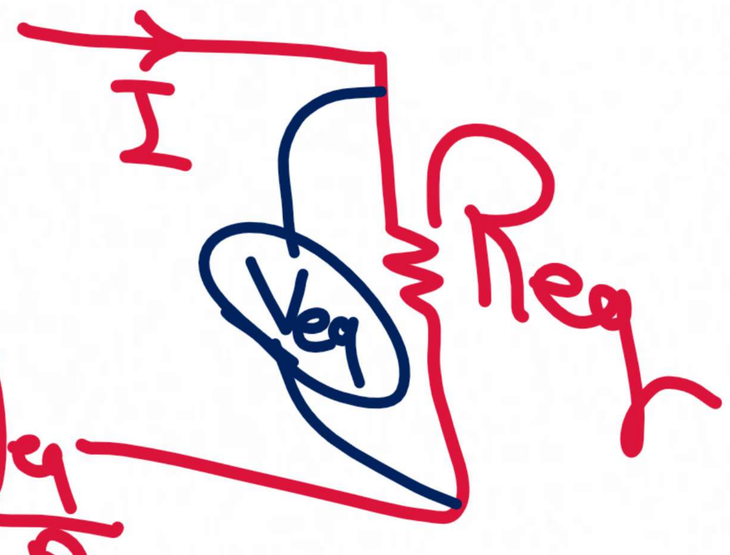




1. $R_{eq} = \left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right]^{-1}$

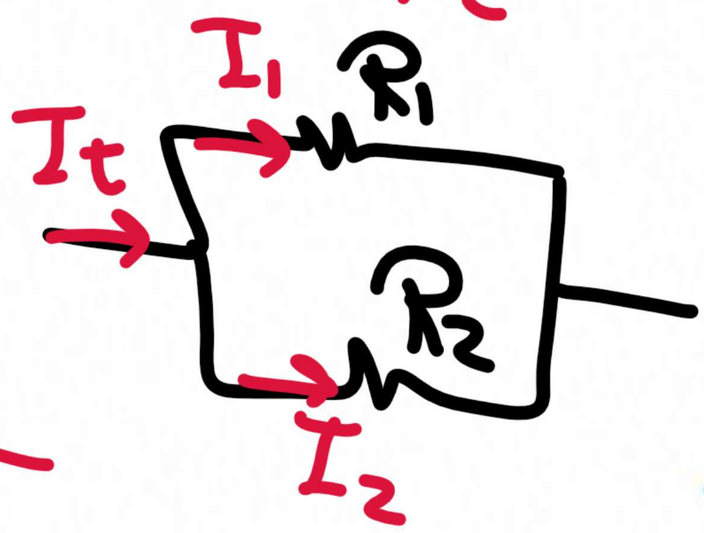
2. $V_{eq} = I_t \times R_{eq}$

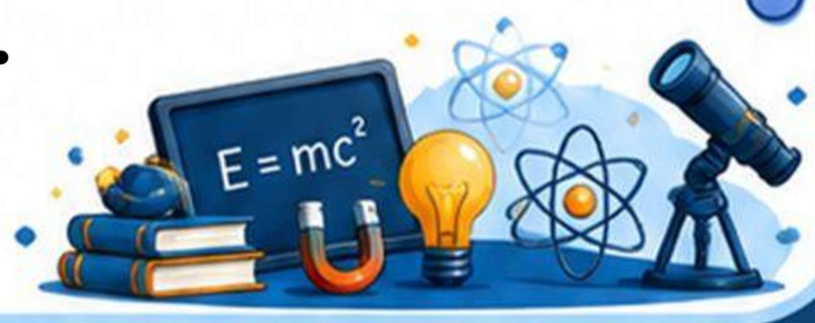
3. $I_1 = \frac{V_{eq}}{R_1} \quad I_2 = \frac{V_{eq}}{R_2} \quad I_3 = \frac{V_{eq}}{R_3}$



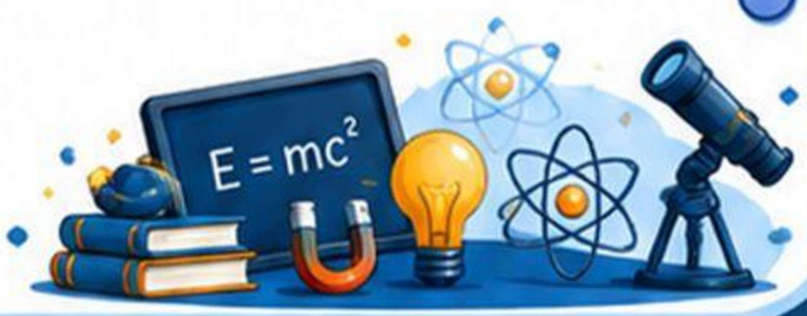
$$I_1 = \frac{I_t \times R_2}{R_1 + R_2}$$

$$I_2 = \frac{I_t \times R_1}{R_1 + R_2}$$



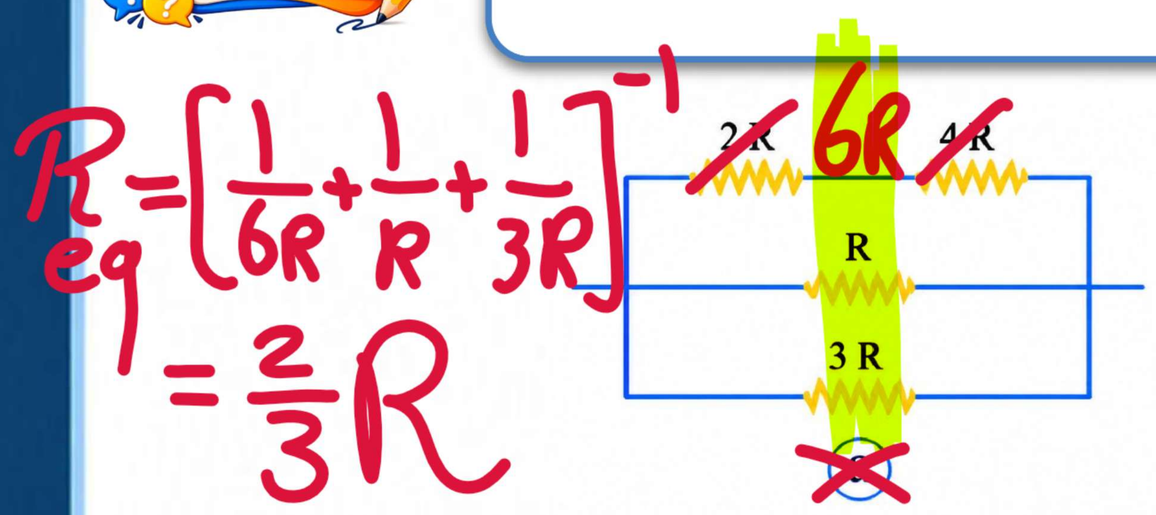


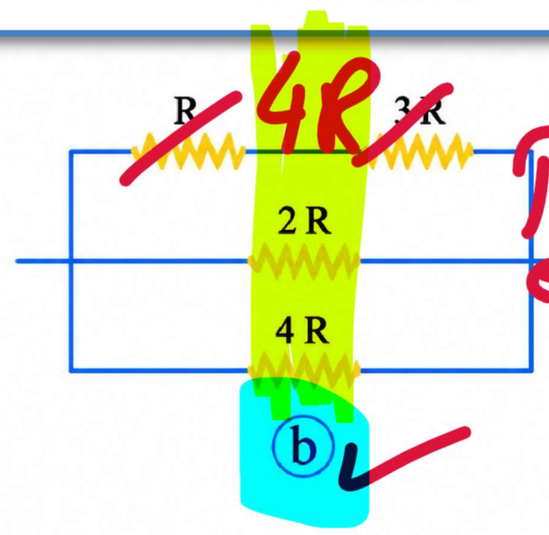
يلا بينا...
نتأكد من فهمك

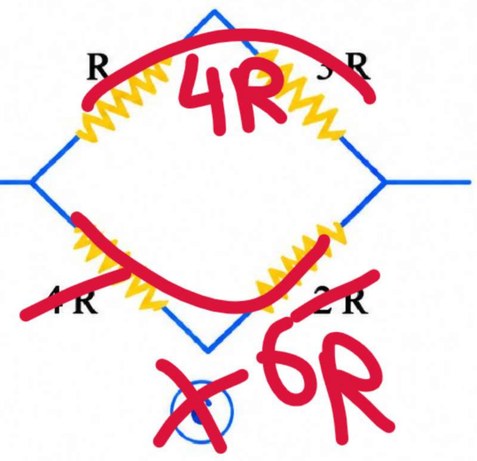

E = mc²

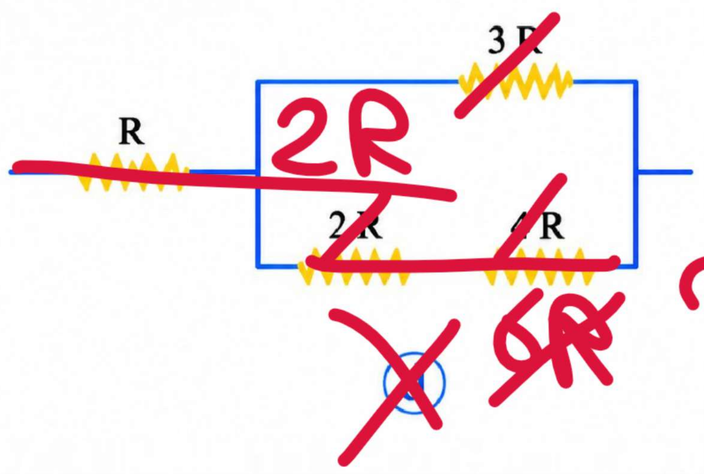
Which combination of the following resistors gives an equivalent resistance or value R?

(Exp. May 21)





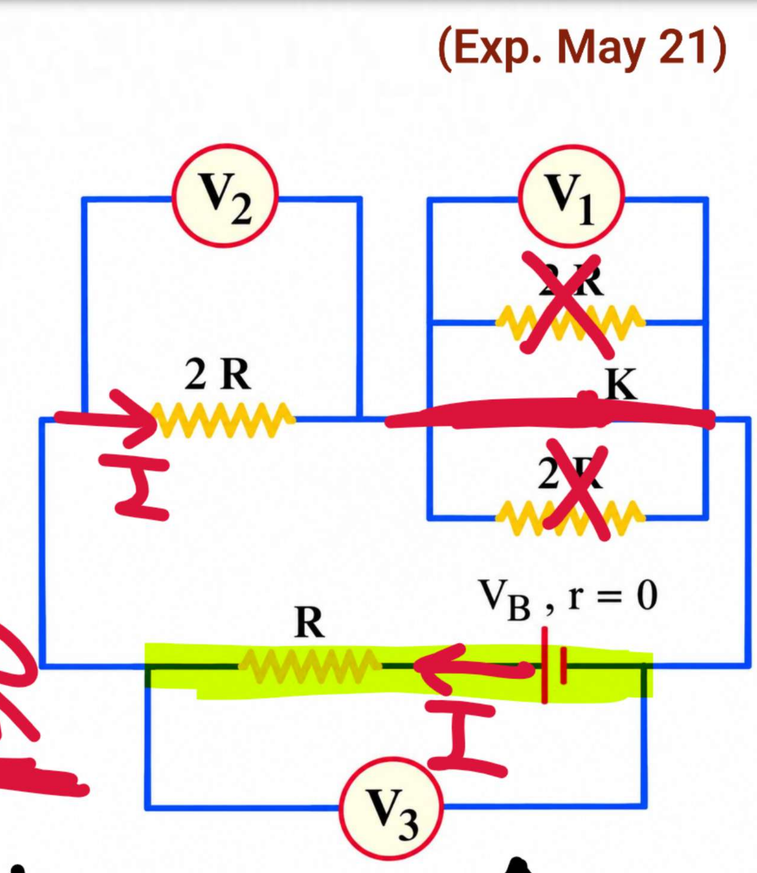


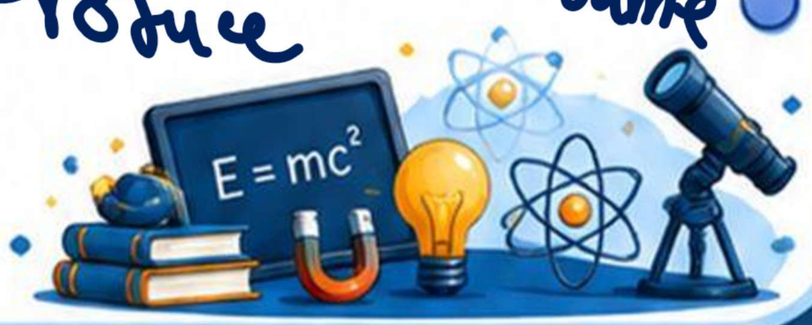


In the opposite electric circuit, when closing the switch (K), which of the following rows shows what happens to the readings of the voltmeters V_1, V_2 and V_3 correctly?

(Exp. May 21)

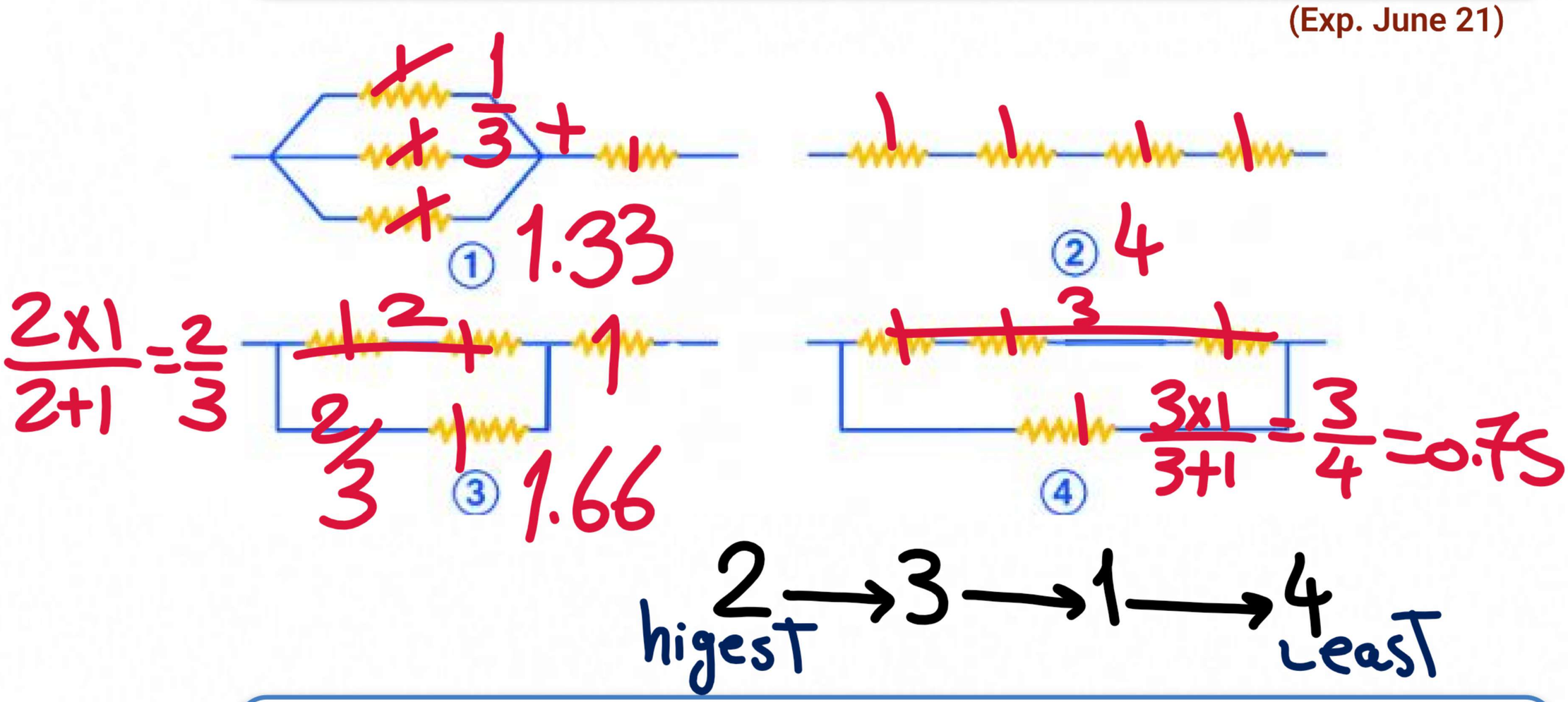
	V_1	V_2	V_3
a	Becomes zero	Increases	Decreases
	Increases	Increases	Decreases
	Becomes zero	Decreases	Increases
	Increases	Increases	Increases





Four equal resistors are connected together as in the following figures, so the correct arrangement of the figures from the highest equivalent resistance to the least one is …… .

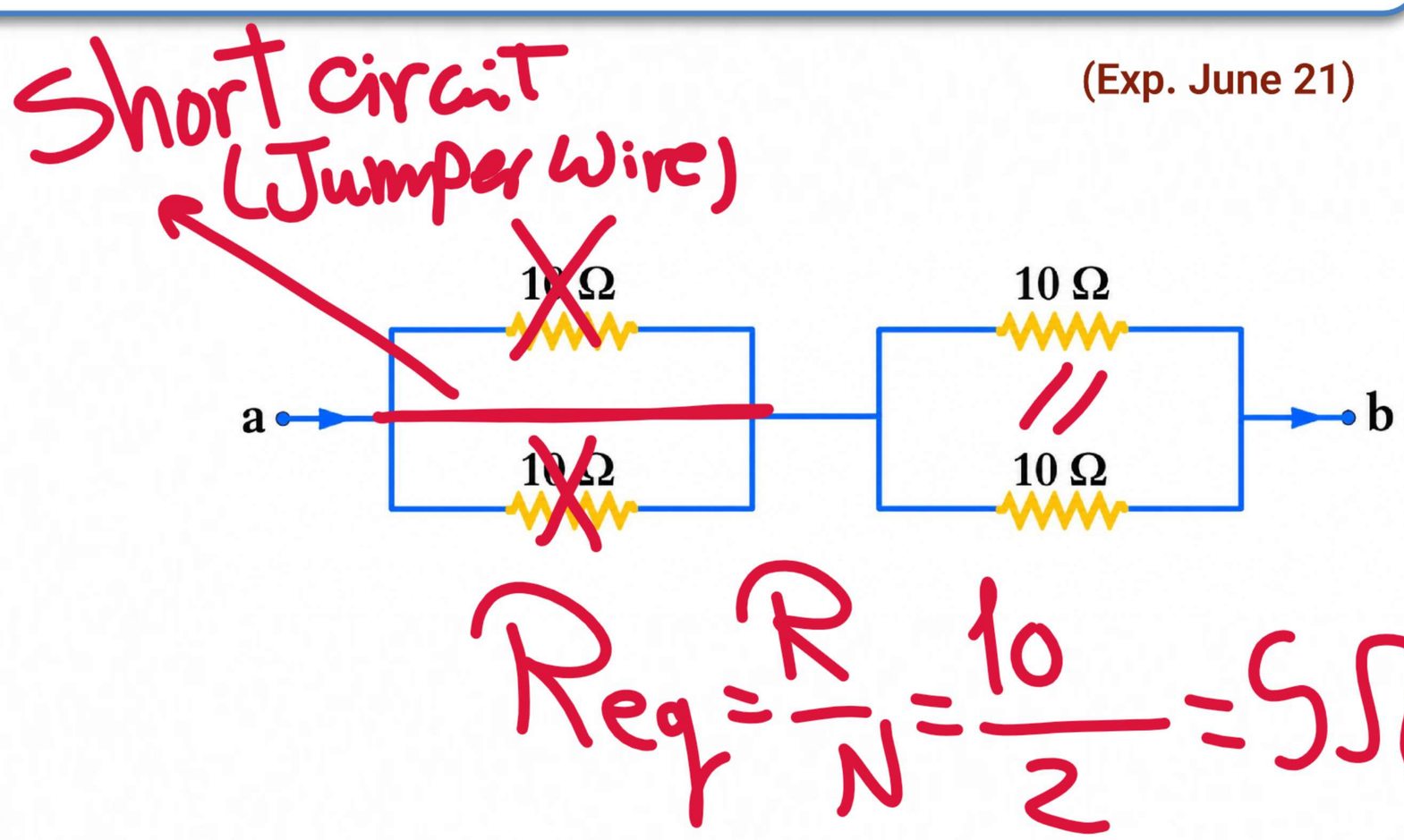
(Exp. June 21)

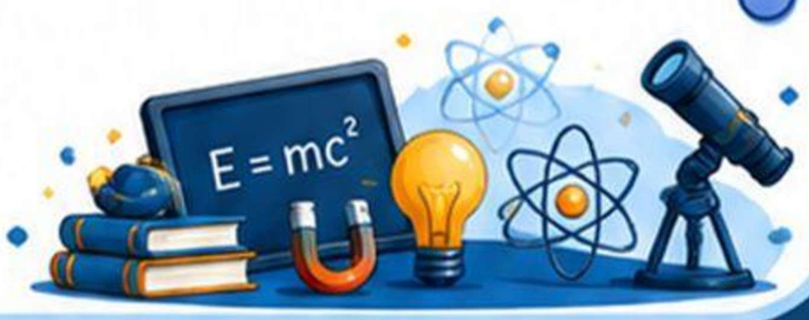


In the opposite electric circuit, the equivalent resistance between points a and b equals ……. .

(Exp. June 21)

- 5 Ω
- 10 Ω
- 20 Ω
- 40 Ω



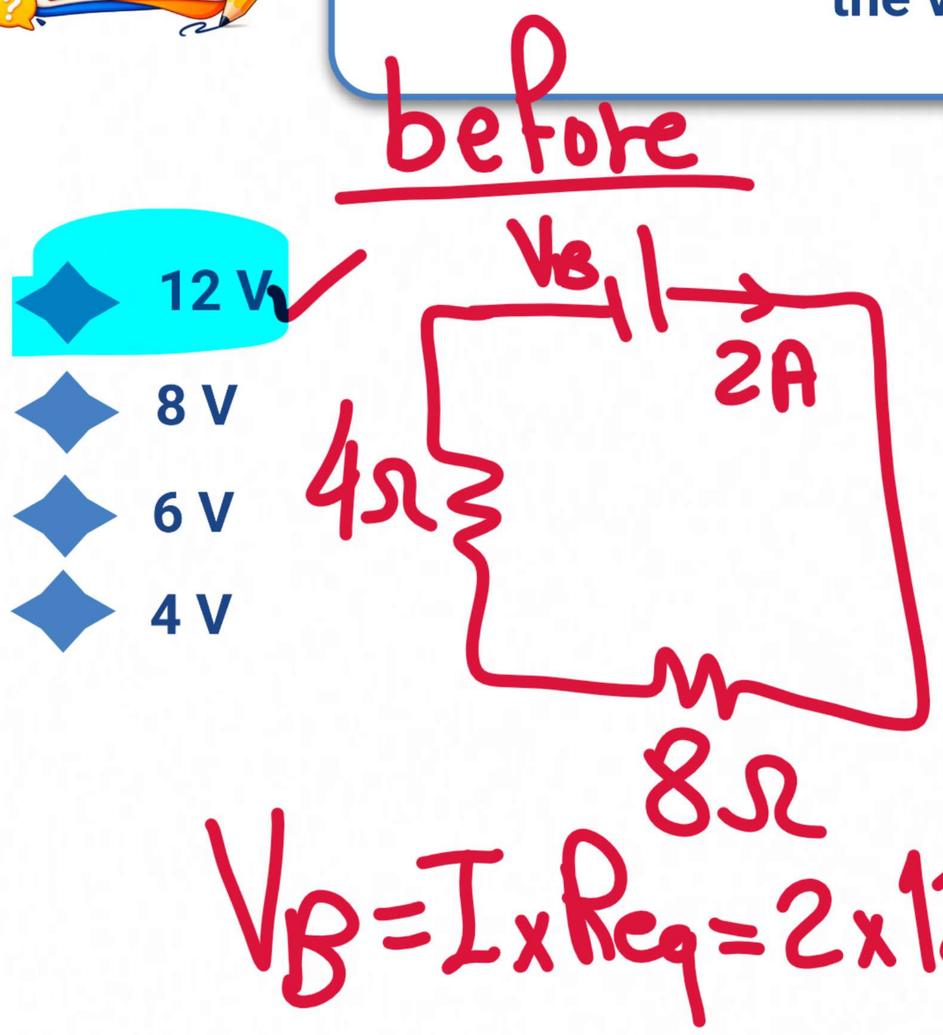


Questions

In the opposite electric circuit, when the key (K) is closed, the voltmeter reading equals ……. .

(Exp. June 21)

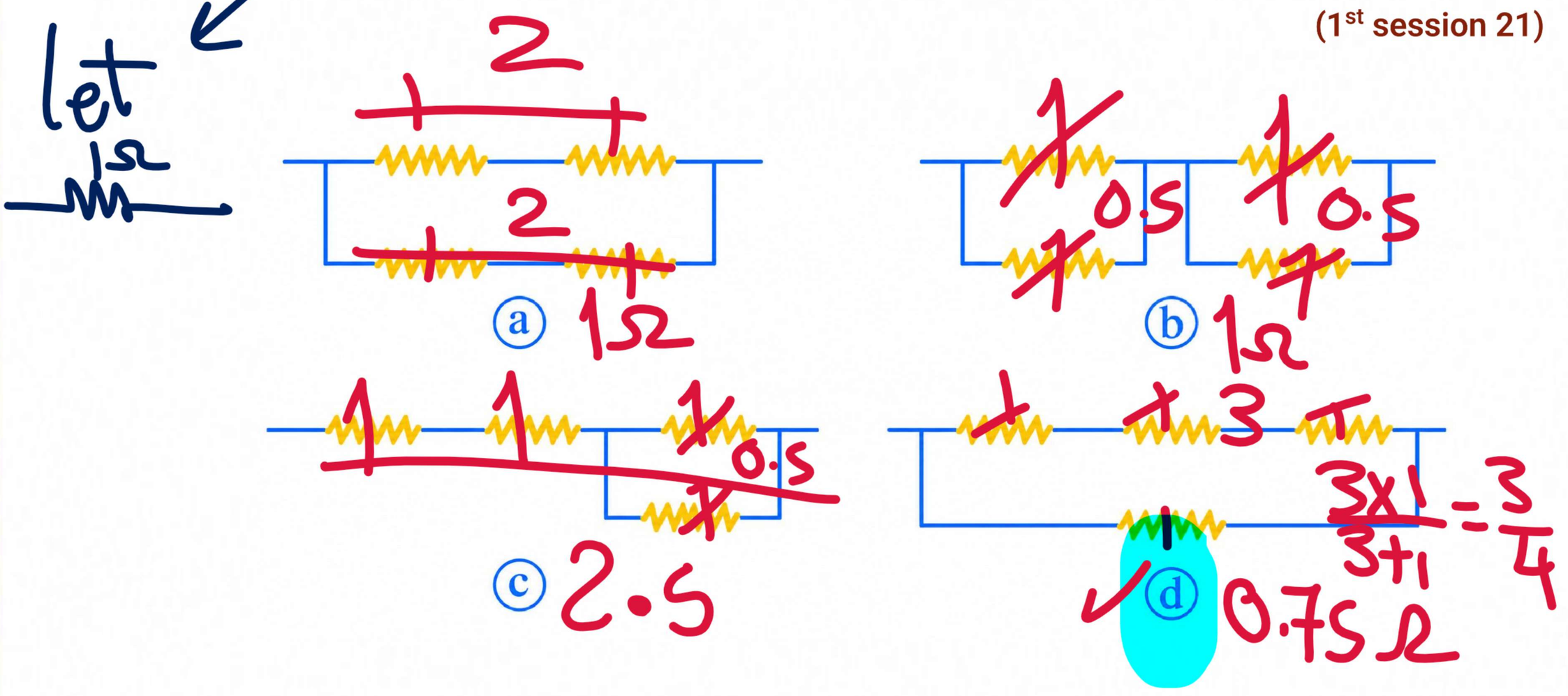
12 V
8 V
6 V
4 V

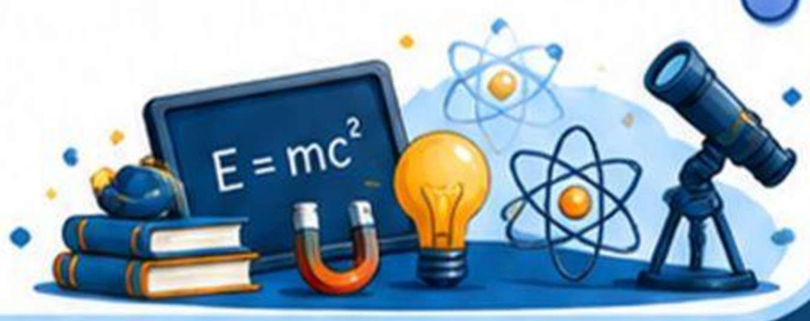


Questions

Four equal resistors are connected together as in the following figures, which figure gives the least equivalent resistance?

(1st session 21)





Questions

In the opposite electric circuit, when the key (K) is closed, the ammeter reading becomes ..... .

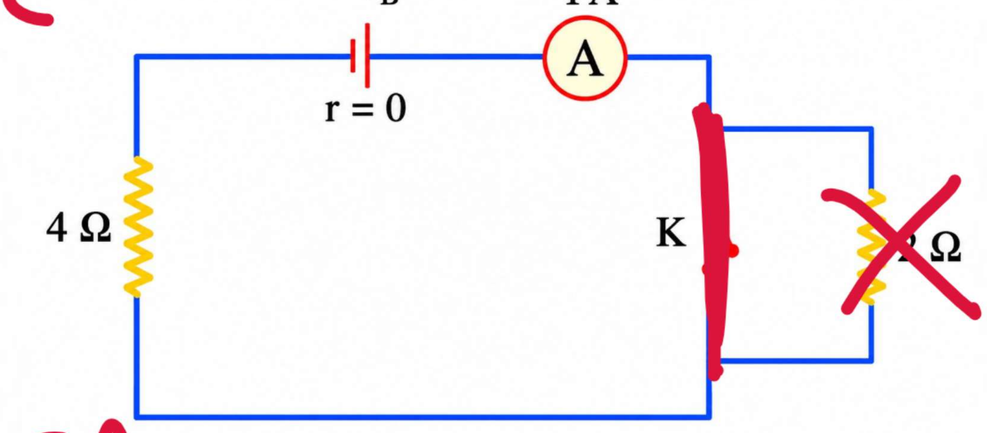
(1st session 21)

- 0.5 A
- 1.5 A ✓
- 2 A
- 0.75 A

$\frac{I_{before}}{I_{After}} = \frac{R_{After}}{R_{before}}$ $I = \frac{V_B}{R_{eq}}$

$\frac{1}{I} = \frac{4}{6}$

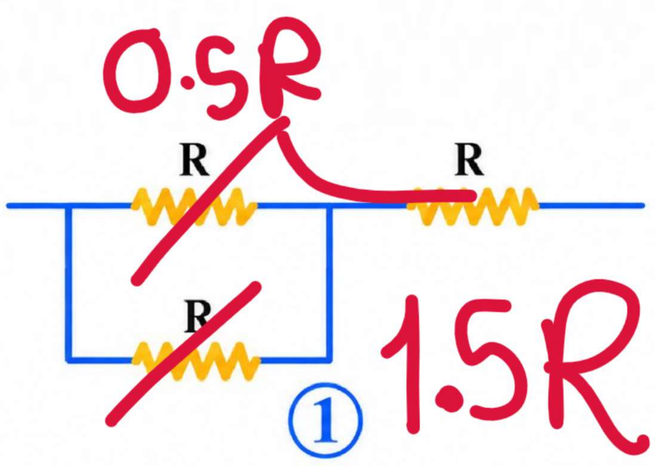
$I = \frac{6}{4} = 1.5A$

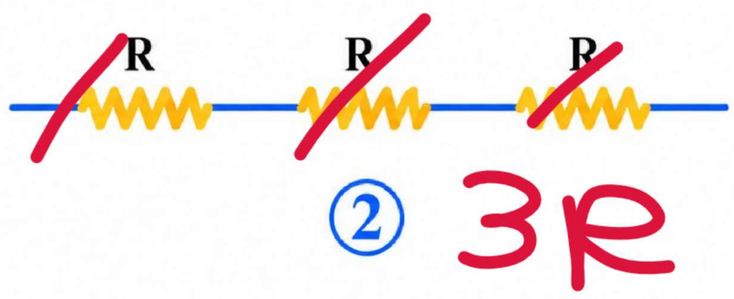


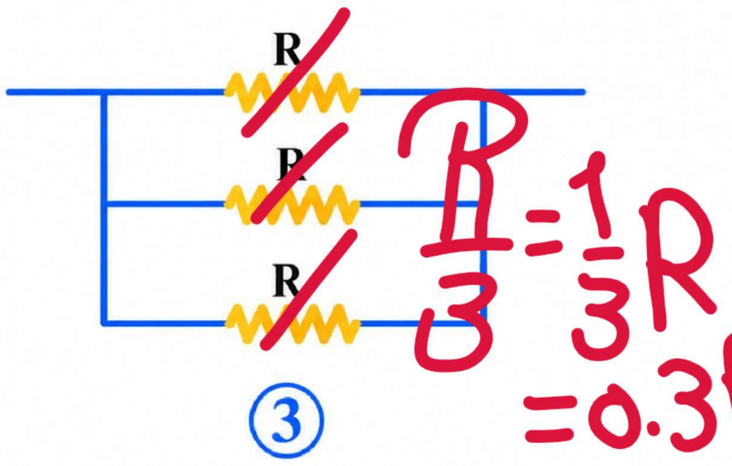
Questions

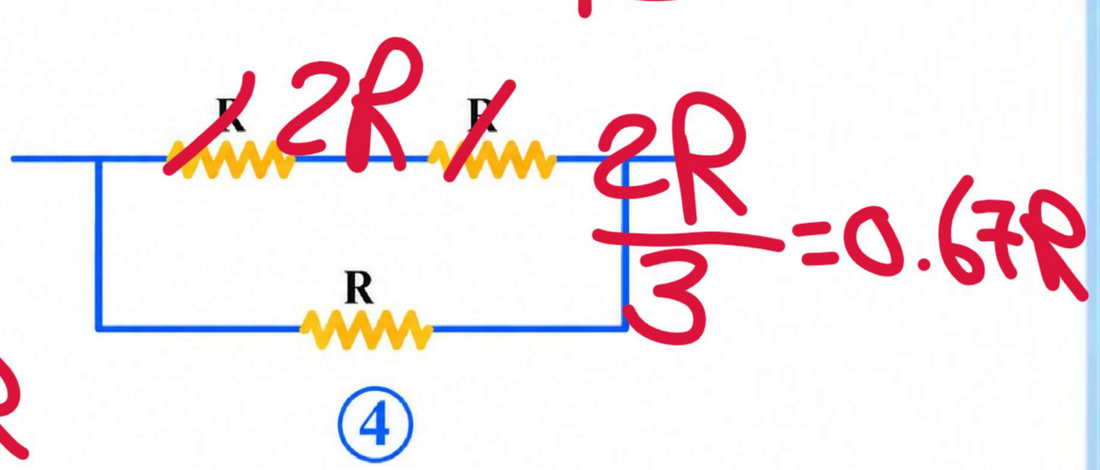
Arrange the figures according to their equivalent resistance ascendingly given that the resistors are identical.

(2nd session 21)



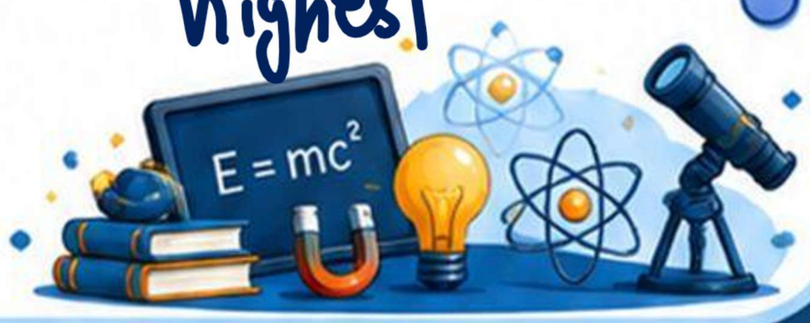






3 → 4 → 1 → 2

least — highest



Questions

In the shown three electric circuits (1), (2) and (3). Which one from the following relations is correct?

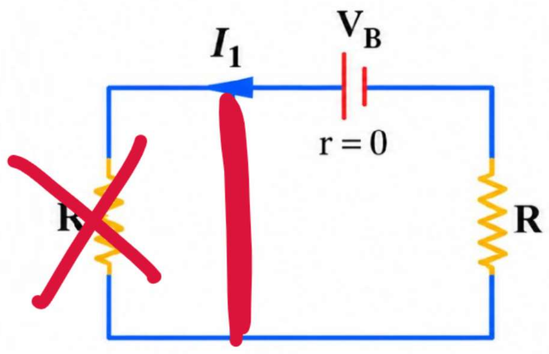
(2nd session 21)

let $V_B = 1V$

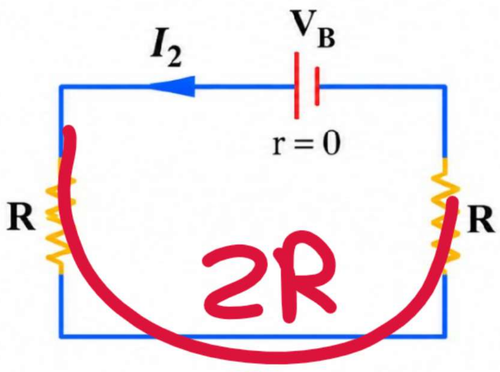
$R = 1\Omega$

$I_1 > I_3 > I_2$

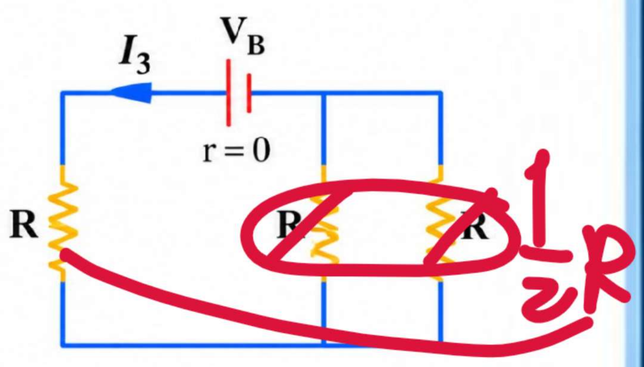
- ~~$I_1 = I_2$~~
- ✓ $I_1 > I_3$
- ~~$I_2 > I_3$~~
- ~~$I_3 > I_1$~~



① $I_1 = \frac{V_B}{R} = 1$



② $I_2 = \frac{V_B}{2R} = 0.5$



③ $I_3 = \frac{V_B}{1.5R} = 0.67$

Questions

A uniform cross-sectional copper wire was shaped as a rectangle {kyxm} as in the figure, its length (ky) is double its width (mk), which two terminals should be connected to the electric source to obtain the greatest electric resistance?

(1st session 22)

$R = \frac{\rho l}{A}$

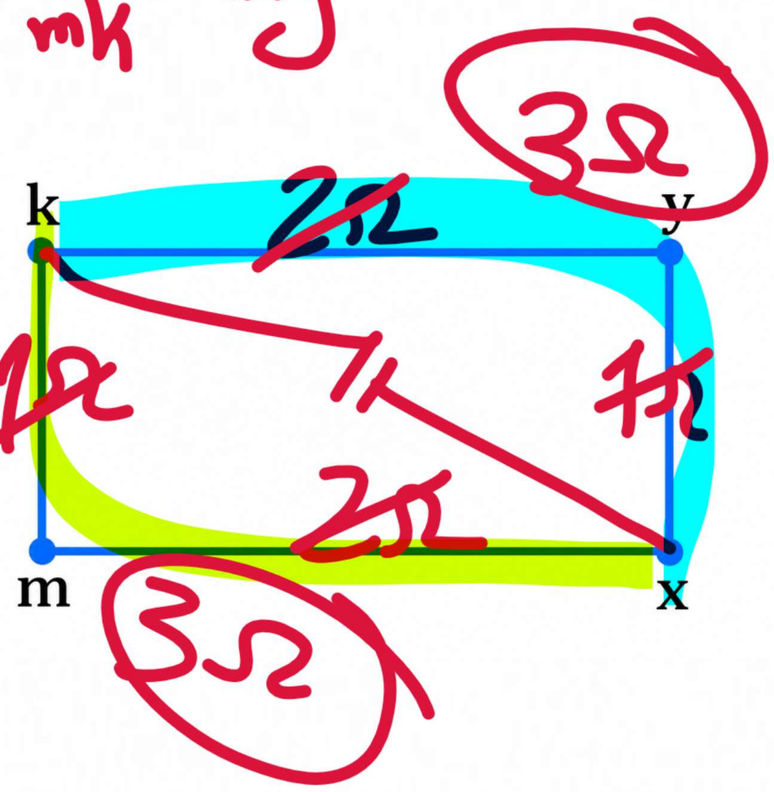
$L_{ky} = l_{mk}$

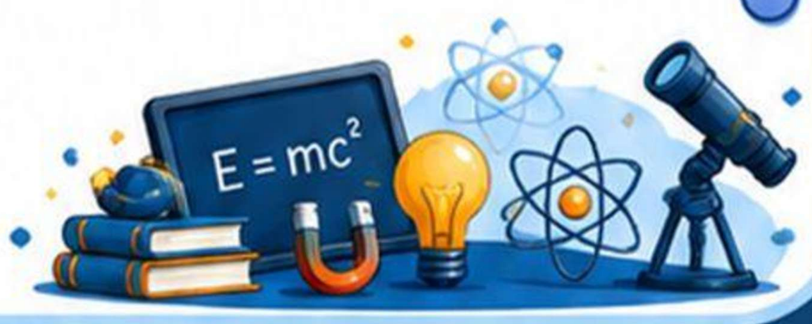
$R_{kx} > R_{ky} > R_{mk} = R_{xy}$

$R \propto l$

L ↑ between two points

- m , k → $\frac{1 \times 5}{1+5} = \frac{5}{6}$
- k , y → $\frac{4 \times 2}{4+2} = \frac{8}{6} = \frac{4}{3} = 1.3$
- x , y → $\frac{5}{6}$
- ✓ k , x → 1.5Ω





Questions

Two electric resistors, the resistance of the first is three times as that of the second. When they are connected in parallel, they give an equivalent resistance of 3 Ω, then when they are connected in series gives …… .

(1st session 22)

- 12 Ω
- 16 Ω ✓
- 8 Ω
- 4 Ω

$R_1 = 3R \quad R_2 = R$

$\frac{3R \times R}{3R + R} = 3$

$R = 4\Omega$

$R_{eq} = 3R + R$

$= 4R$

$= 4 \times 4 = 16\Omega$

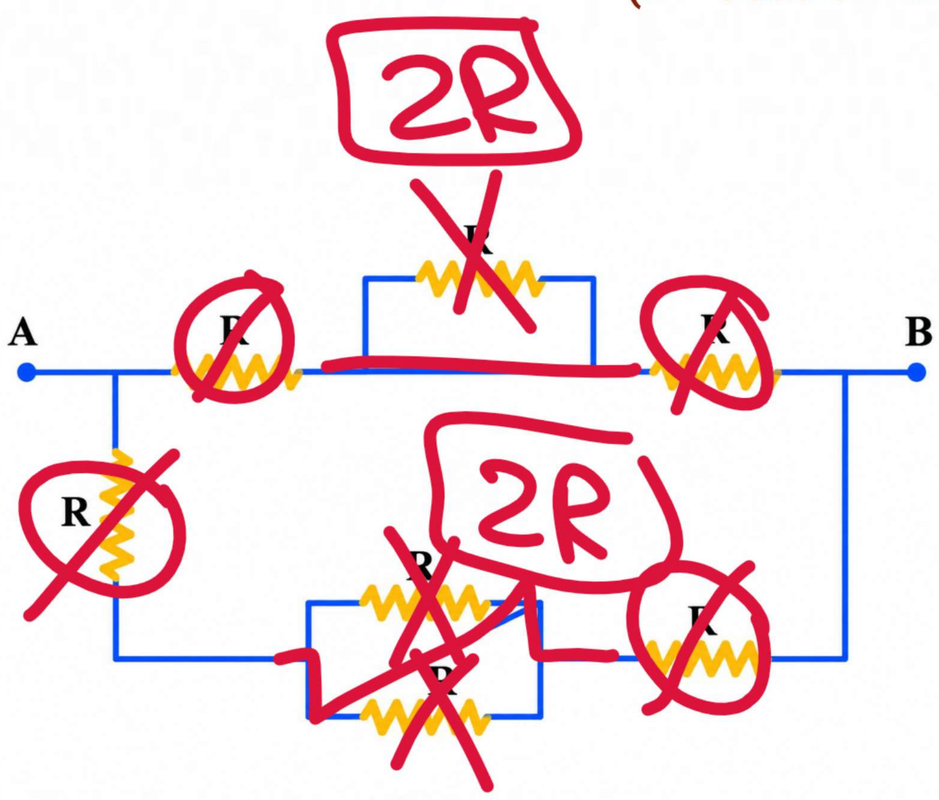
Questions

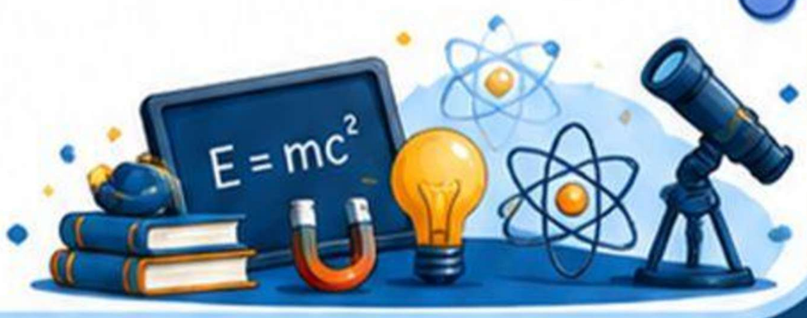
The figure shows a part of a closed electric circuit that contains a group of identical resistors. The equivalent resistance between the two points (A) and (B) equals …….. .

(2nd session 22)

- 6R/5
- 5R/4
- 3R/2
- R ✓

$R_{eq} = \frac{R}{N} = \frac{2R}{2} = R$





Questions

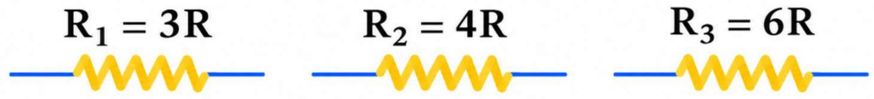
You have three resistors R_1, R_2 and R_3 as shown in the figure. When they are connected in parallel, they give an equivalent resistance 4 Ω, then when they are connected in series, they give an equivalent resistance …… .

$$R_{eq} = \left[\frac{1}{3R} + \frac{1}{4R} + \frac{1}{6R}\right]^{-1} = 4$$

$$R = 3\Omega$$

(2nd session 22)

- 9 Ω
- 27 Ω
- 13 Ω
- 39 Ω ✓



$$R_{eq} = 3R + 4R + 6R = 13R$$
$$= 13 \times 3 = 39\Omega$$

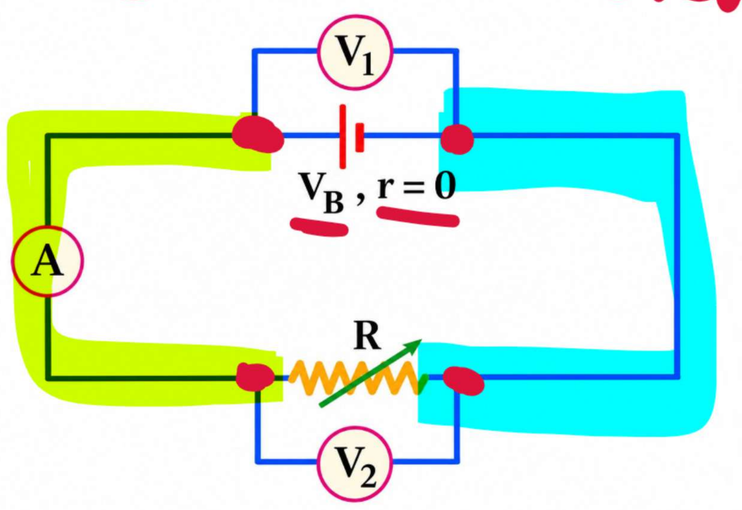
Questions

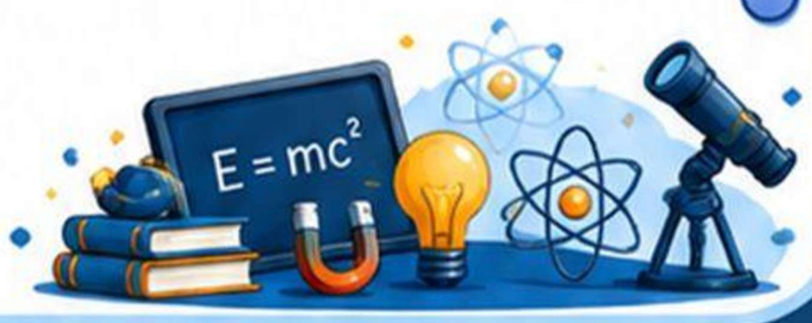
In the electrical circuit in front of you, when the value of resistance (R) increases. Then, the reading of V_1 and the reading of V_2 …… .

V_1	V_2
does not change ✓	does not change ✓
~~increases~~	increases
increases	does not change
does not change	increases

$$V_1 = V_2 = V_B \longrightarrow \text{Constant}$$

(1st session 22)





Mr.Abdelwahab

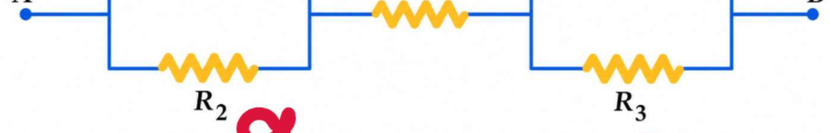
Questions

In the opposite figure, which of the following choices makes the resistance between A and B equals 5 Ω?

R_1 (Ω) //	R_2 (Ω)	R_3 (Ω) //	R_4 (Ω)
✗ 2	✗ 9	✗ 8	✗ 2.5
✓ 1	9	2	8
✗ 1	2	8	9
✗ 8	1	9	5

$\frac{2\times9}{2+9}+2.5+\frac{2.5\times8}{2.5+8}\neq5$ (Exp.23)

2 2s



9 8

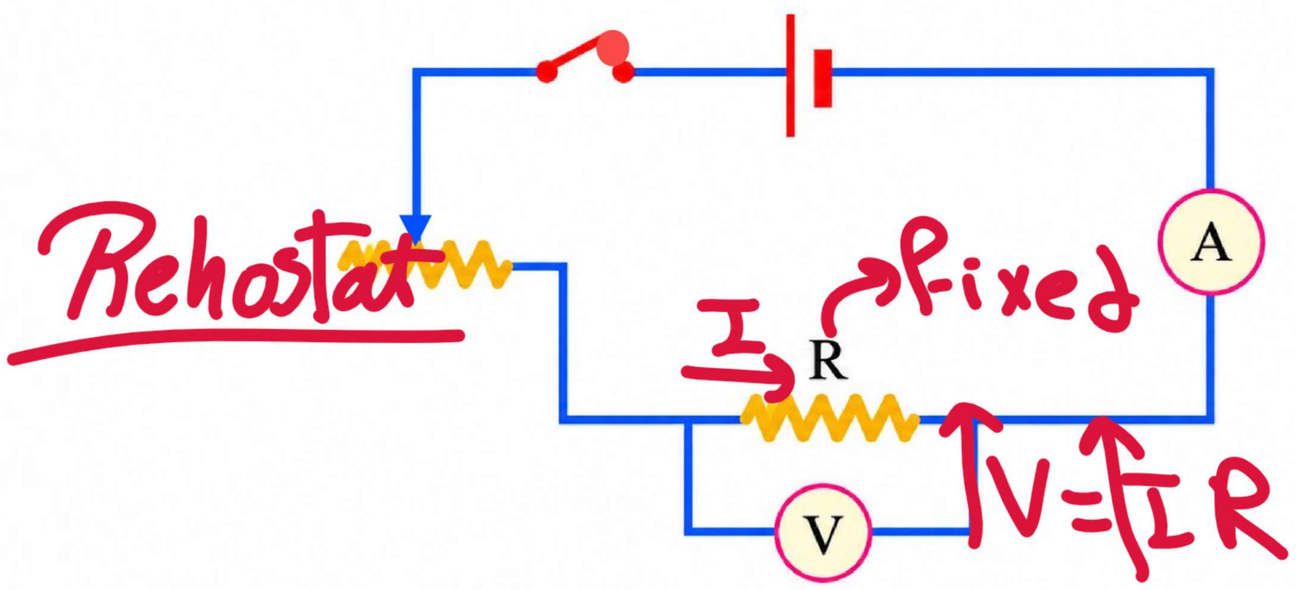
$\frac{1\times9}{1+9}+2.5+\frac{8\times2}{8+2}=5\Omega$

Questions

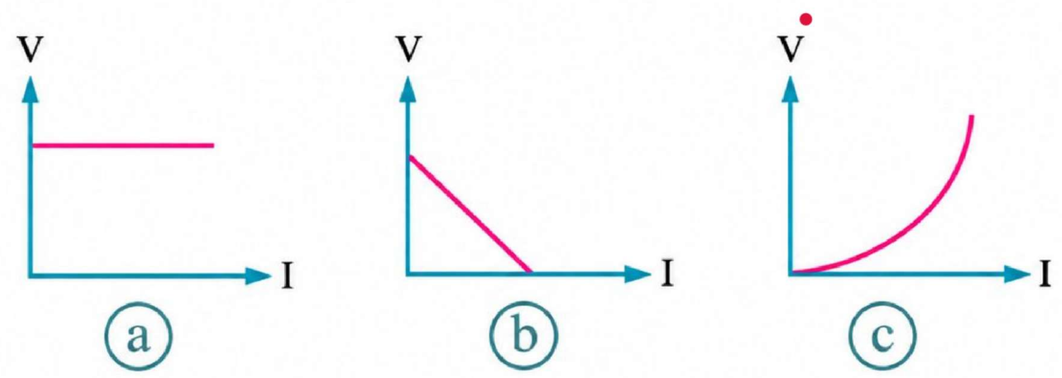
Which graph represents the correct relation between the electric potential difference (V) across the two terminals of the fixed resistance (R) and the ammeter reading (I) at the same temperature?

(1st session 23)

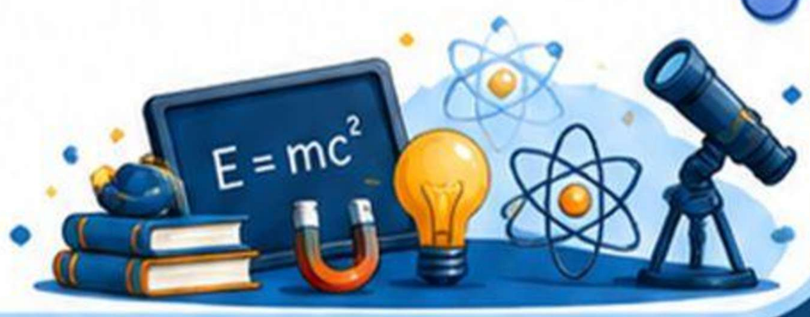
Pe



Fixed



Slope = V/I = R

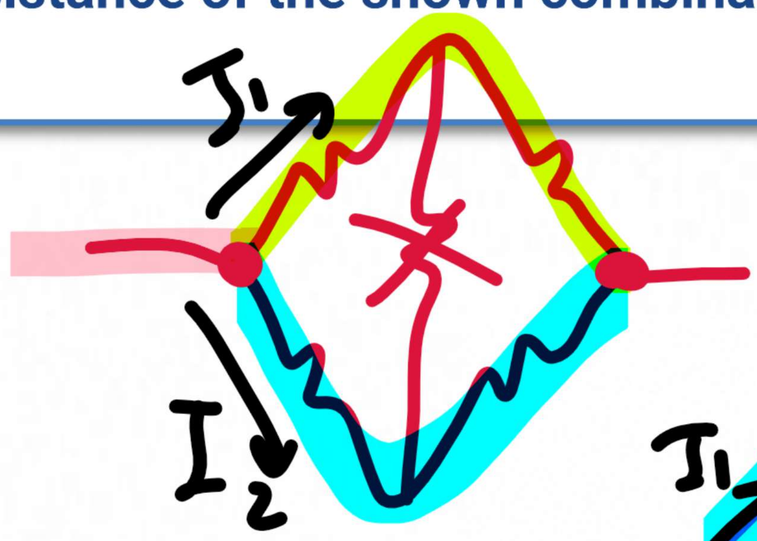


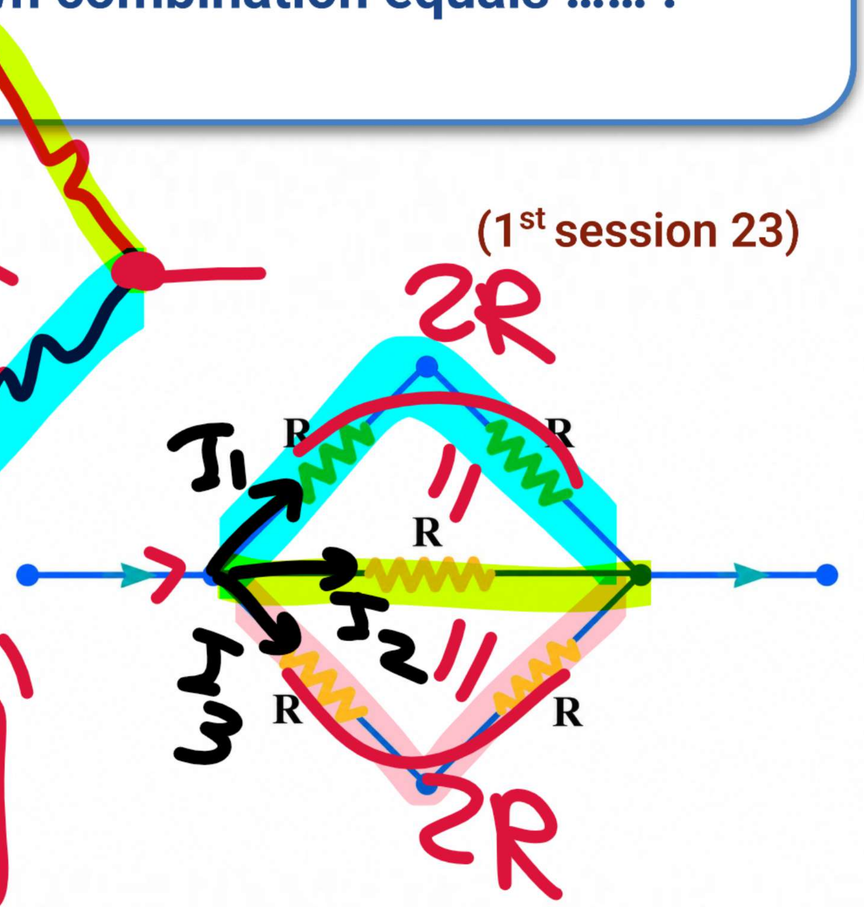
Questions

The figure represents a part from a closed electric circuit. The equivalent resistance of the shown combination equals …… .

(1st session 23)

- R
- R/2
- 3R/5
- 2R





$$R_{eq} = \left[\frac{1}{2R} + \frac{1}{R} + \frac{1}{2R}\right]^{-1}$$
$$= 0.5R$$

Questions

In the electric circuit, if the voltmeter reading when the slider of the rheostat is at (A) was 12 V and when the slider is moved to point (B), the voltmeter reading was 3 V. The value of the resistance taken from the rheostat at (B) is ……. .

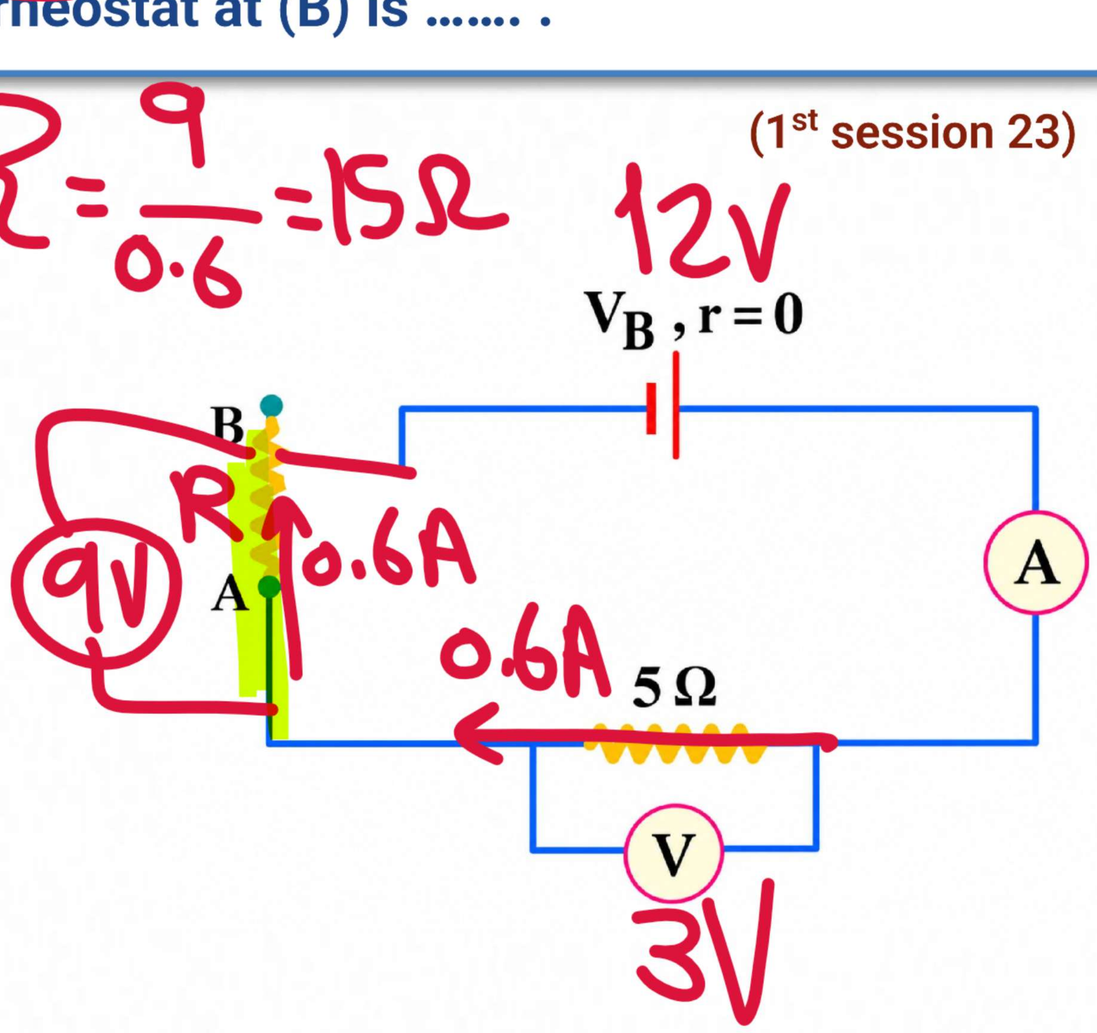
(1st session 23)

- 25 Ω
- 30 Ω
- 15 Ω
- 20 Ω

before

$V_B = 12V$

$R = \frac{9}{0.6} = 15\Omega$



Mr.Abdelwahab

Questions

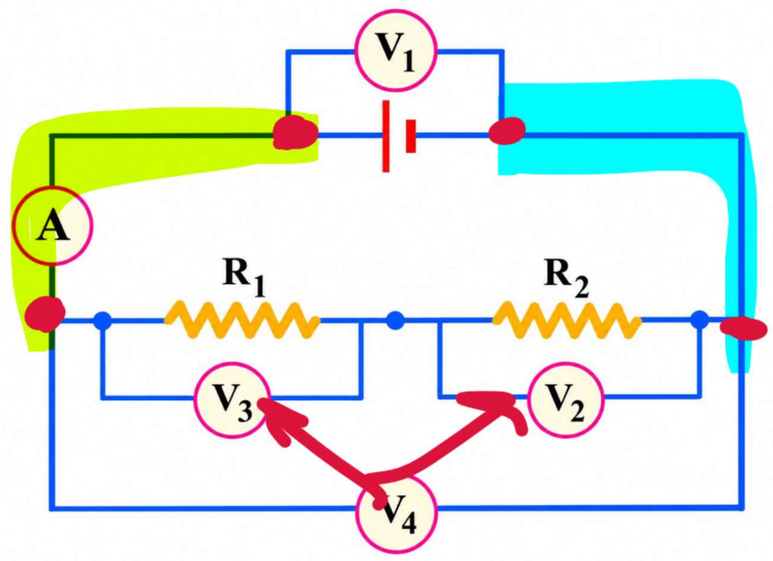
In the shown electric circuit:
Which readings of those voltmeters are equal? …… .

(2nd session 23)

$V_1 // V_4$

- V_2, V_3
- V_2, V_4
- V_2, V_1
- V_1, V_4 ✓

$V_1 = V_4 = V_2 + V_3$



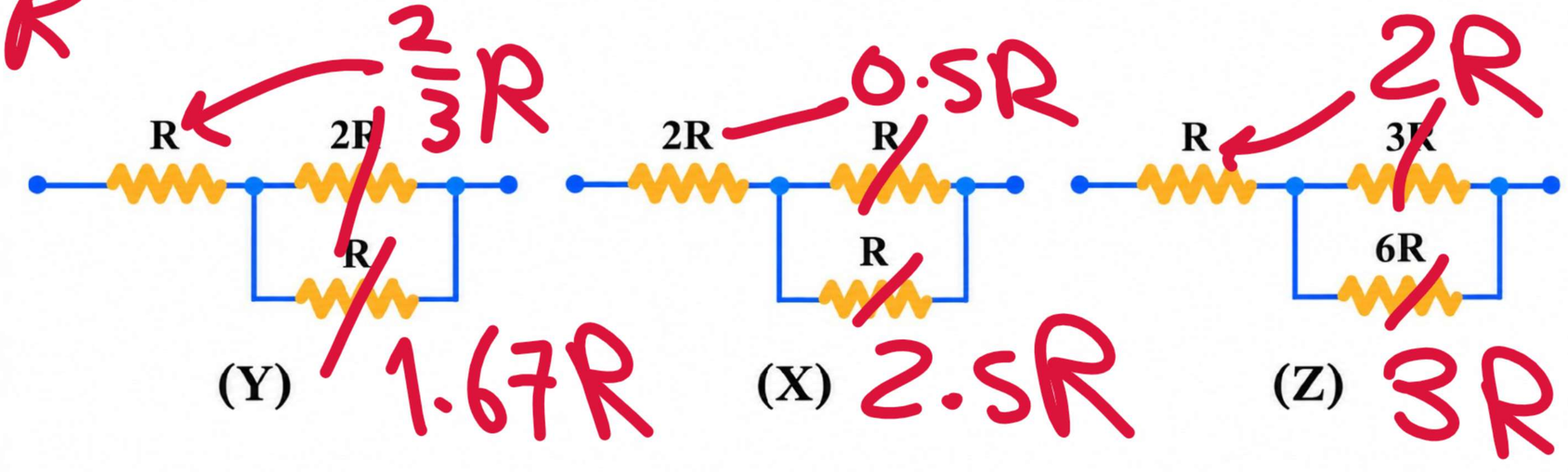
Questions

The figures show a number of resistors connected to each other.
Which of the following expressions is correct concerning the equivalent resistance of each combination?

$\frac{2 \times 1}{2+1} = \frac{2}{3}R$

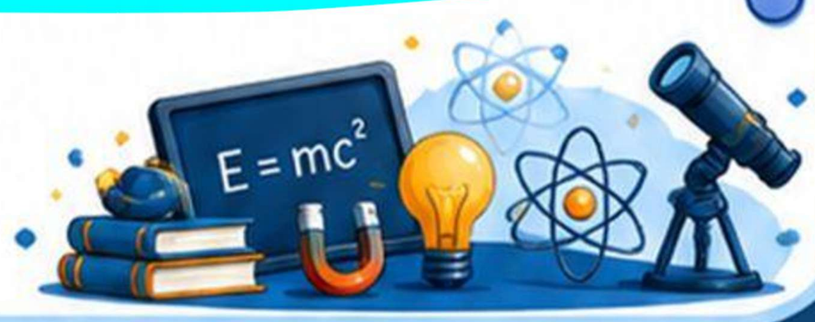
$\frac{3 \times 6}{3+6} = 2$ (1st session 24)

$\frac{5}{3}R$



- ✗ The equivalent resistance of combination (X) equals that of combination (Y).
- ✗ The equivalent resistance of combination (X) is less than that of combination (Y).
- ✗ The equivalent resistance of combination (Z) is less than that of combination (X).
- ✓ The equivalent resistance of combination (Z) is more than that of combination (Y).

$R_z > R_x > R_y$



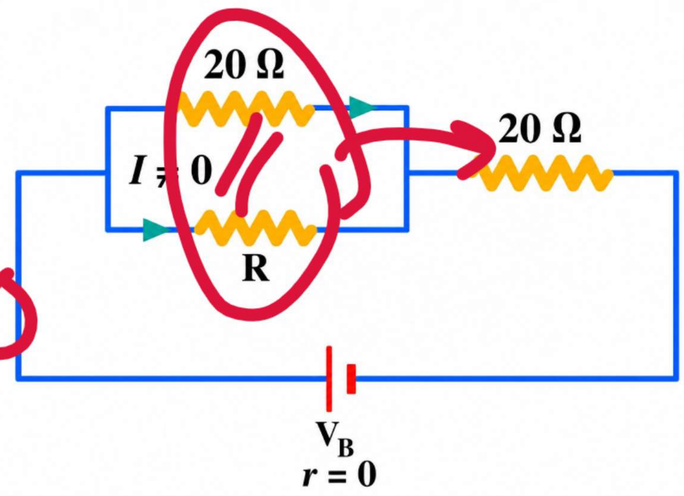
Questions

From the opposite electric circuit:
Which choice is possible to express the total resistance of the circuit?

(2nd session 24)

$\frac{20 \times R}{20+R} < 20$

- 19 Ω ✗
- 25 Ω ✓
- 15 Ω ✗
- 40 Ω ✗



$20 < \frac{20R}{20+R} + 20 < 40$

$20 < R_{eq} < 40$

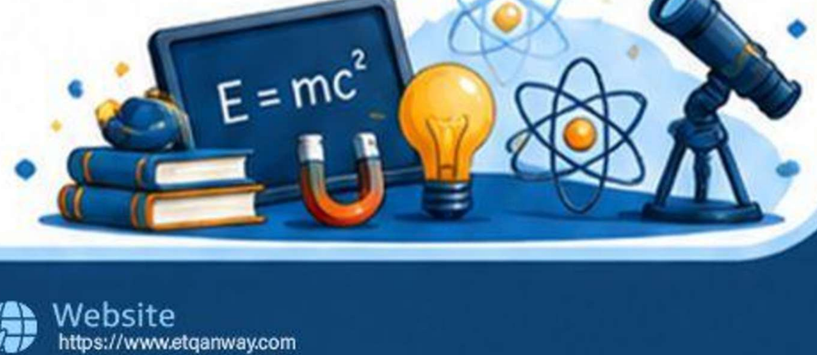
Questions

When two different resistors are connected together in parallel, their equivalent resistance will …… .

$R_{eq} < R_{smallest}$ $R_{eq} = \frac{R_1 \times R_2}{R_1 + R_2}$

(Exp.25)

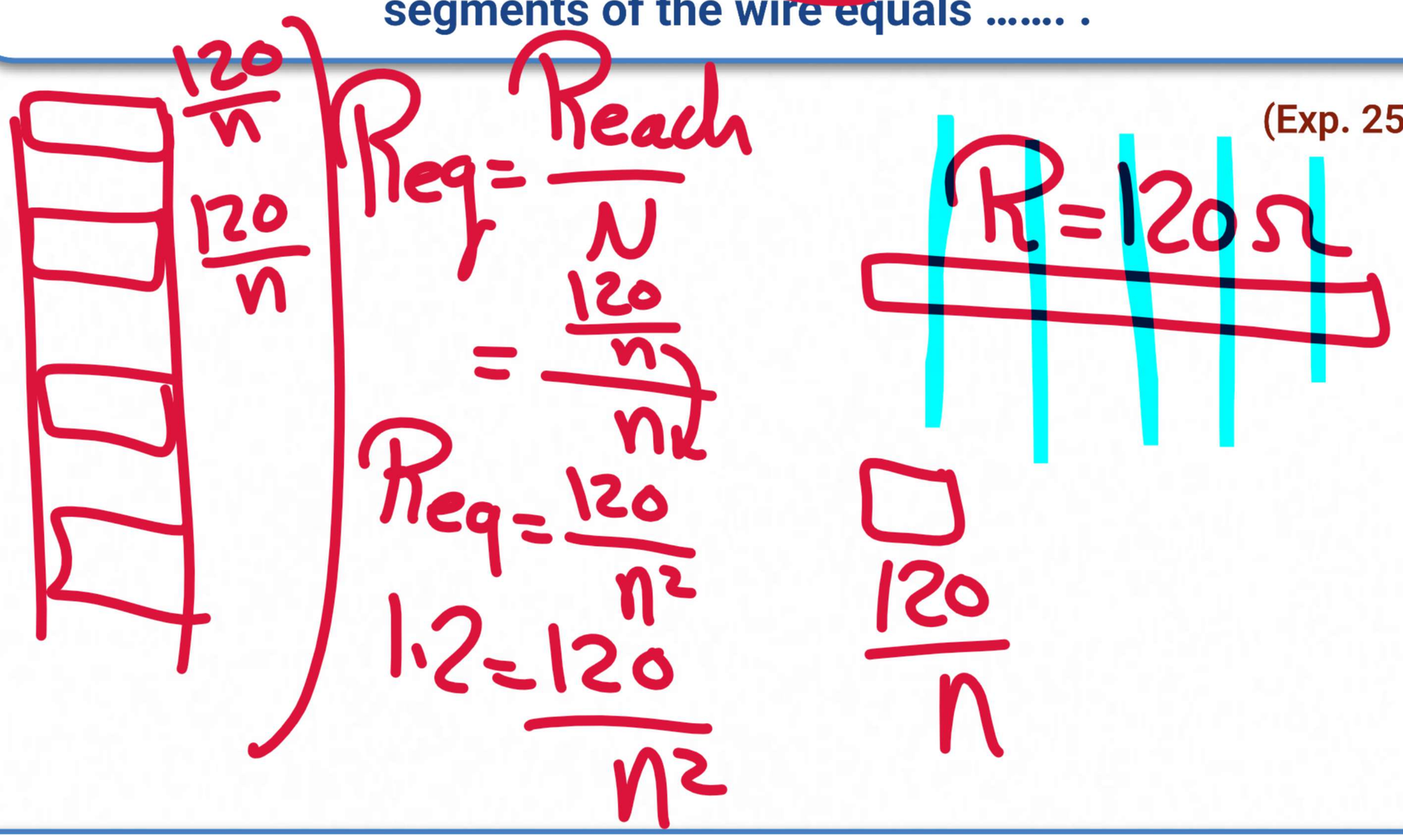
- be equal to the sum of the resistors
- have an average value between the values of the two resistors
- be less than the smaller resistance ✓
- be greater than the larger resistance



Questions

A uniform wire of resistance 120 Ω is cut into a number of equal segments and when these wire segments are connected together in parallel, their total resistance becomes 1.2 Ω, so the number of these segments of the wire equals ……. .

6
10 ✓
12
24



(Exp. 25)

Questions

Four identical bulbs N, M, P and Q, the resistance of the filament of each is R, are connected to different resistors as shown in the opposite figure, then the brightness is the same for …… .

$I_1 = \frac{I \times 3}{5}$

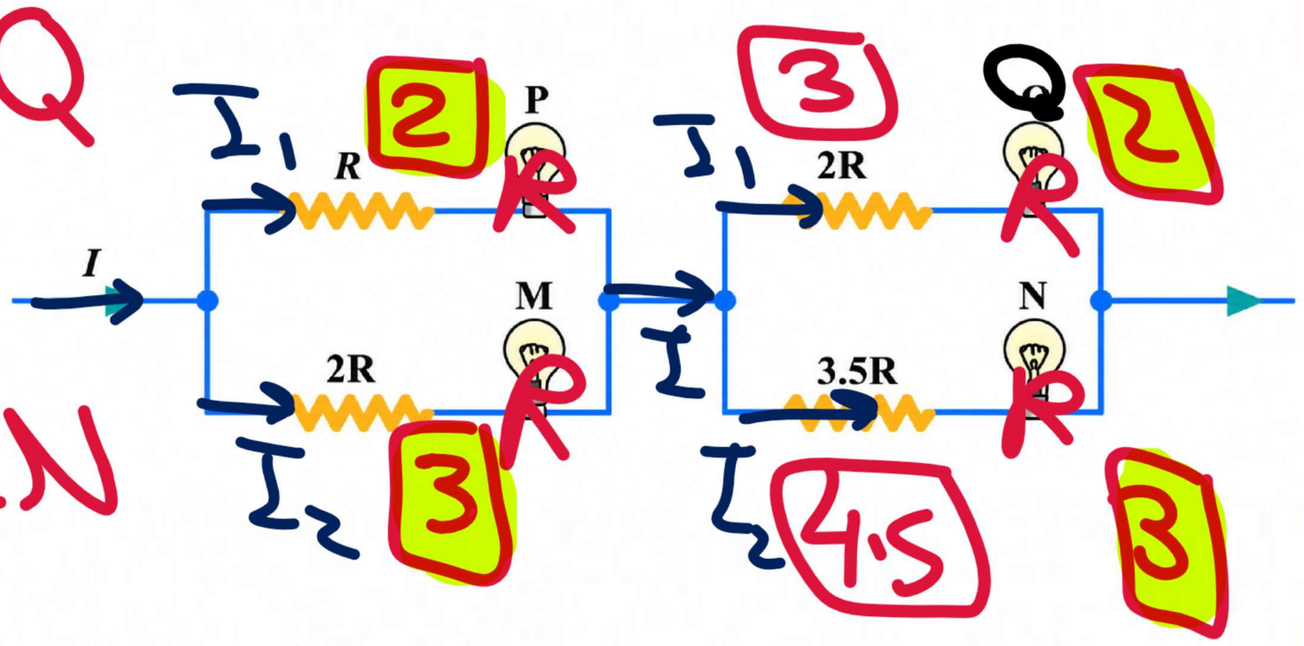
Power → V, I

(Exp.25)

both bulbs Q and M
both bulbs M and N ✓
both bulbs P and N
all the bulbs

P = Q

M = N

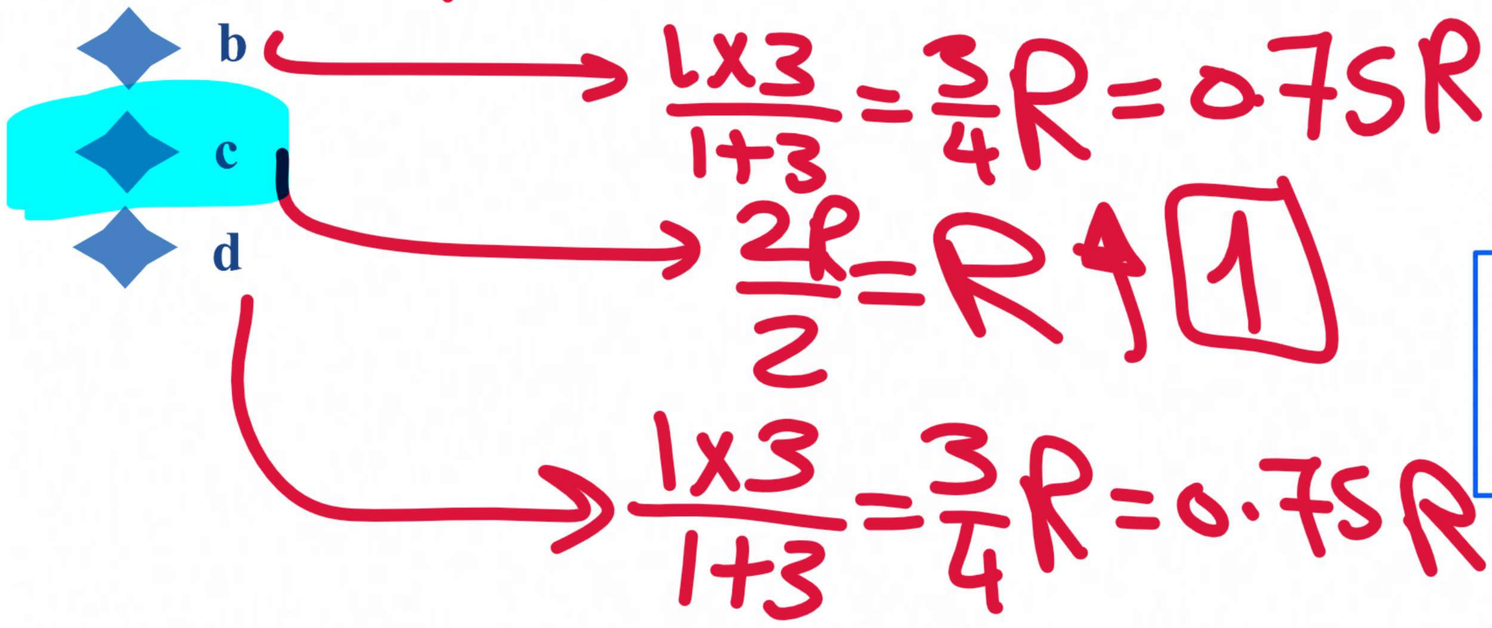


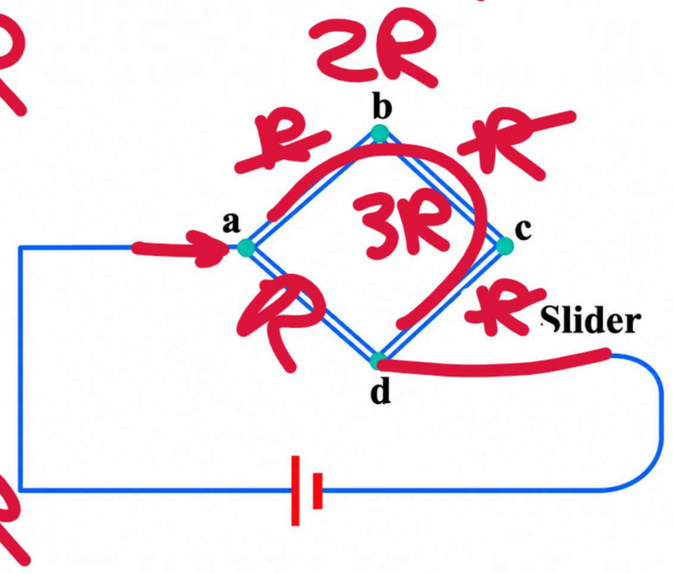
Questions

A wire of uniform cross-section is bent to form a square abcd, of which each side has a resistance R, one of the poles of a battery is connected to point a on the frame. Then the point at which the other pole of the battery is connected so that the lowest current intensity passes in the circuit is……

(Exp. 25)

- a
- b
- c
- d





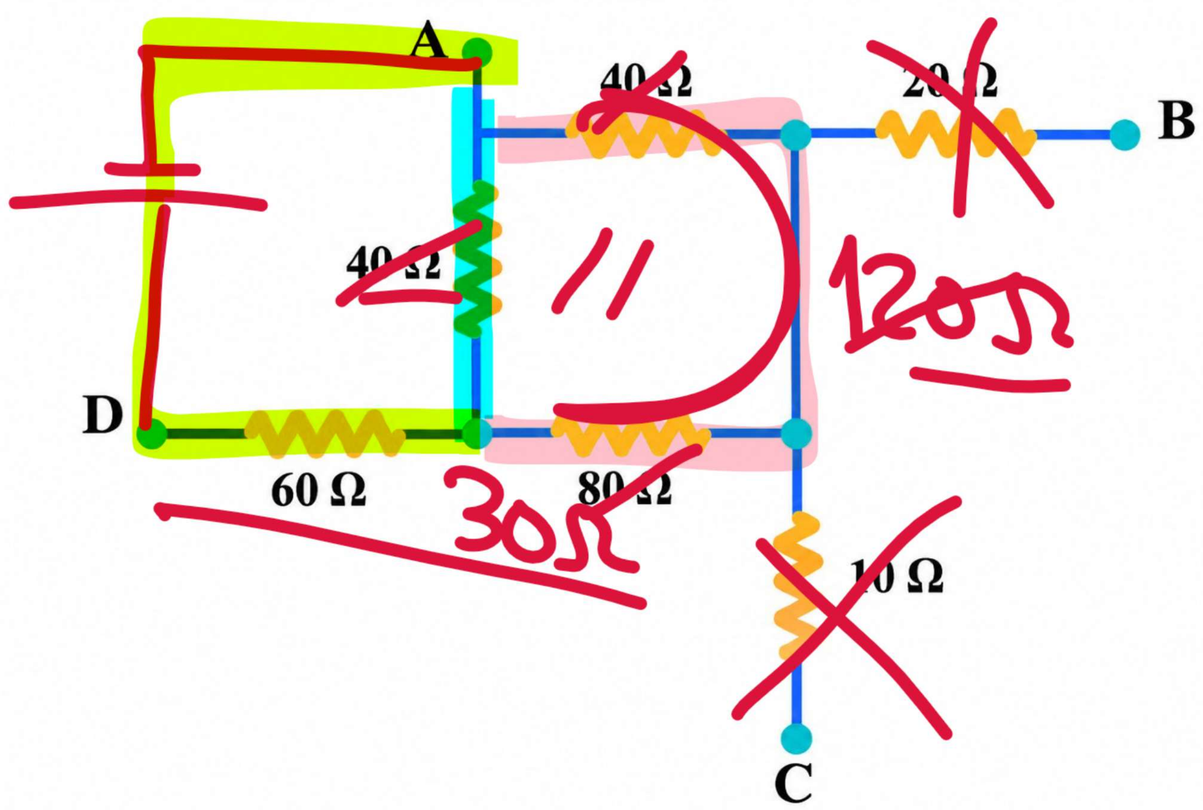
Questions

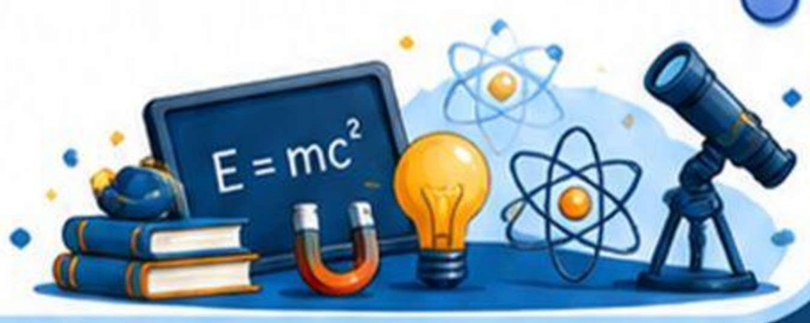
Resistors are connected as shown in the opposite figure. To which two points of the following should a battery terminals be connected in order to get the lowest equivalent resistance ?

(Exp.25)

- B,C 30Ω ✓
- A, B 50Ω
- C, D 110Ω
- A, D 90Ω

Req = 20+10 = 30Ω



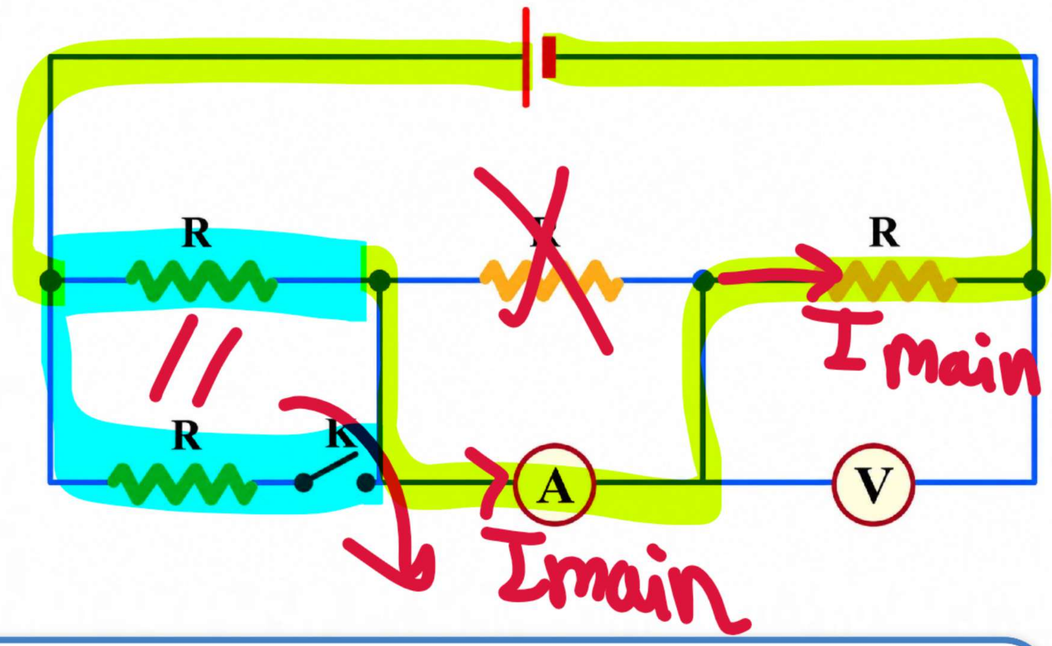


Questions

In the opposite electrical circuit, if the voltmeter resistance is infinite and the ammeter resistance is negligible, what happens to the reading of each of the ammeter and the voltmeter respectively when switch (K) is closed?

(Exp. 25)

- Increases, increases.
- Decreases, decreases.
- Decreases, increases.
- Increases, decreases.

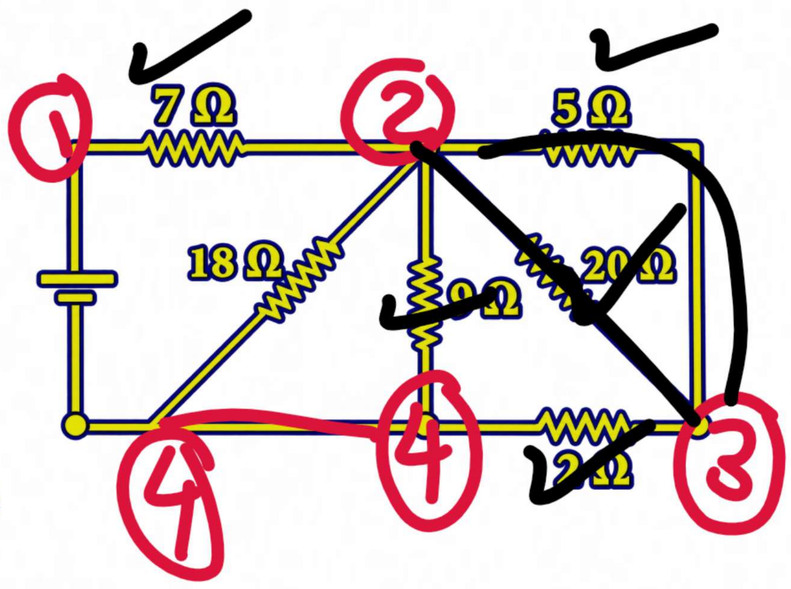


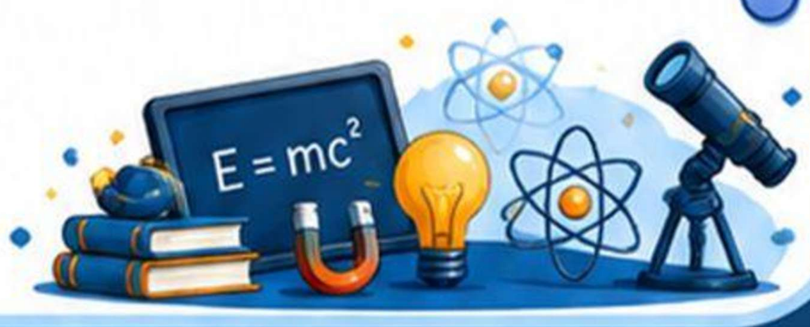
Questions

The figure represents a closed electric circuit. The equivalent external resistance of the circuit equals …….. .

(First Session 2026)

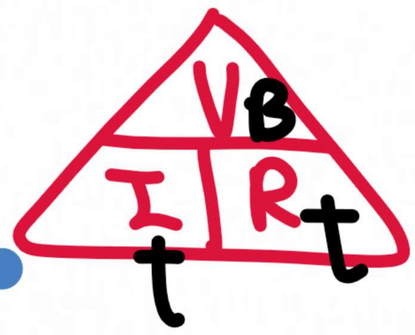
- 19 Ω
- 10 Ω
- 13 Ω
- 14 Ω





Lesson Three: Ohm's Law for the Closed Circuit

Ohm's Law for a Closed Circuit:



$$I_t = \frac{V_B}{R_t} = \frac{V_B}{R_{eq}+r}$$

When switch open

$I = 0$

$V_1 = V_B$

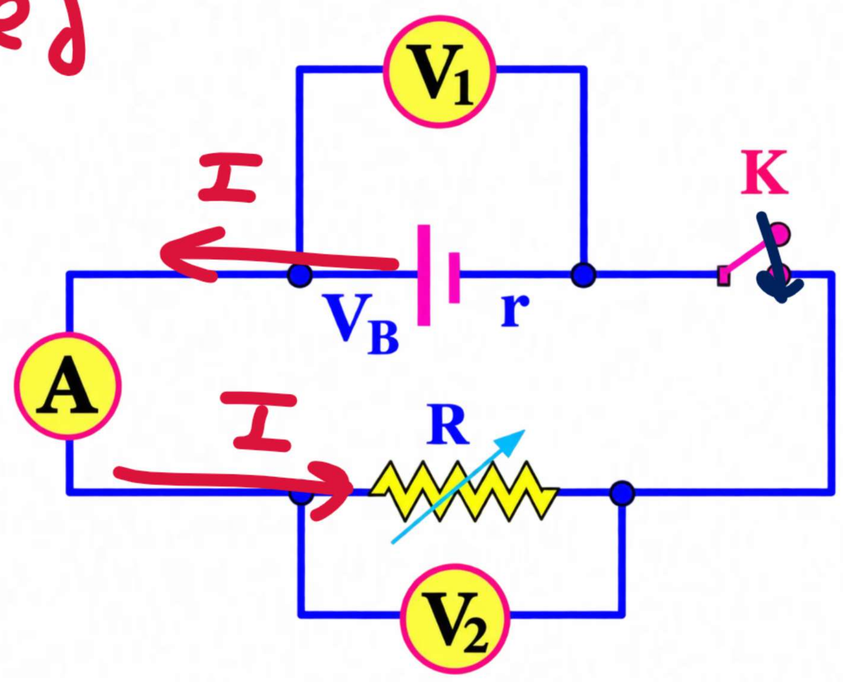
$V_2 = zero$

switch closed

$$I = \frac{V_B}{R_{eq}+r}$$

$V_1 = V_B - Ir$

$V_2 = I \times R$

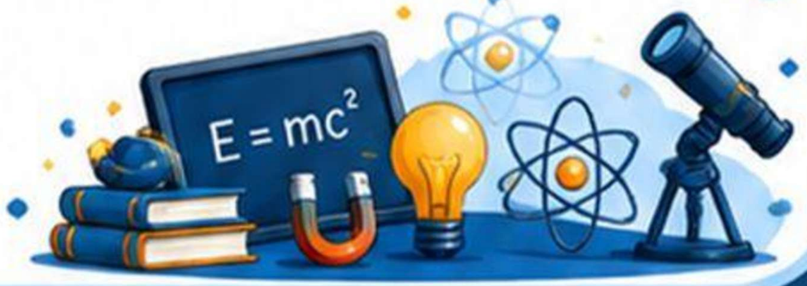


Battery Efficiency:

$$\frac{V}{V_B} = \frac{I R_{eq}}{I R_t} = \frac{R_{eq}}{R_{eq}+r}$$

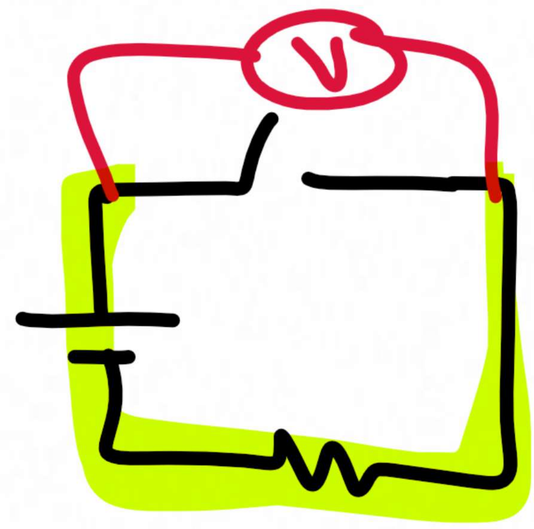
$0.8 = 80\%$

$= \frac{80}{100}$



Mr.Abdelwahab

Ohm's Law for a Closed Circuit:



- K is open $I=0$

$V_3 = \text{Zero}$ $V_1 = V_B$ $V_2 = V_B$

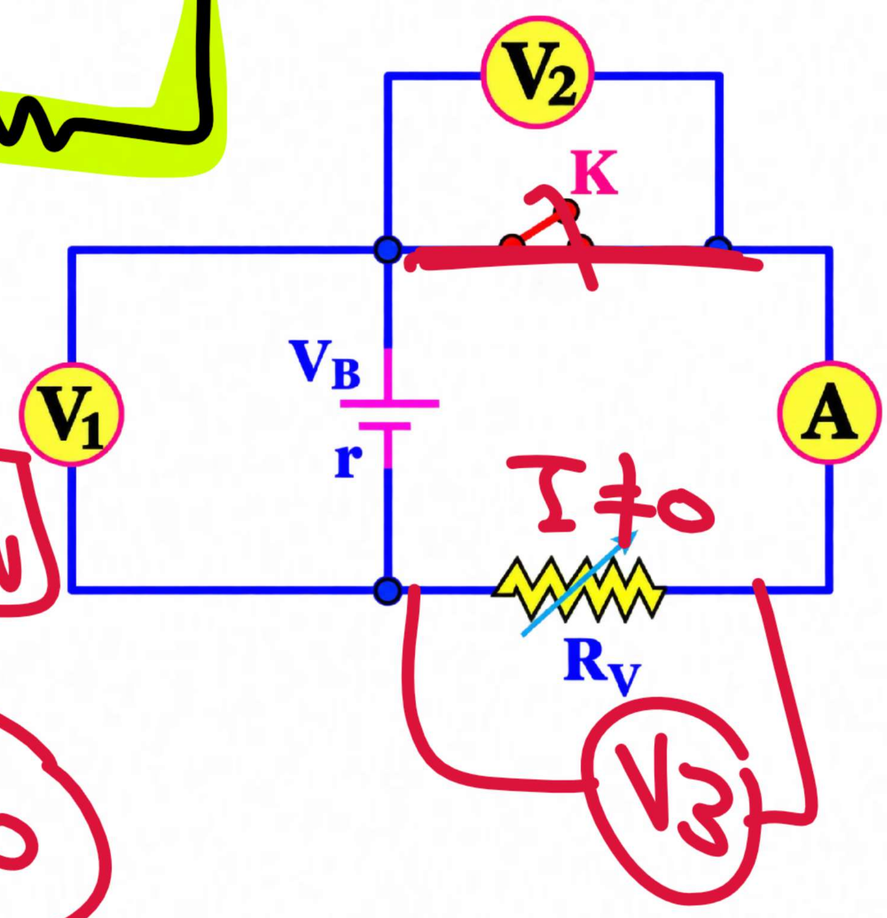
- K is closed

$$I = \frac{V_B}{R_V + r}$$

$V_3 = I R_V$

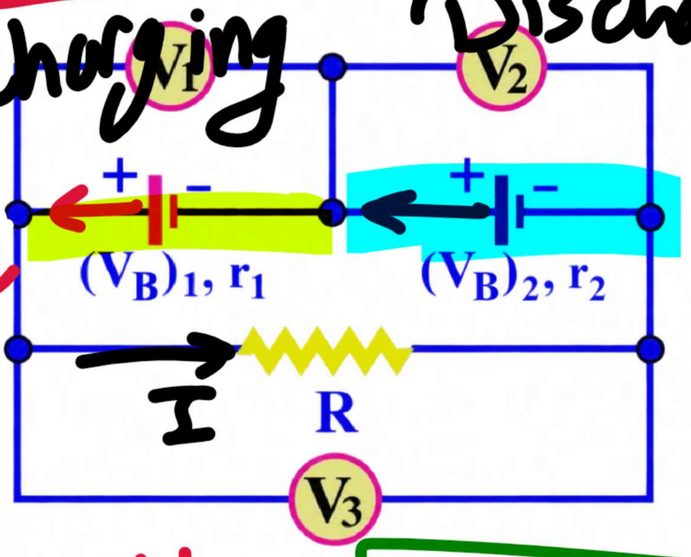
$V_1 = V_B - Ir$

$V_2 = \text{Zero}$



Support

Discharging



Discharging

Produce

$$I = \frac{V_{B_1} + V_{B_2}}{R + r_1 + r_2}$$

$V_1 = V_{B_1} - Ir_1$

$V_2 = V_{B_2} - Ir_2$

$V_3 = IR$

$V_1 + V_2 = V_3$

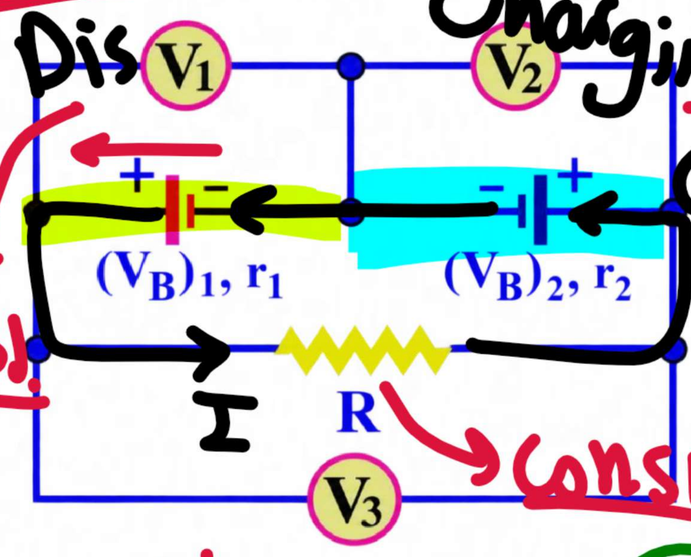
Produce

Consume

Destroy

$V_{B_1} > V_{B_2}$

Dis Charging



Proj

Consume

Consume

$$I = \frac{V_{B_1} - V_{B_2}}{R + r_1 + r_2}$$

$V_1 = V_2 + V_3$

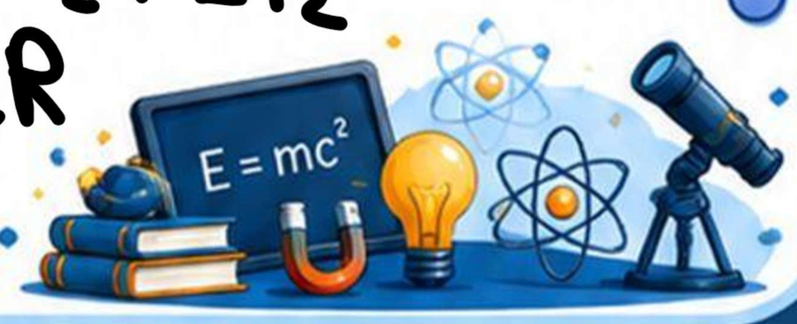
P

C

$V_1 = V_{B_1} - Ir_1$

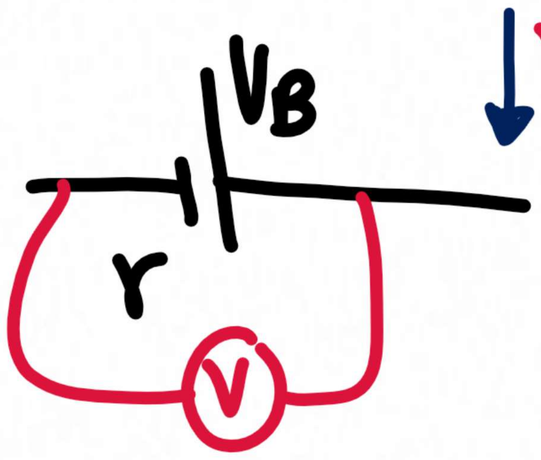
$V_2 = V_{B_2} + Ir_2$

$V_3 = IR$

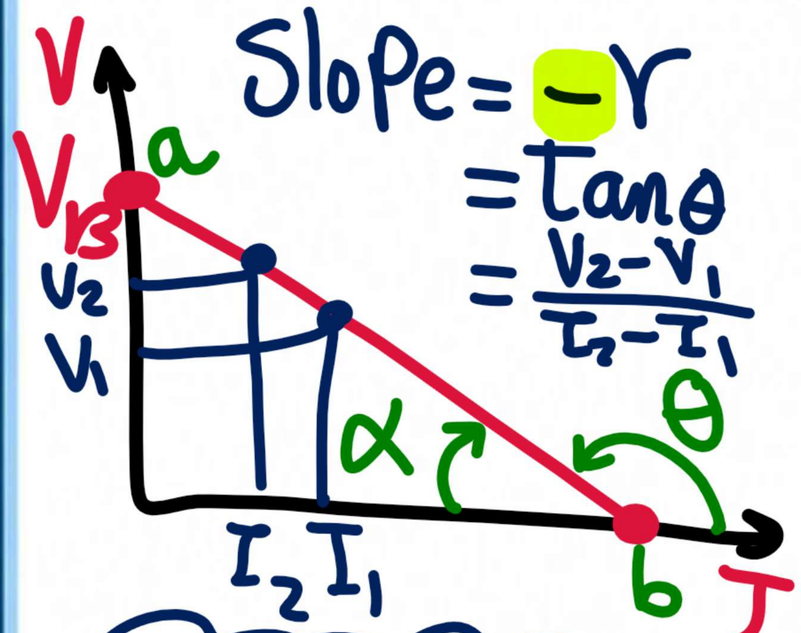


Mr.Abdelwahab

Ohm's Law for a Closed Circuit:



$V = V_B - Ir$

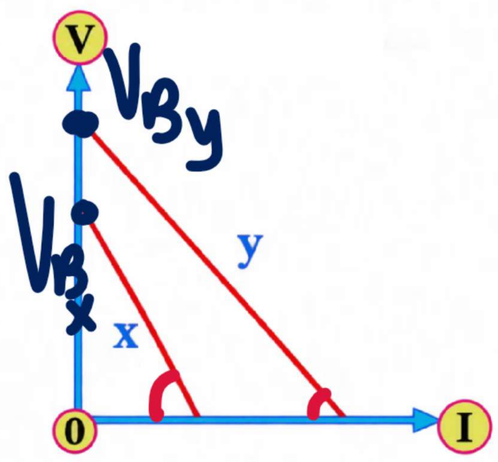


$Slope = -r$

$= \tan\theta$

$= \frac{V_2 - V_1}{I_2 - I_1}$

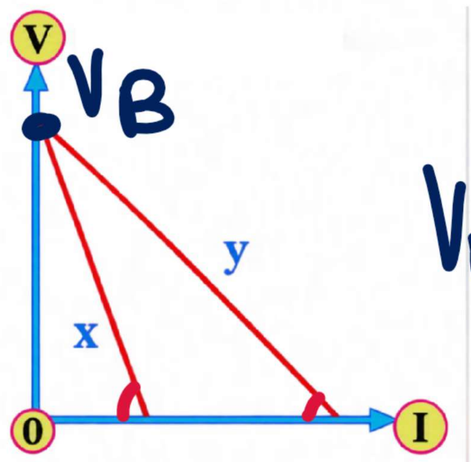
$r = \frac{a}{b} = \tan\alpha$



$V_{By} > V_{Bx}$

$\alpha_x > \alpha_y$

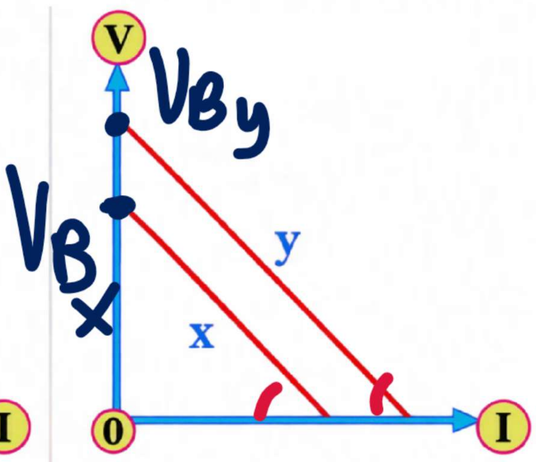
$r_x > r_y$



$V_{Bx} = V_{By}$

$\alpha_x > \alpha_y$

$r_x > r_y$



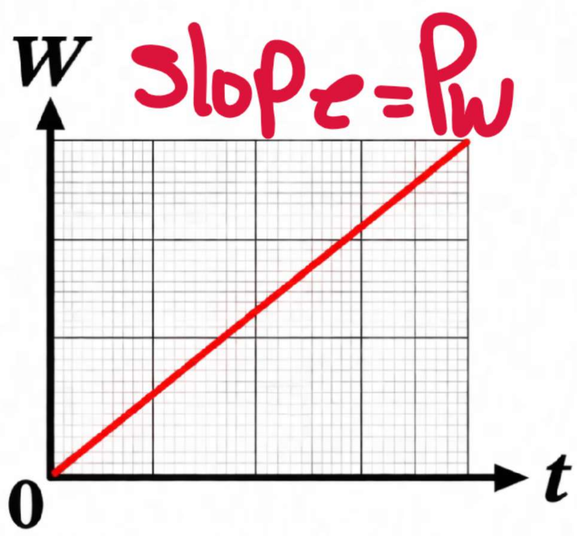
$V_{By} > V_{Bx}$

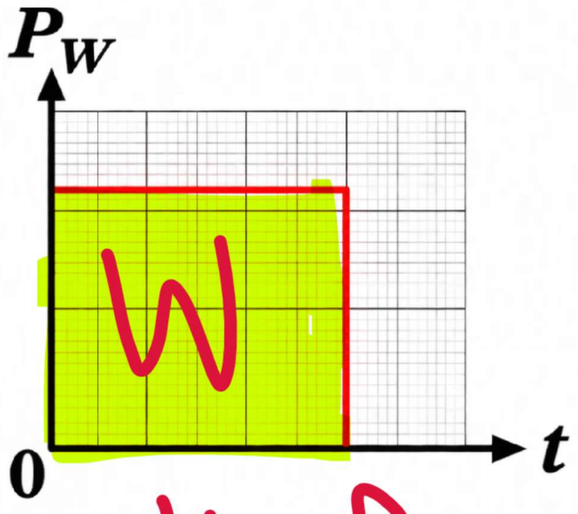
$\alpha_x = \alpha_y$

$r_x = r_y$

Power

Brightness
illumination
Glowing

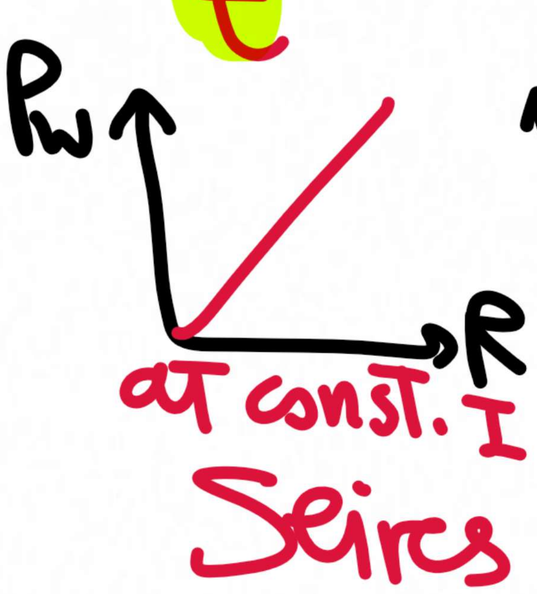


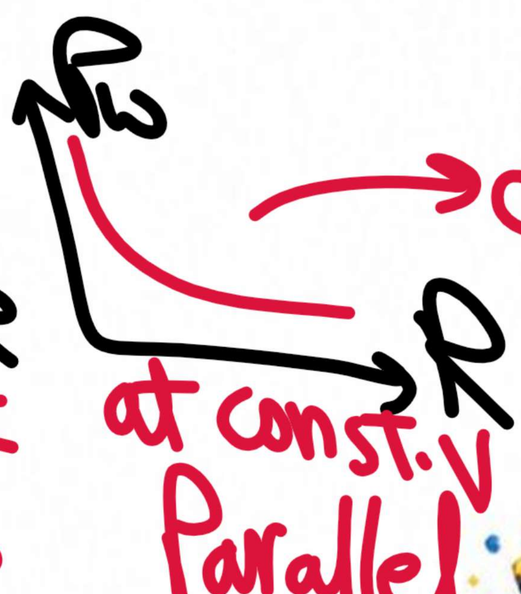


$P_W = \frac{W}{t} = \frac{VQ}{t} = VI = I^2R = \frac{V^2}{R}$

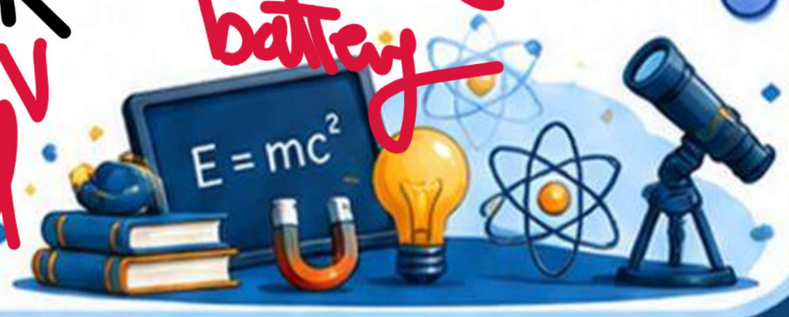
$W = P_W \times t$

= Area under graph

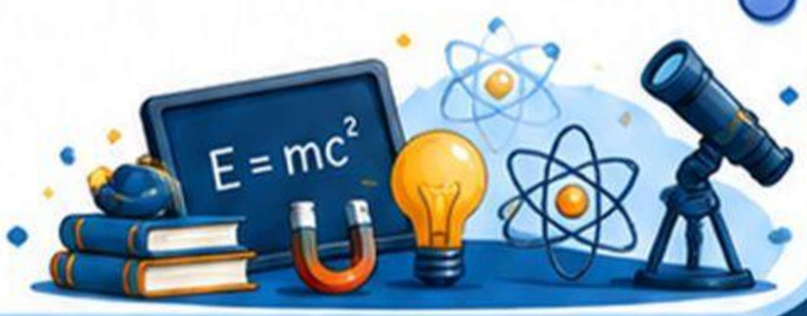




Connected To same battery



يلا بينا...
نتأكد من فهمك


E = mc²

An electric cell of unknown emf is connected to resistance R_1, hence the intensity of the passing current through the cell equals 0.5 A as in figure (1) and when replacing resistance R_1 by another resistance R_2, the intensity of the passing current through the cell becomes 0.3 A as in figure (2), so the emf of the cell equals ….. .

(Exp. May 21)

- 1.2 V
- 1.5 V
- 2 V
- 3 V ✓

$V_B = I_t[R_{eq}+r]$

$V_B = 0.5[5+r]$

$V_B = 0.3[9+r]$

$0.5[5+r] = 0.3(9+r)$

$r = 1\Omega$

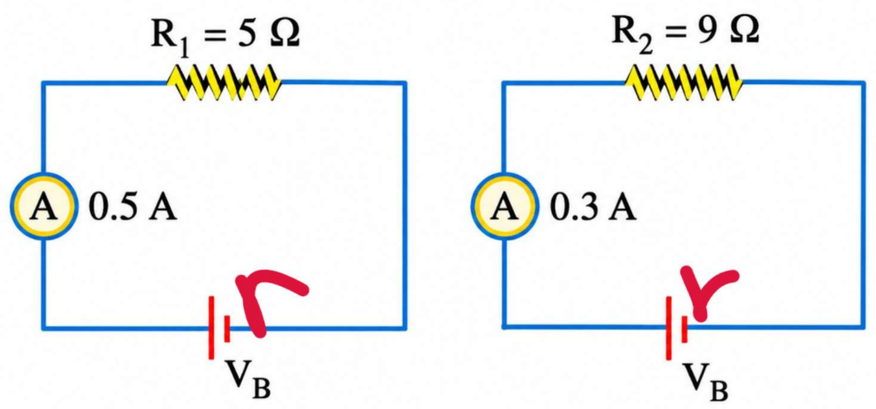


Figure (1) Figure (2)

$V_B = 0.3(9+1) = 3V$

Each of the four electric circuits contains an ammeter. What is the correct arrangement of the readings A_1, A_2, A_3 and A_4 of the four ammeters?

(Exp.25)

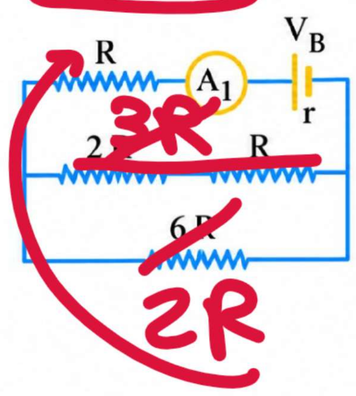
$I_m = \frac{V_B}{R_{eq}+r}$

$R_1 > R_2 > R_4 > R_3$

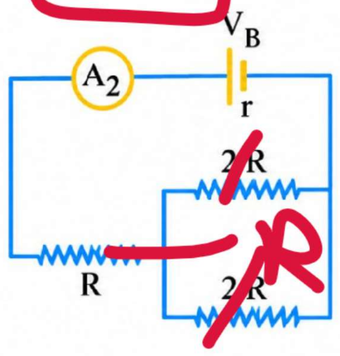
$A_1 < A_2 < A_4 < A_3$

- ✗ $A_4 < A_2 < A_1 < A_3$
- ✗ $A_4 < A_3 < A_1 < A_2$
- ✗ $A_3 < A_4 < A_2 < A_1$
- ✓ $A_1 < A_2 < A_4 < A_3$

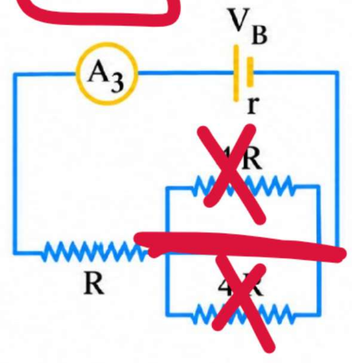
3R



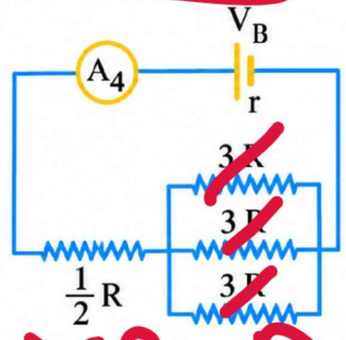
2R



R



1.5R



$\frac{0.5R+R}{1.5R}$

Questions

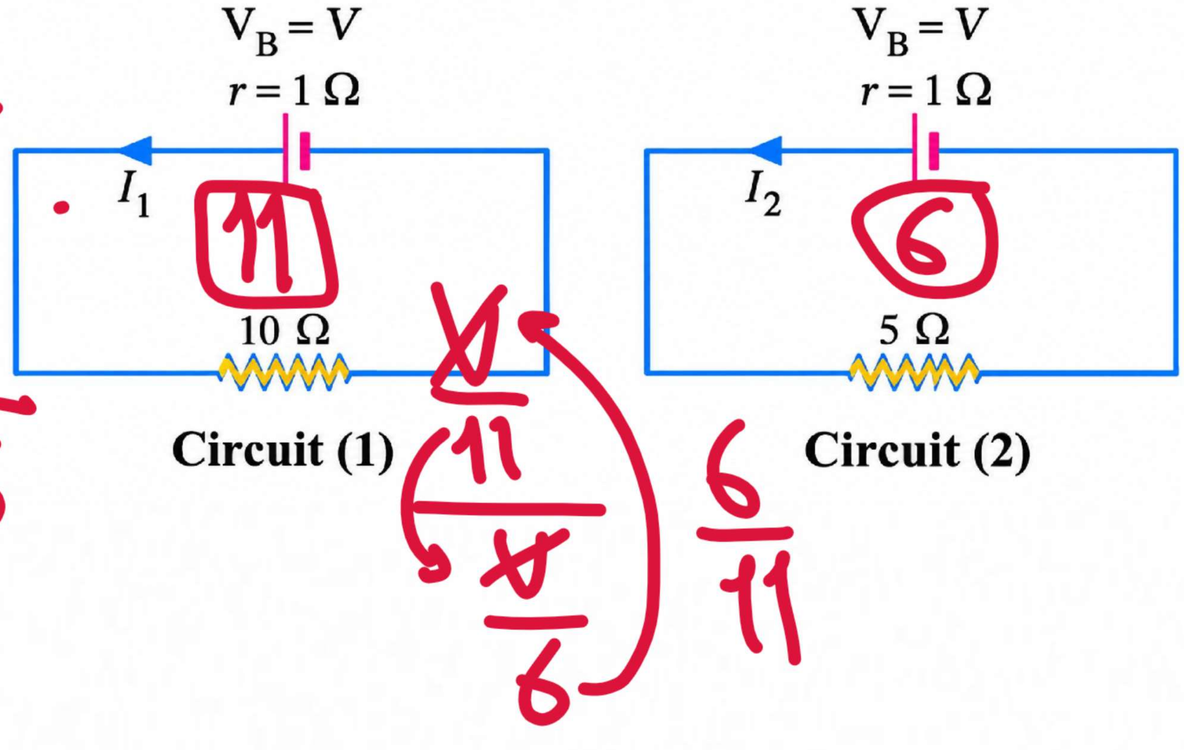
In the opposite figure, the ratio (I_1 / I_2) equals ..... .

(Exp.June 21)

- 6/11 ✓
- 11/6
- 1/2
- 1/1

$$I_1 = \frac{V_B}{R_{eqtr}} \qquad \frac{I_1}{I_2} = \frac{R_{t_2}}{R_{t_1}} = \frac{6}{11}$$

$$I_1 = \frac{V}{10+1} = \frac{V}{11} \qquad I_2 = \frac{V}{5+1} = \frac{V}{6}$$



Questions

In the opposite closed electric circuit, the ratio of (V_1 / V_2) = ...

Discharging

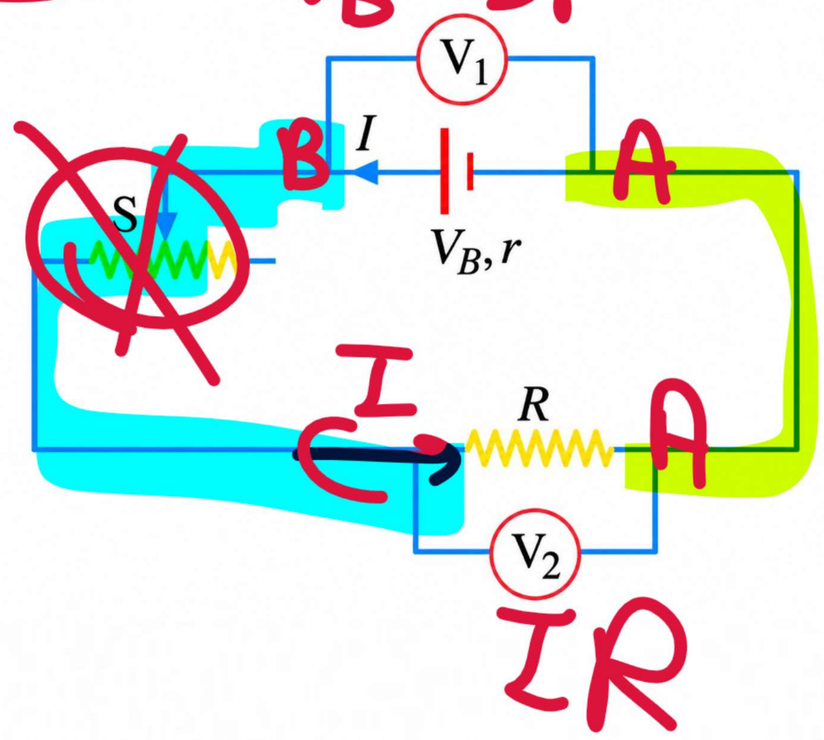
$V_B \rightarrow$ Produce

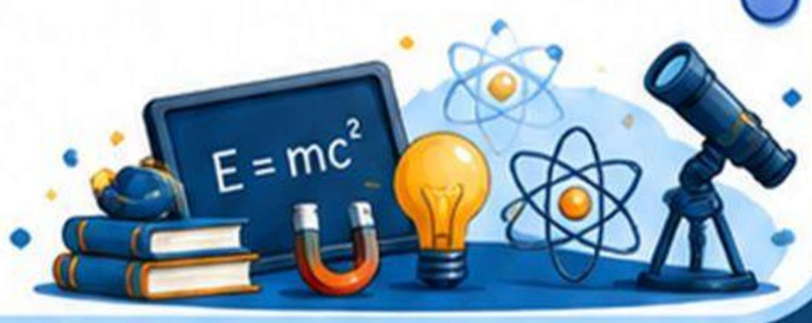
- ~~$\frac{IR}{V_B + V_2}$~~
- ~~$\frac{IR - Ir}{V_2 - V_B}$~~
- $\frac{V_B - Ir}{IR}$ ✓
- ~~$\frac{V_B + Ir}{IR}$~~

$V_1 \neq V_2$

(1st session 21)

Produce $V_B - Ir$ Consume



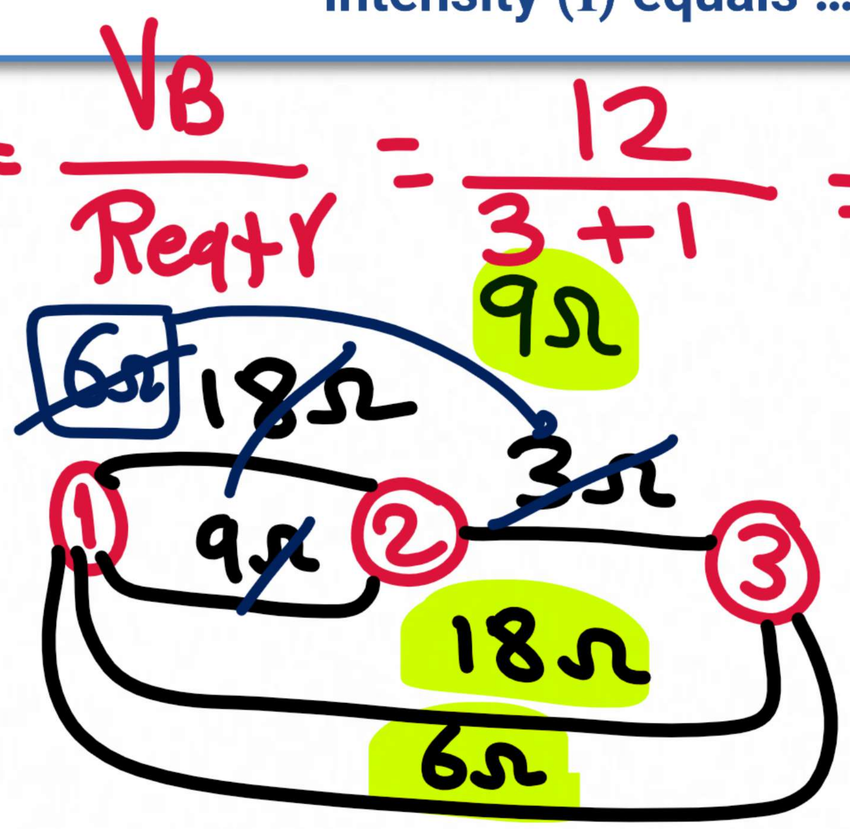


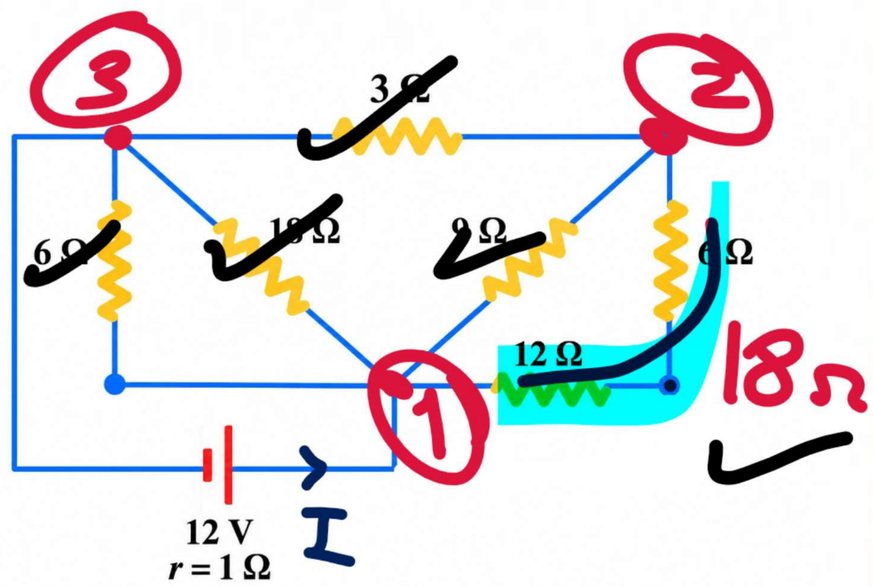
Questions

In the shown electric circuit, the electric current intensity (I) equals ..... .

(1st session 21)

- 4 A
- 0.76 A
- 0.83 A
- 3 A ✓





$I_m = \frac{V_B}{Req+r} = \frac{12}{3+1} = 3A$

$\frac{18 \times 9}{18+9} = 6\Omega$

$Req = \left[\frac{1}{9} + \frac{1}{18} + \frac{1}{6}\right]^{-1} = 3\Omega$

Questions

In the opposite closed electric circuit, when the value of resistance S is increased,

(2nd session 21)

$R_v \uparrow \; Req \uparrow \; I_m \downarrow$

$V_1 = V_B - I_m[R+r]$

- the readings of both (V_1) and (V_2) increase ✓
- the reading of (V_1) increases while of (V_2) decreases
- the reading of (V_1) decreases while of (V_2) increases
- the readings of both (V_1) and (V_2) decrease

$V_2 = V_B - Ir$

$V_1 = I R_v$

In the opposite electric circuit: If the reading of voltmeter V_3 equals 0.8 V. Which choice correctly represents the readings of V_1 and V_2?

$V_{B_1}+Ir_1$ $V_{B_2}-Ir_2$

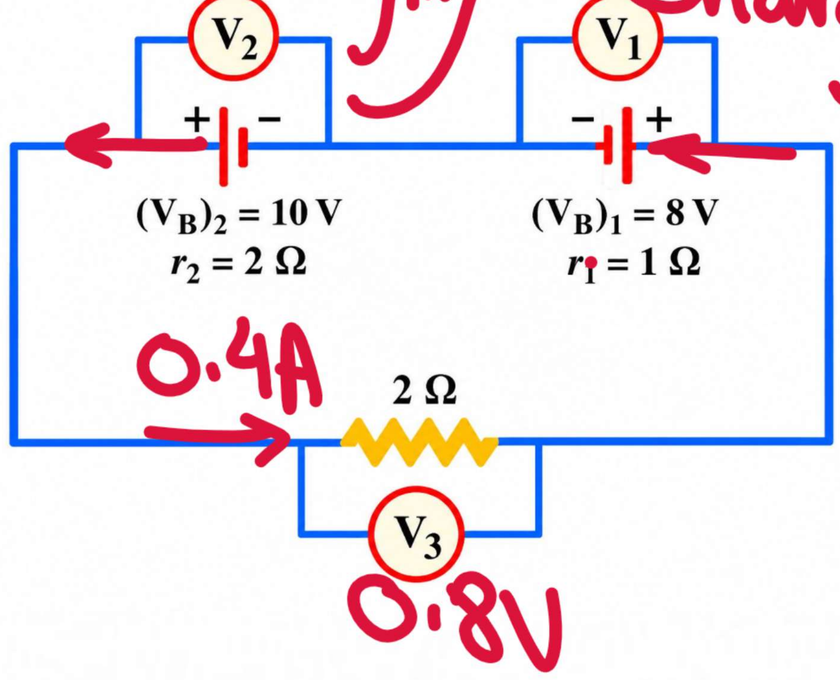
V_1	V_2
~~10 V~~	6 V ✗
8.4 V ✓	9.2 V ✓
~~7.6 V~~	9.2 V ✓
~~4 V~~	8 V ✗

$8+0.4\times1$ $10-0.4\times2$

V_B-Ir Discharging

V_B+Ir Charging

(2nd session 21)



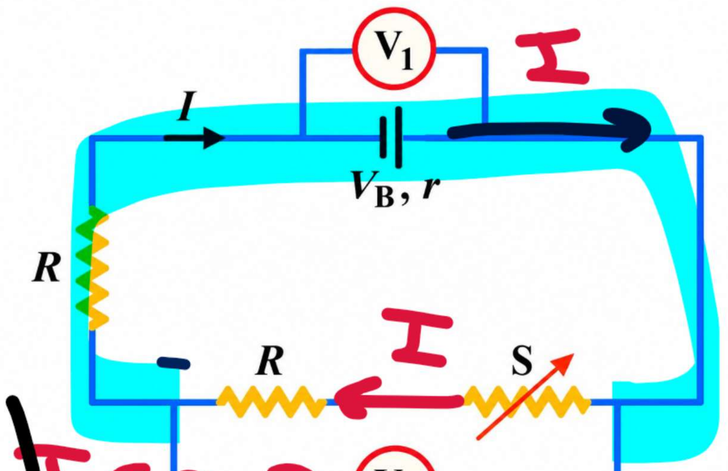
In the shown electric circuit: On increasing the value of the variable resistance (S). Which choice correctly represents the change in the readings of voltmeters (V_1) and (V_2)?

$R_v\uparrow R_{eq}\uparrow I_m\downarrow$

$V_1=V_B-Ir\uparrow$

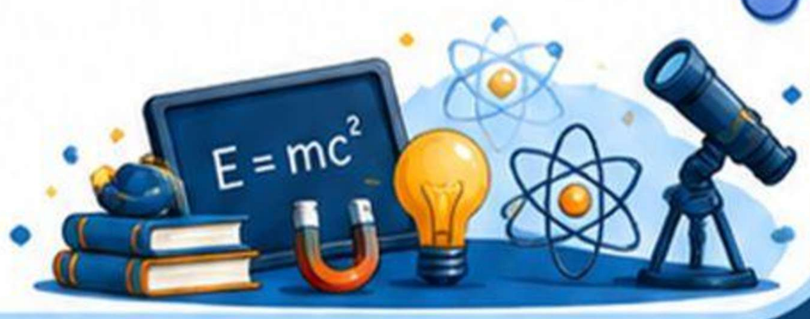
(1st session 22)

V_1	V_2
Increases ✓	Increases ✓
Remains constant ✗	Increases ✓
Decreases ✗	Remains constant ✗
Decreases ✗	Decreases ✗



$V_2=V_B-I(R+r)\uparrow$

$V_2=I(R_v+R)\downarrow$



If the voltmeter reads 18 V when the switch (K) is opened, but reads 15 V when the switch is closed, so the internal resistance of the cell = ……. .

(1st session 22)

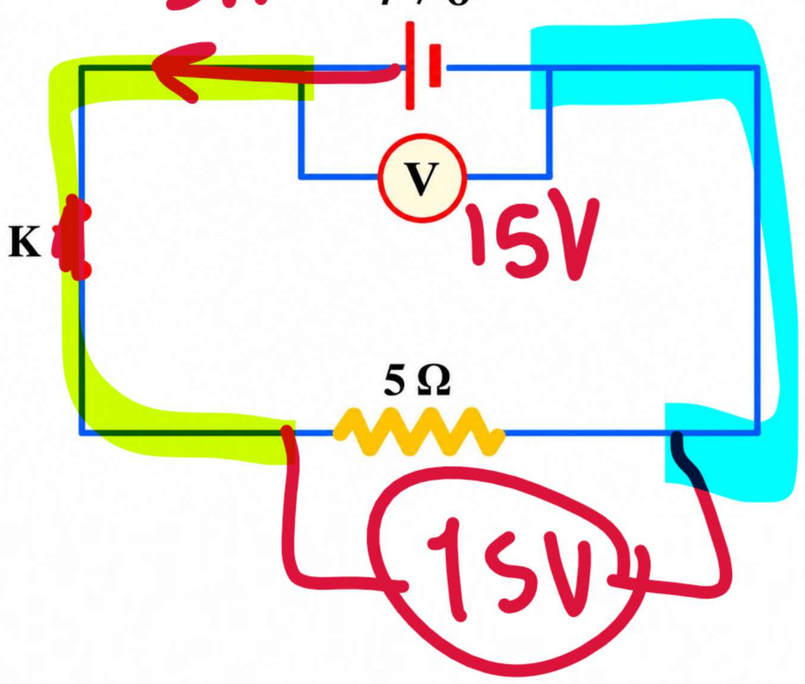
$V_{open} = V_B = 18V$

- 2 Ω
- 3 Ω
- 1 Ω ✓
- 4 Ω

$V_{Closed} = V_B - Ir$

$15 = 18 - 3 \times r$

$r = 1\Omega$



In the electric circuit shown, the voltmeter reads 14 V when the switch (K) is opened, but reads 8 V when the switch (K) is closed. Then the internal resistance of the battery = ……. .

(2nd session 22)

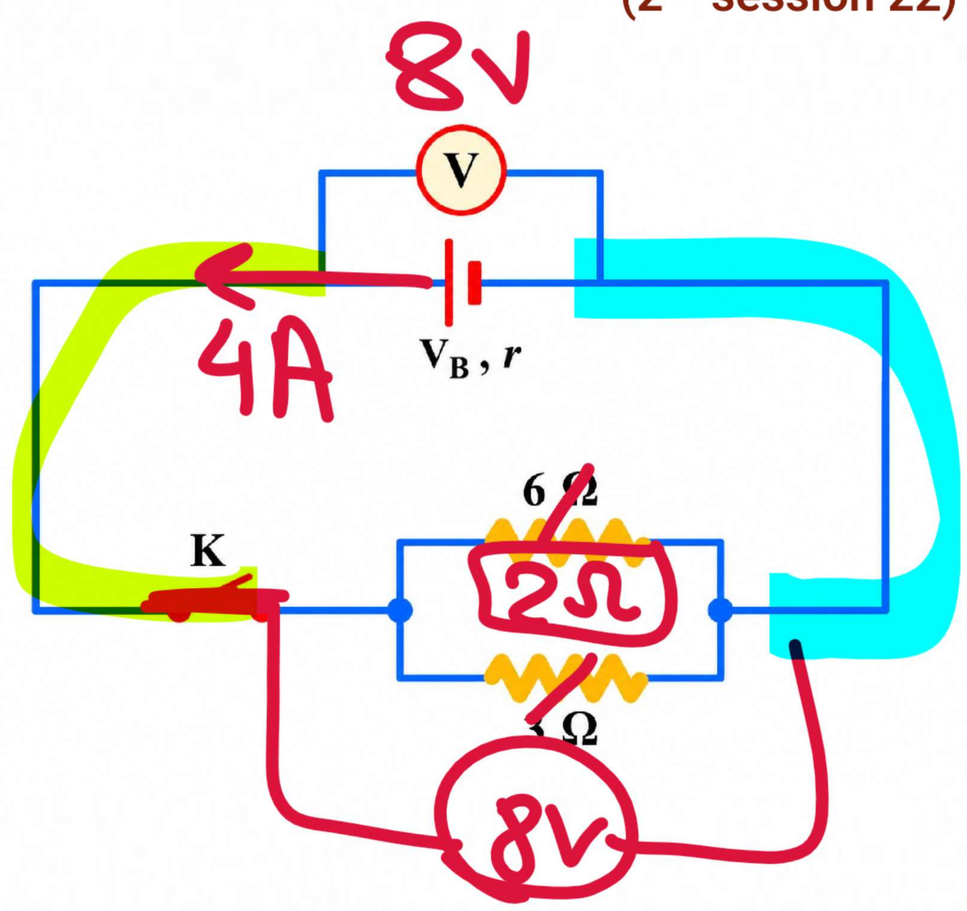
$V_{open} = V_B = 14V$

$V_{Closed} = V_B - Ir$

$8 = 14 - 4 \times r$

$r = 1.5\Omega$

- 12.5 Ω
- 0.5 Ω
- 1.5 Ω ✓
- 0.25 Ω



Mr.Abdelwahab

Questions

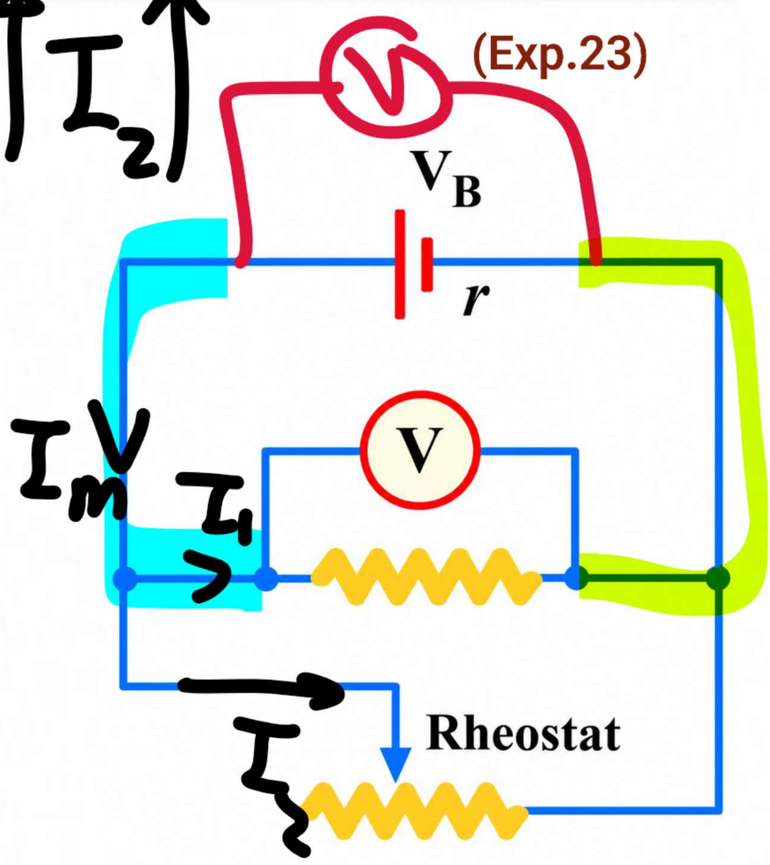
According to the opposite circuit, which of the following represents what will happen to the reading of the voltmeter by changing the value of resistance which is taken from rheostat?

The value of the resistance taken from the rheostat	The reading of the voltmeter
Decreases ✓	Decreases ✓
Decreases	Increases
Increases	Decreases
Increases	Doesn't change

$R_v \downarrow \quad R_{eq} \downarrow \quad I_m \uparrow \quad I_2 \uparrow$

$V = V_B - I_m r$

(Exp.23)



Questions

The opposite graph shows the relation between the terminal voltage of a battery (V) that has an internal resistance 0.5 Ω connected with a closed electric circuit and the current intensity (I) that passes through it, then the value of the electromotive force of the battery is ……. .

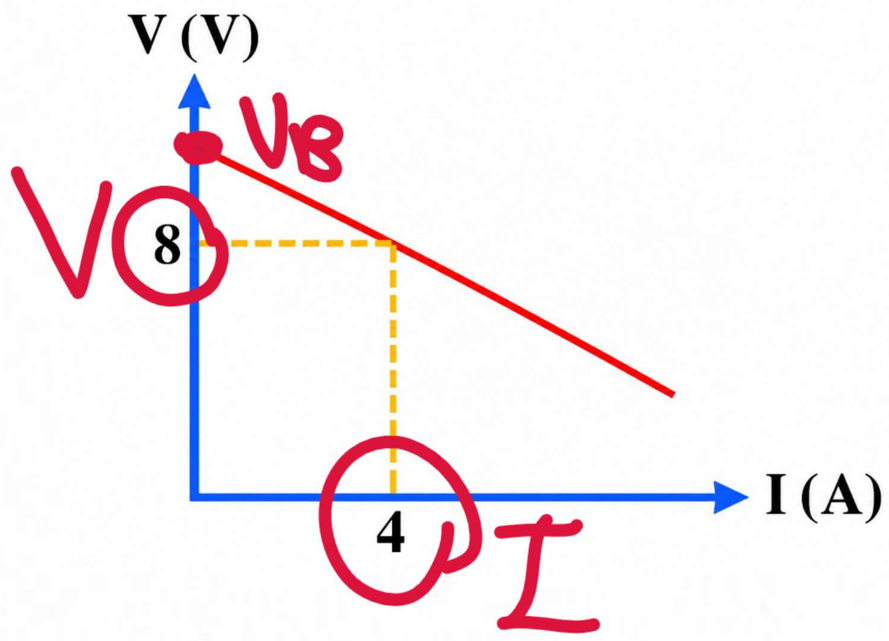
- 8 V
- 10 V ✓
- 9 V
- 12 V

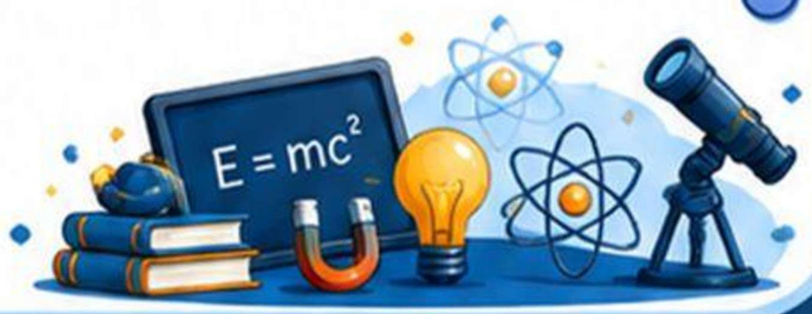
$V = V_B - I_m r$

$8 = V_B - 4 \times 0.5$

$V_B = 10\,V$

(Exp. 23)





Questions

In the shown closed electric circuit:
We can conclude that …… .

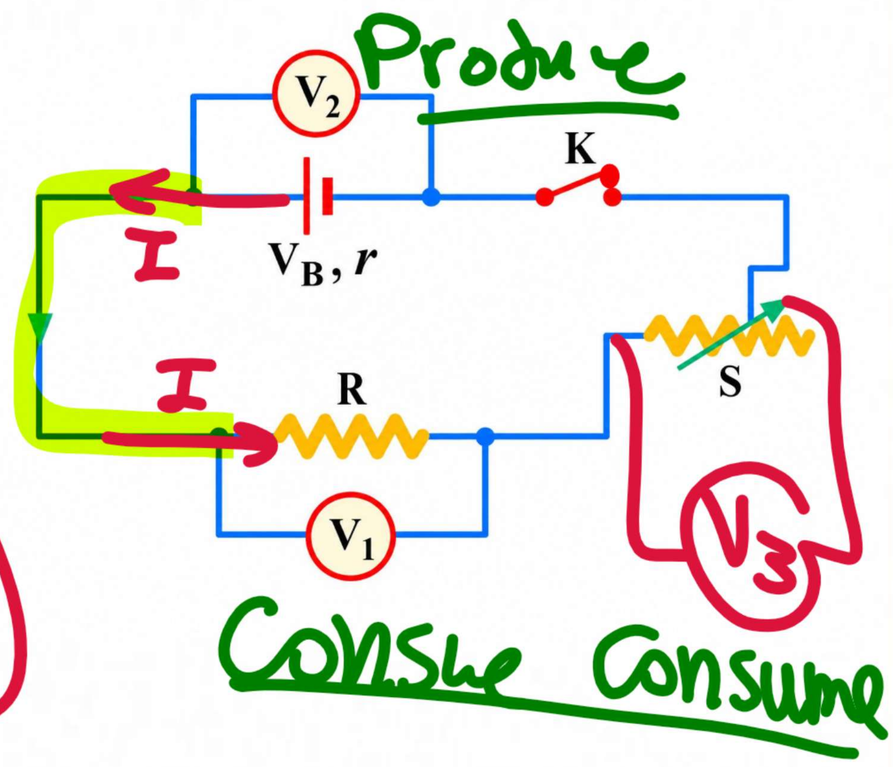
- $V_2 < V_B$ ✓
- $V_1 > V_B$ ✗
- $V_2 = V_B$ ✗
- $V_1 = V_B$ ✗

$V_2 = V_B - Ir \rightarrow V_B = V_2 + Ir$ (2nd session 23)

$V_1 = IR$

$V_2 = V_1 + V_3$

$V_B > V_2 > V_1$



Questions

The figure illustrates a cell with internal resistance (r), when it is connected to 2 different circuits (each individually), the ammeter reading is (1.2 A) in the first circuit and (1 A) in the second circuit. So the value of the internal resistance of the cell = ……. .

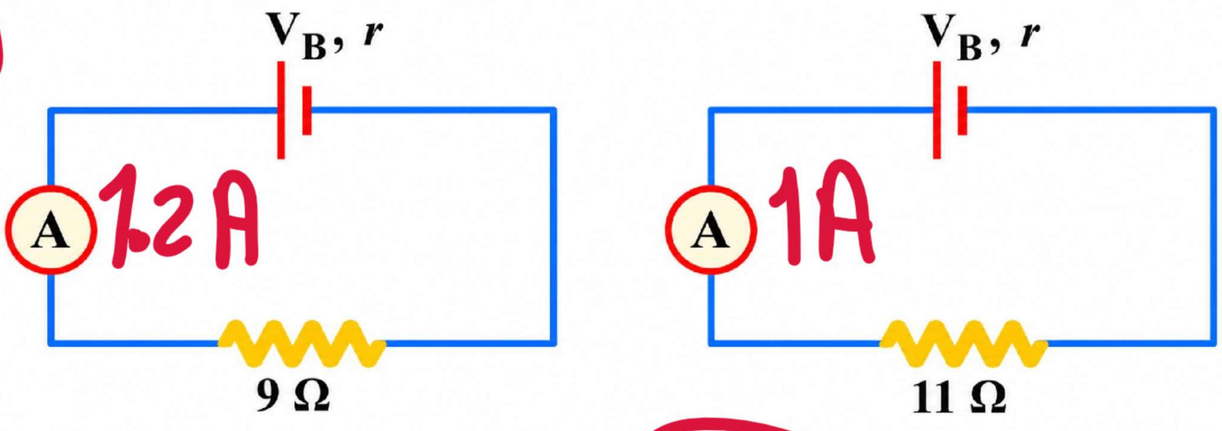
$V_B = I_t[R_{eq} + r]$

(2nd session 23)

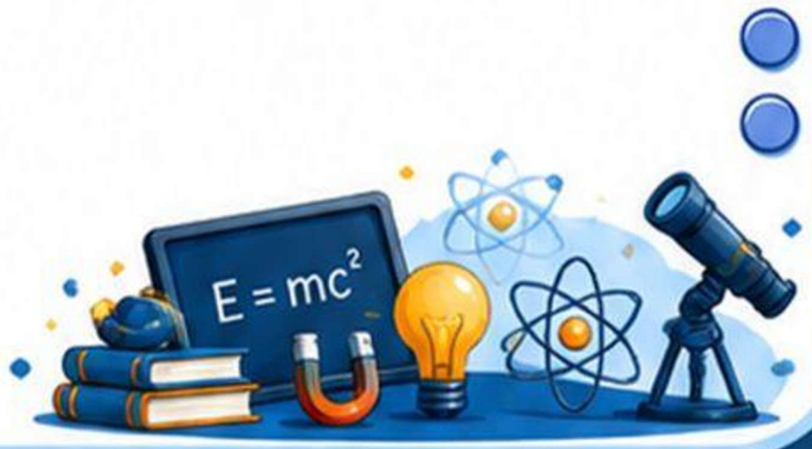
$V_B = 1.2[9 + r]$

$V_B = 1[11 + r]$

- 1.5 Ω
- 2 Ω
- 0.5 Ω
- 1 Ω ✓



$1.2[9 + r] = 1[11 + r] \rightarrow r = 1\ \Omega$

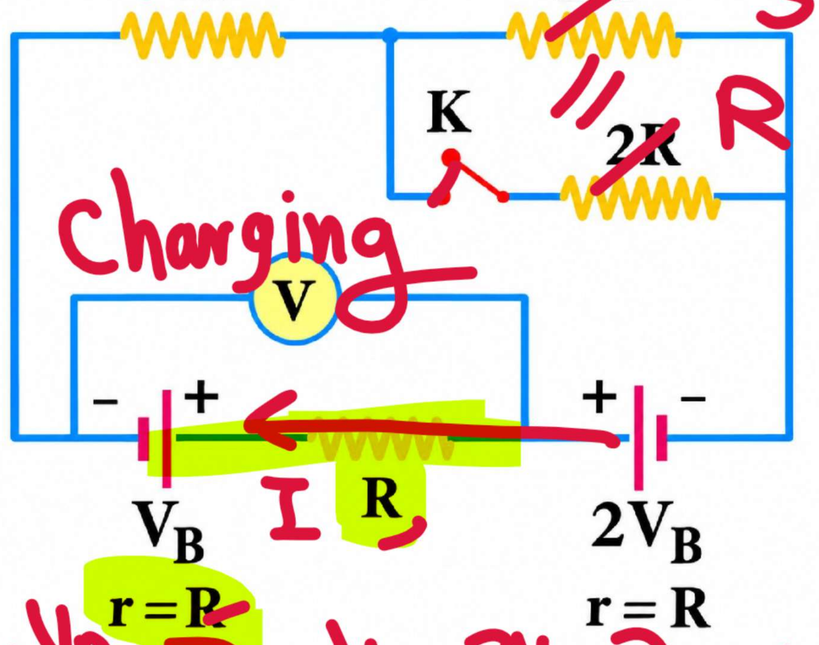


Questions

In the shown electric circuit, which choice is correct?

(1st session 24)

Reading of voltmeter when K is opened	Reading of voltmeter when K is closed
(4/3) V_B	(6/5) V_B
(4/3) V_B	(7/5) V_B
(7/6) V_B	(6/5) V_B
(7/6) V_B	(7/5) V_B

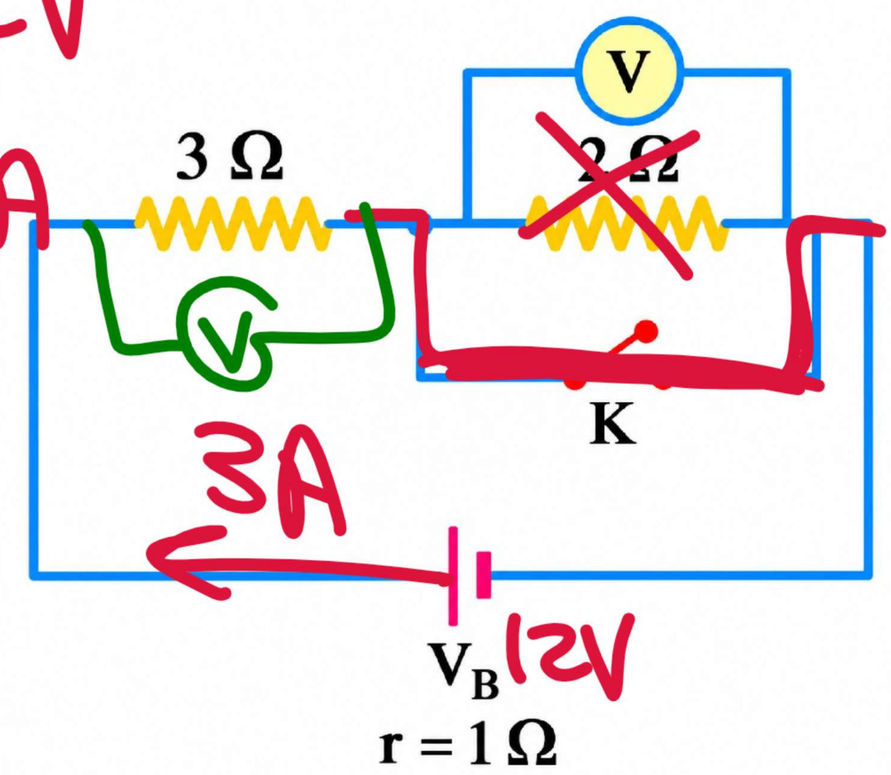


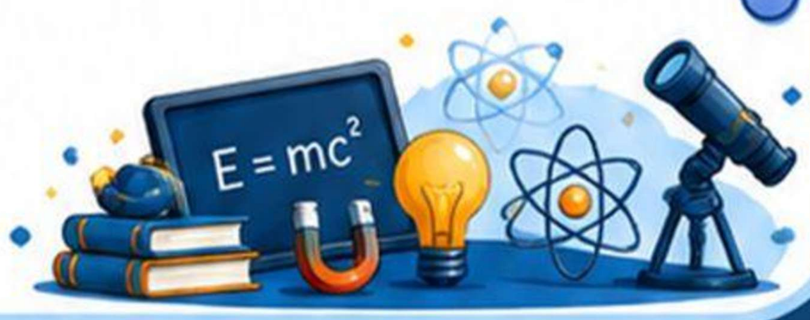
Questions

The opposite figure represents an electric circuit, if the reading of the voltmeter, when the key is open = 4 V Then the potential difference between the terminals of the 3 Ω resistor when the key is closed = … .

(2nd session 24)

- 4 V
- 8 V
- 6 V
- 9 V





Questions

In the opposite electric circuit, if the internal resistance of the battery is 0.5 Ω and the reading of the voltmeter when switch K is open is 21 V and when it is closed, is 19.5 V, then the intensity of the electric current in the circuit and the value of resistance R respectively are ..... ,.....

(Exp. 25)

- 2 A, 10 Ω
- 2 A, 13 Ω
- 3 A, 10 Ω
- 3 A, 13 Ω

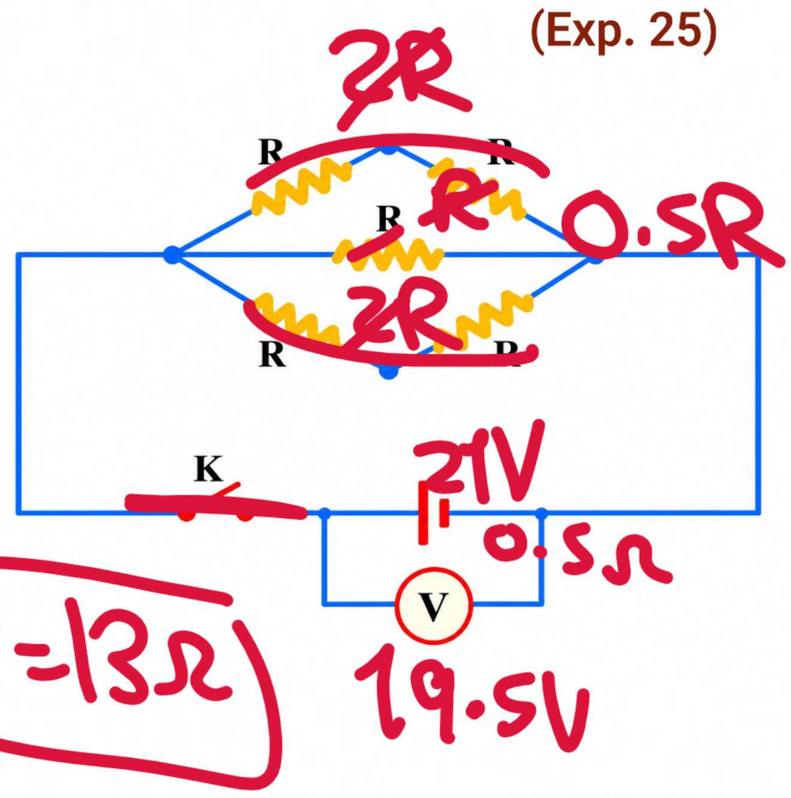
$V = V_B - Ir$

$19.5 = 21 - I \times 0.5$

$I = 3A$

$I = \frac{V}{R_{eq}} \rightarrow 3 = \frac{19.5}{0.5R} \rightarrow R = 13\Omega$

$I = \frac{V_B}{R_{eq+r}}$



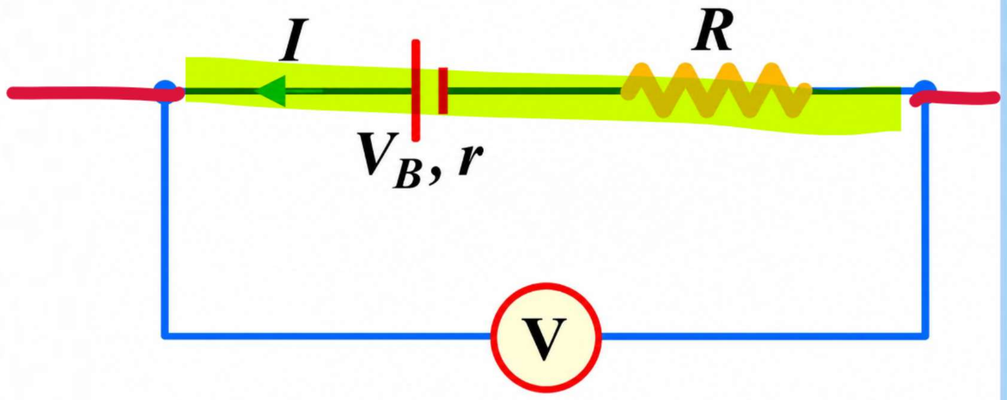
Questions

The opposite figure represents section of electric circuit, so the reading of the voltmeter (V) can be calculated from the relation ..... .

(Exp. 25)

- $V = V_B - I(R+r)$
- $V = V_B - I(R-r)$
- $V = V_B + I(R+r)$
- $V = V_B + I(R-r)$

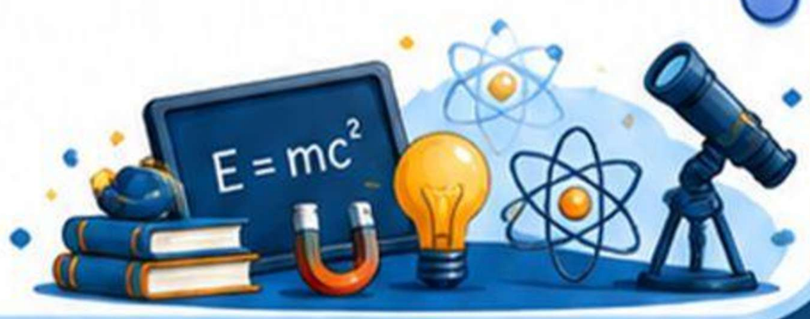
Discharging



$V = V_B - I(R+r)$

Produce

Consume



Questions

In the electrical circuit shown in the opposite figure, if the potential difference between points a and b is 15.2 V, then the value of r and the potential difference between points c and d respectively are ..... ,.....

$V=V_B+Ir=7+0.5\times1.4$

$=7.7V$

- 1.4 Ω , 7.7 V
- 1.4 Ω , 7.8 V
- 0.7 Ω , 7.7 V
- 2.8 Ω , 8.4 V

$V=V_B-Ir$

$15.2=16-I\times1.6$

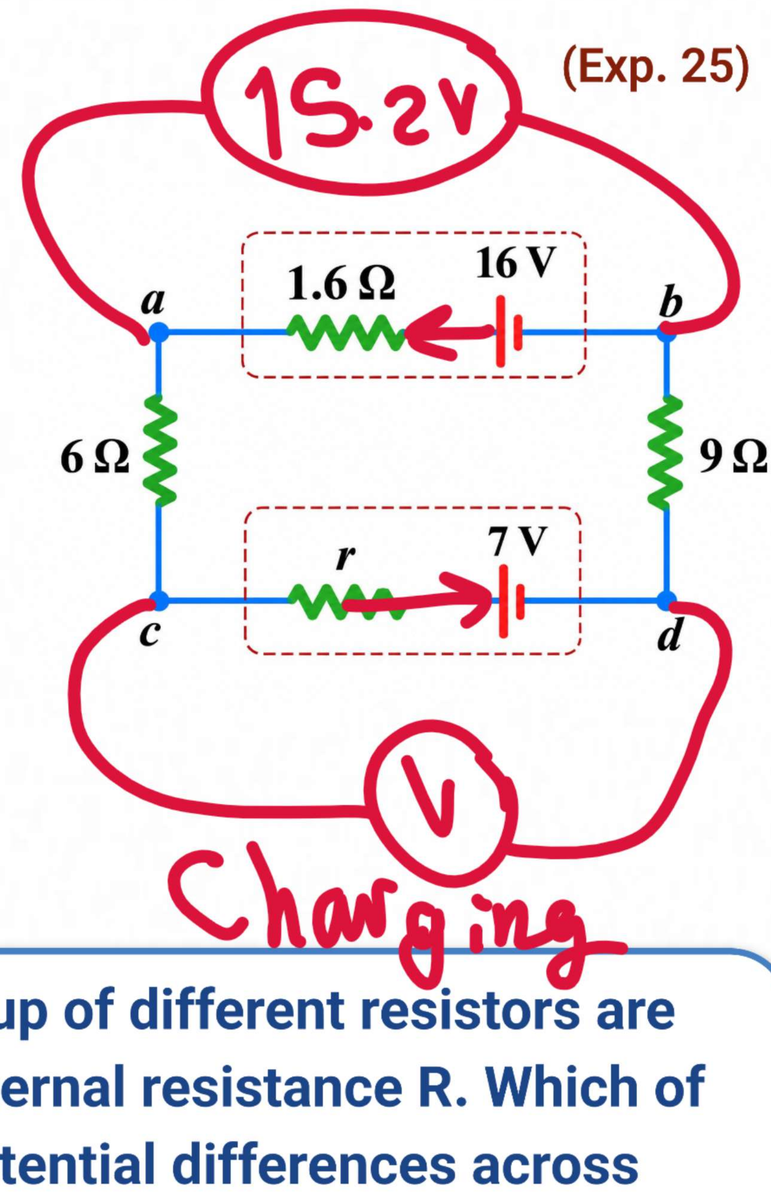
$1.6I=0.8$

$I=0.5A$

$I=\frac{16-7}{9+6+1.6+r}=0.5$

$r=1.4\Omega$

(Exp. 25)



Questions

In the opposite electrical circuit, a group of different resistors are connected together with a battery of internal resistance R. Which of the following relationships for the potential differences across the external resistors is correct ?

Let $R=1\Omega$

$V_B=12V$

$I=\frac{V_B}{R_{eq}+r}=\frac{12}{3+1}=3A$

- $V_{3R}=3V_R=3V_{2R}=1.5V_{4R}$
- $V_{3R}=3V_R=1.5V_{2R}=0.75V_{4R}$
- $V_{3R}=2V_R=V_{2R}=V_{4R}$
- $V_{3R}=2V_R=3V_{2R}=1.5V_{4R}$

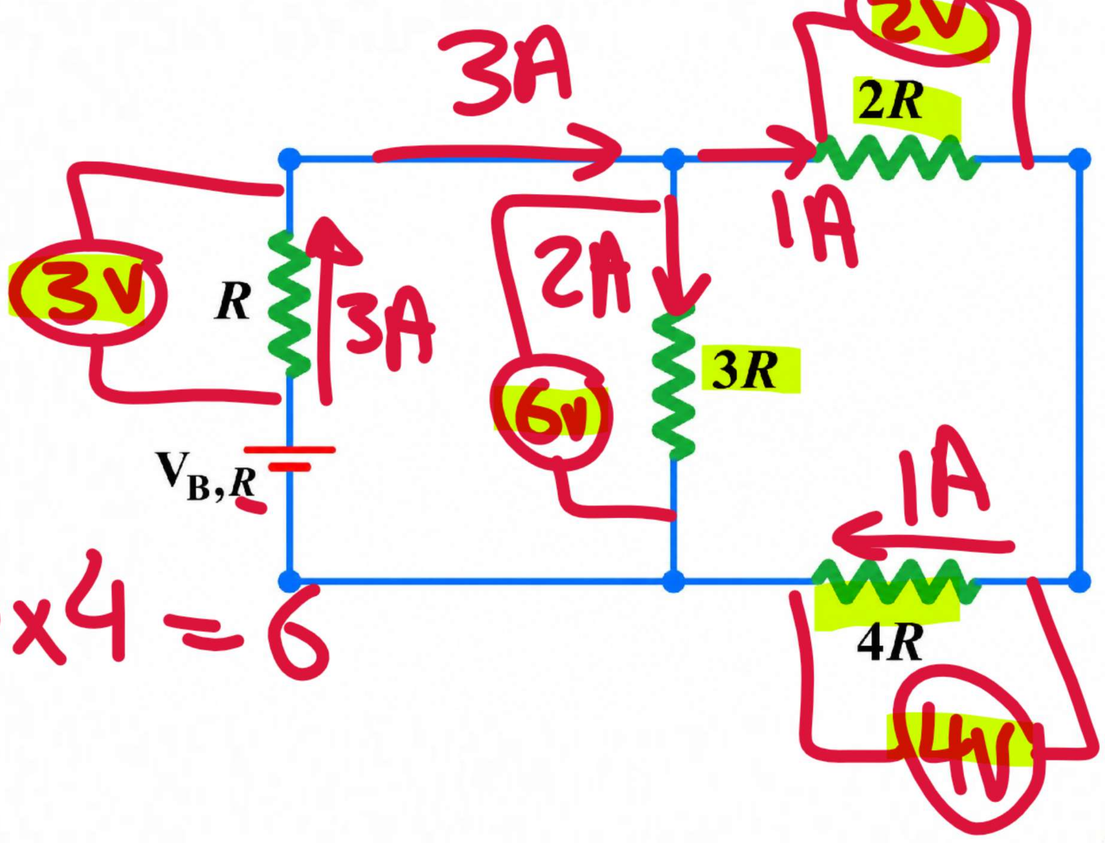
$6=3\times3$

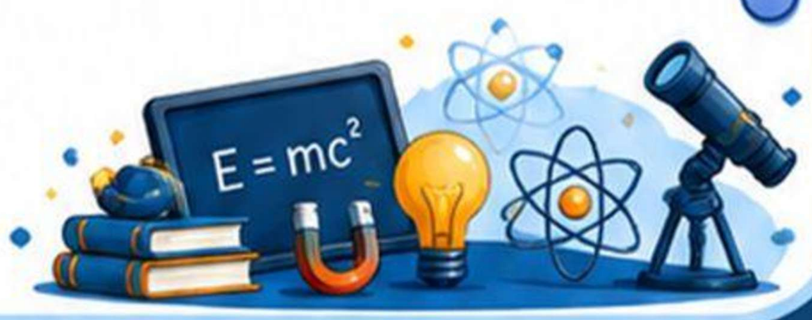
$6=3\times3$

$6=2\times3$

$6=2\times3=3\times2=1.5\times4=6$

(Exp. 25)





Questions

If the total work done to transfer a quantity of charge of 2 C inside a battery and outside it is 12 J, then this means that.....

(Exp. 25)

- the intensity of the current in the battery is 6 A
- the electrical power produced by the battery is 6 W
- the internal resistance of the battery is 6 Ω
- the electromotive force of the battery is 6 V

First session 2025

$V_B = \frac{W}{Q} = \frac{12}{2} = 6\ V$

$V_B = I[R_{eq} + r]$

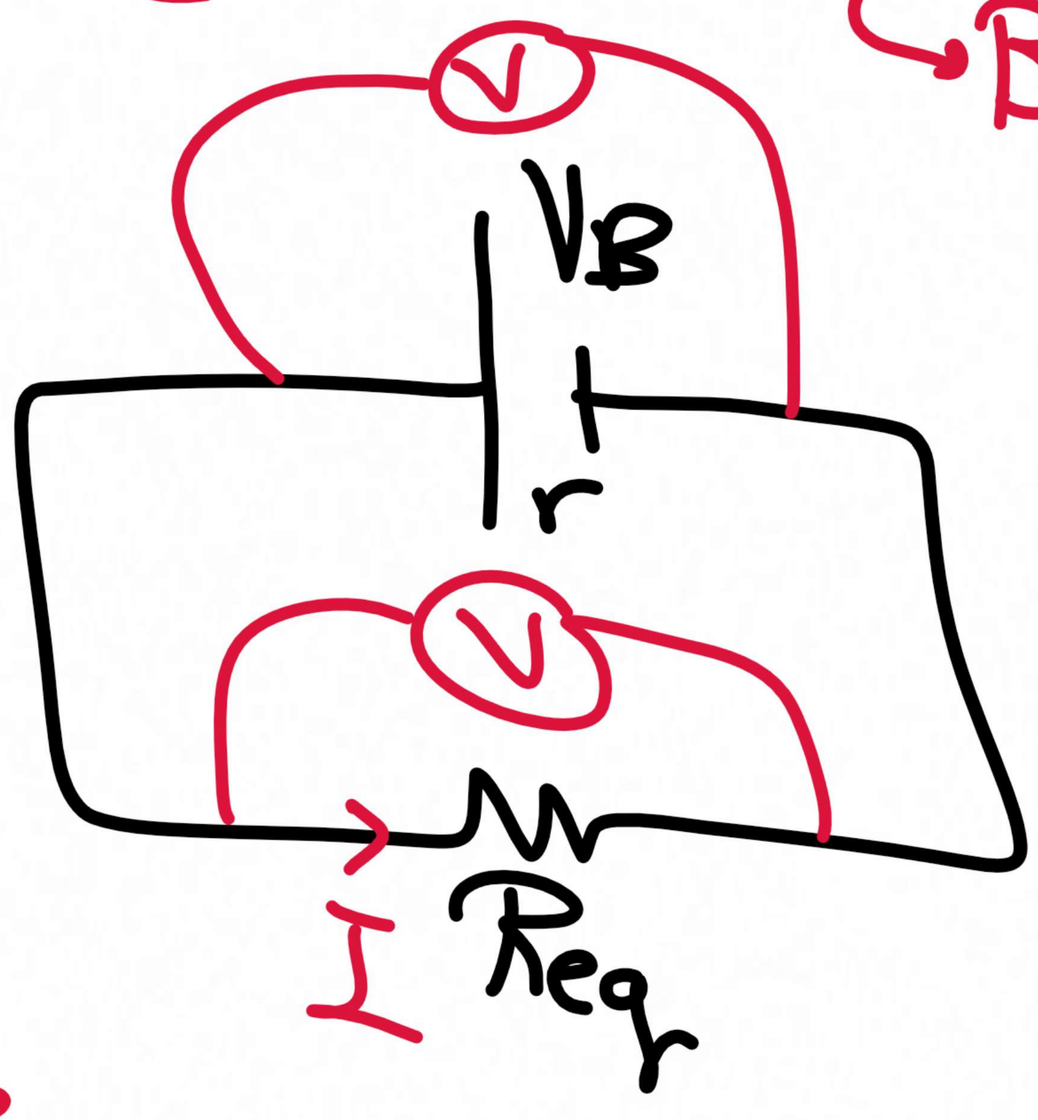
$V = I R_{eq}$

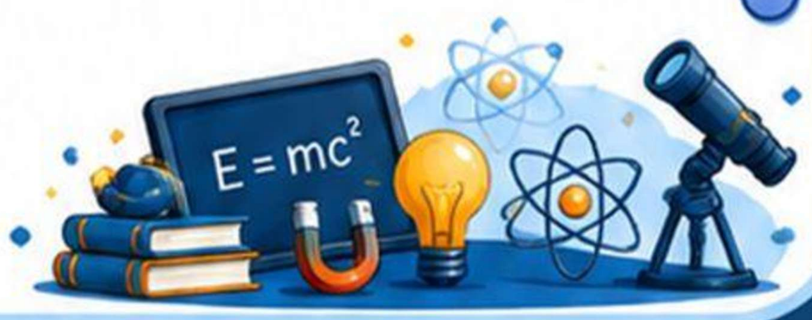
$V_{drop} = I r$

$V_B = I[R_{eq} + r]$

$V_{total} = V + V_{drop}$

outside inside

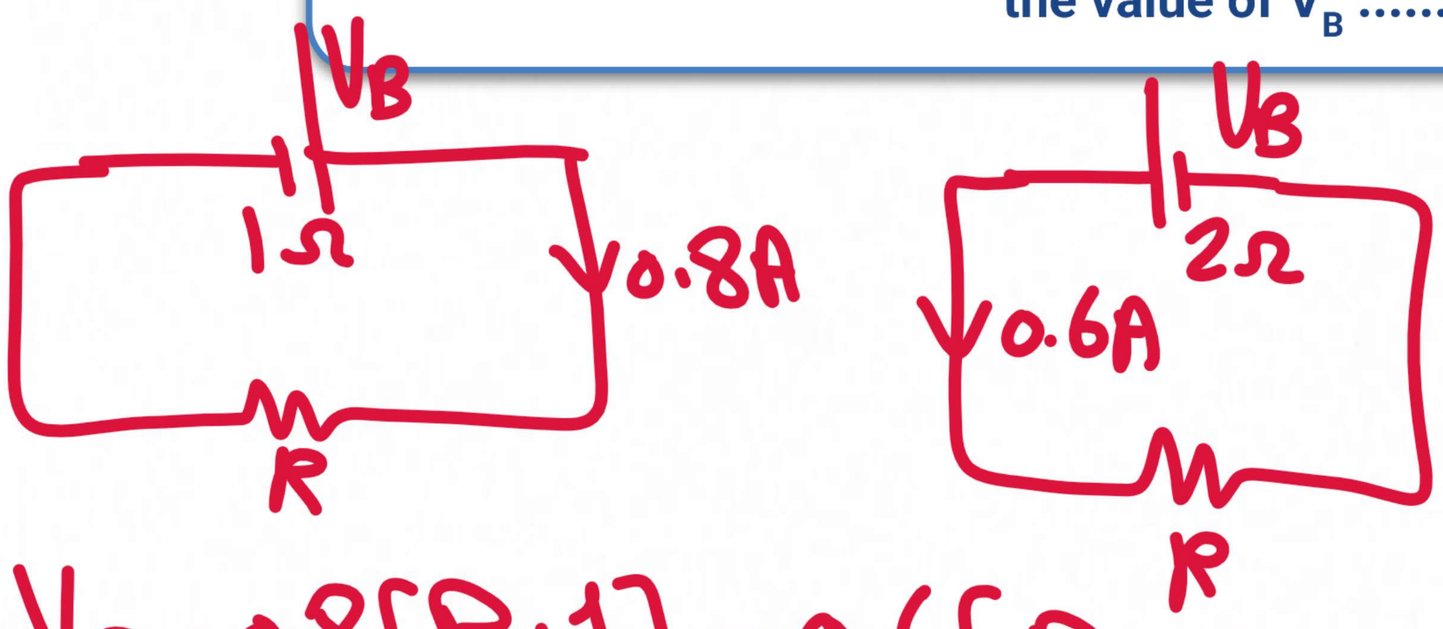




Mr.Abdelwahab

In the shown electric circuit when closing K_1 and K_3 only, a current of 0.8 A passes. And when closing K_2 and K_3 only, a current of 0.6 A passes, calculate the value of V_B .....

(1st session 24)

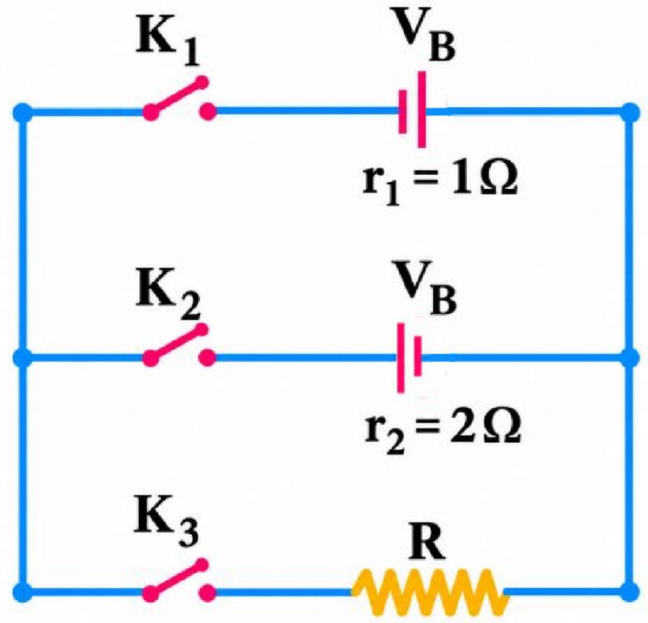


$V_B = 0.8[R+1] = 0.6[R+2]$

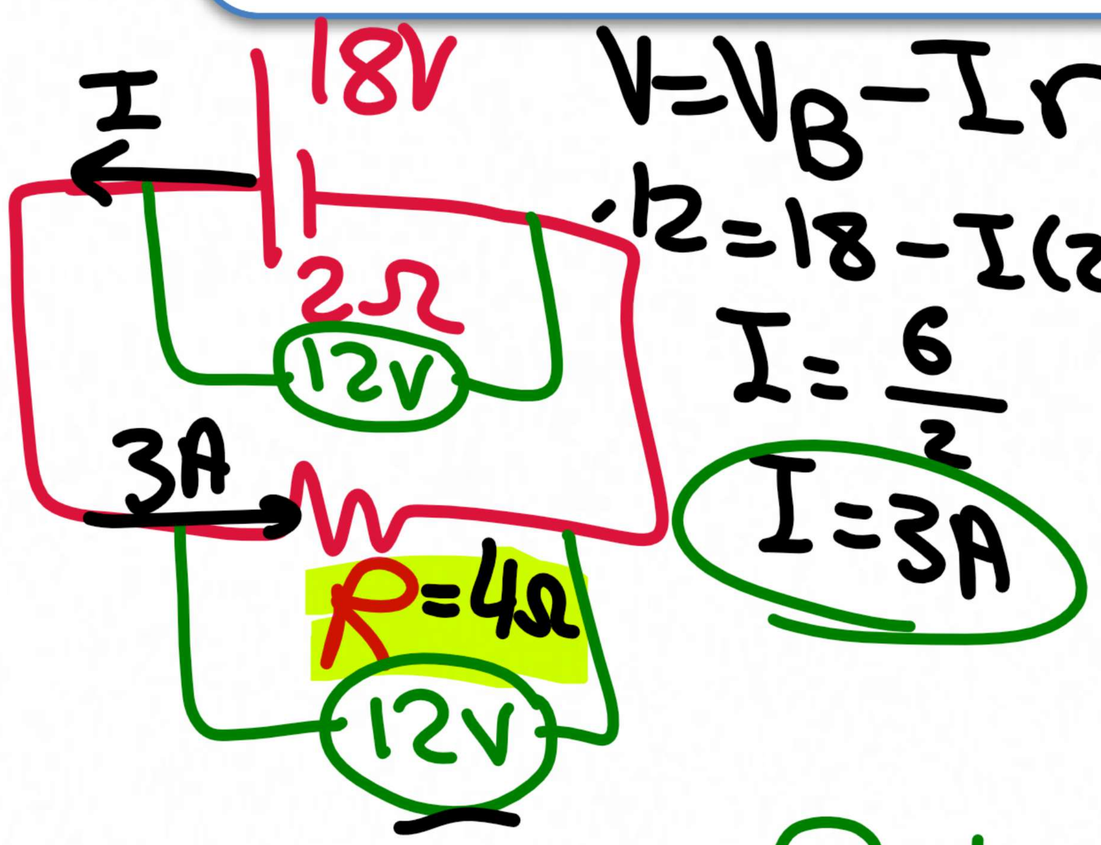
$R = 2\Omega$

$= 0.6[2+2]$

$= 2.4V$ #



battery of electromotive force 18 V and internal resistance 2 ohms is connected with resistance R, if the potential difference between the two terminals of the battery is 12 V. Calculate the electric current intensity in the circuit if a resistance of 12 ohms is connected in parallel with the resistance R?



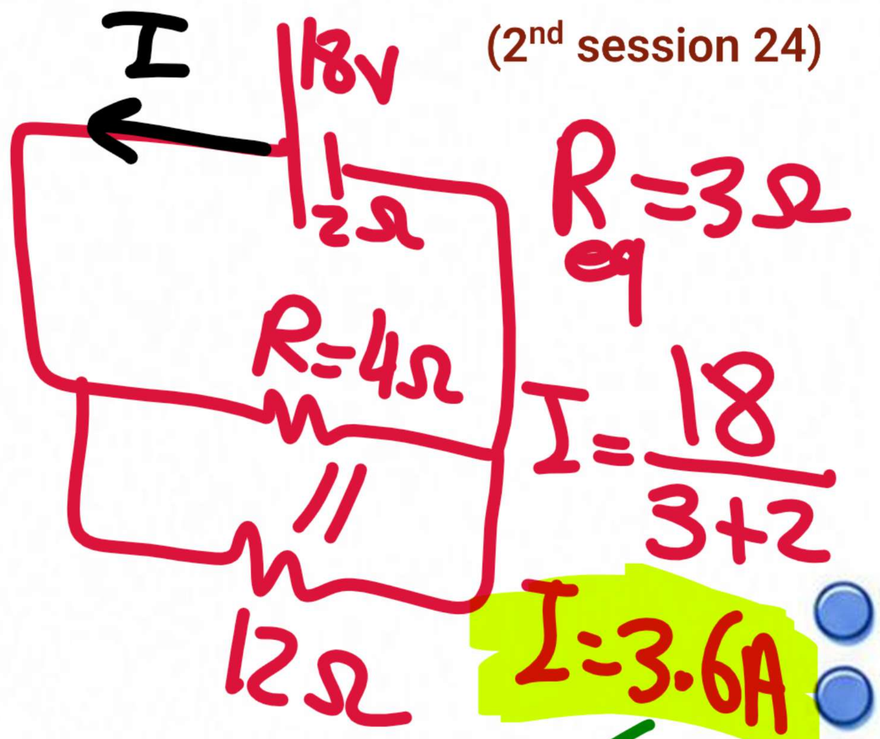
$V = V_B - Ir$

$12 = 18 - I(2)$

$I = \frac{6}{2}$

$I = 3A$

(2nd session 24)

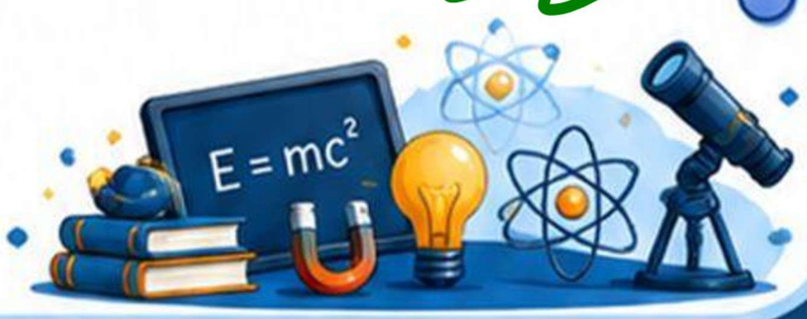


$R_{eq} = 3\Omega$

$I = \frac{18}{3+2}$

$I = 3.6A$

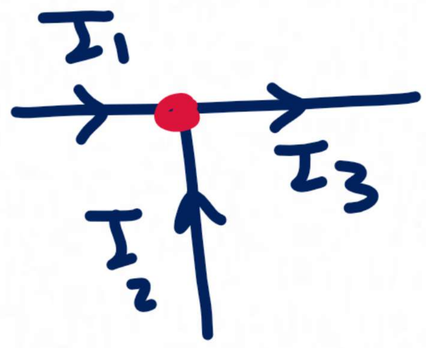
$R_{eq} \downarrow \quad I \uparrow$



Lesson Four: Kirchhoff's Laws

Mr.Abdelwahab

Kirchhoff's Law



Kirchhoff's First Law

Based on law of conservation of electric charges

KCL: $\sum I_{in} = \sum I_{out}$ or $\sum I_{node} = 0$

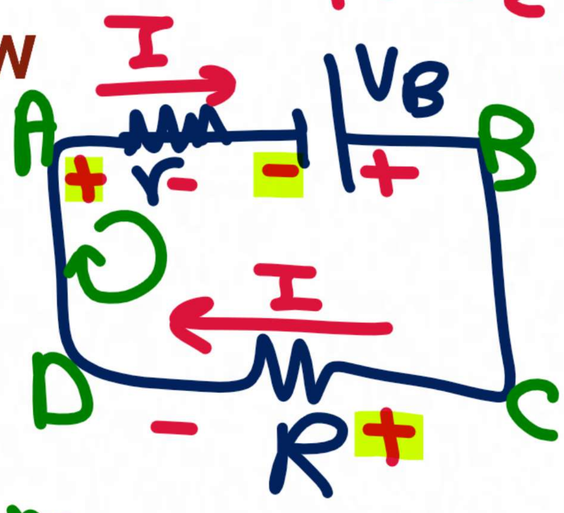
$I_1 + I_2 = I_3$

$I_1 + I_2 - I_3 = 0$

Kirchhoff's Second Law

KVL: $\sum V_{closed\ Loop} = zero$

Based on law of conservation of energy



KVL ABCDA

$Ir - V_B + IR = 0$

$Ir + IR = V_B$

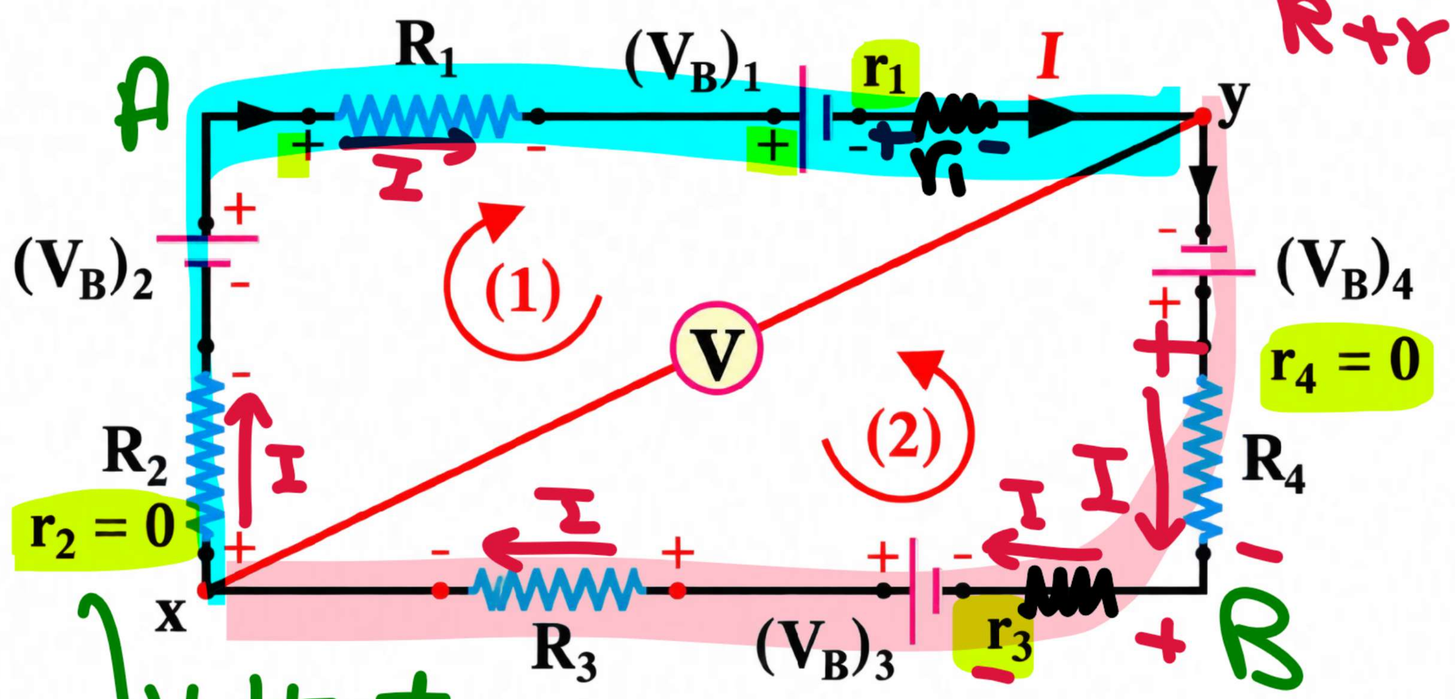
$I(R+r) = V_B$

$I = \frac{V_B}{R+r}$

KVL AyBxA Closed loop

$\sum V_{closed\ Loop} = Zero$

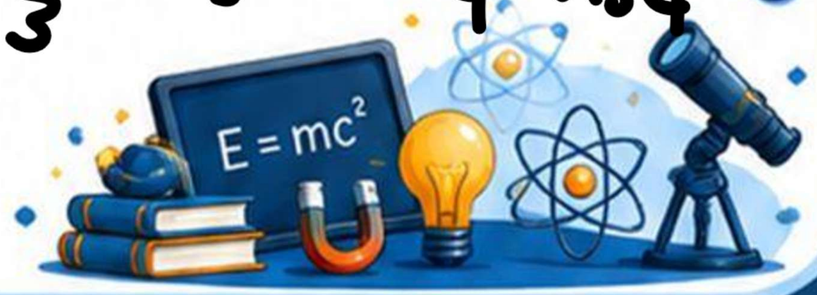
$+IR_1 + V_{B_1} + r_1 I - V_{B_4} + IR_4 + Ir_3 - V_{B_3} + IR_3 + IR_2 - V_{B_2} = 0$



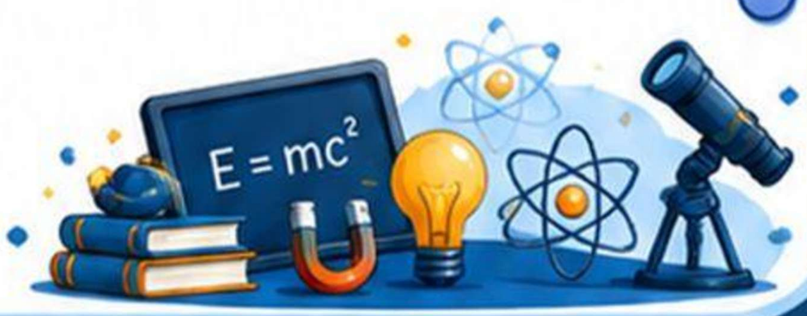
Voltmeter Reading

$V_{xy} = +IR_2 - V_{B_2} + IR_1 + V_B + Ir_1$

$V_{xy} = -IR_3 + V_{B_3} - Ir_3 - IR_4 + V_{B_4}$



يلا بينا...
نتأكد من فهمك
؟


E = mc²

Questions

By applying Kirchhoff's first law at the point (x), So ..... .

KcL

$\Sigma I_{node} = 0$

$\Sigma I_{in} = \Sigma I_{out}$

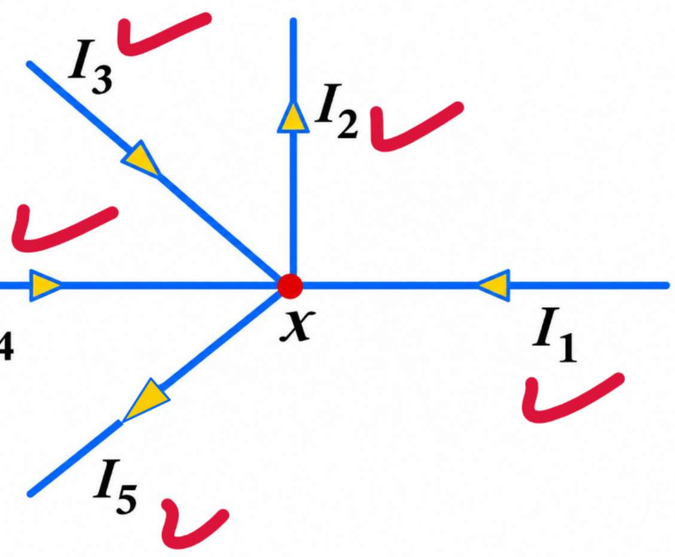
(Exp.June 21)

- $I_1 + I_3 + I_4 + I_2 + I_5 = 0$
- $-I_1 - I_3 - I_4 + I_2 + I_5 = 0$
- $-I_1 - I_3 + I_4 + I_2 + I_5 = 0$
- $I_1 + I_3 + I_4 - I_2 + I_5 = 0$

$I_1 + I_3 + I_4 = I_2 + I_5$

$I_1 + I_3 + I_4 - I_2 - I_5 = 0$

$-I_1 - I_3 - I_4 + I_2 + I_5 = 0$



Questions

From the opposite electric circuit, the correct equation is ..... .

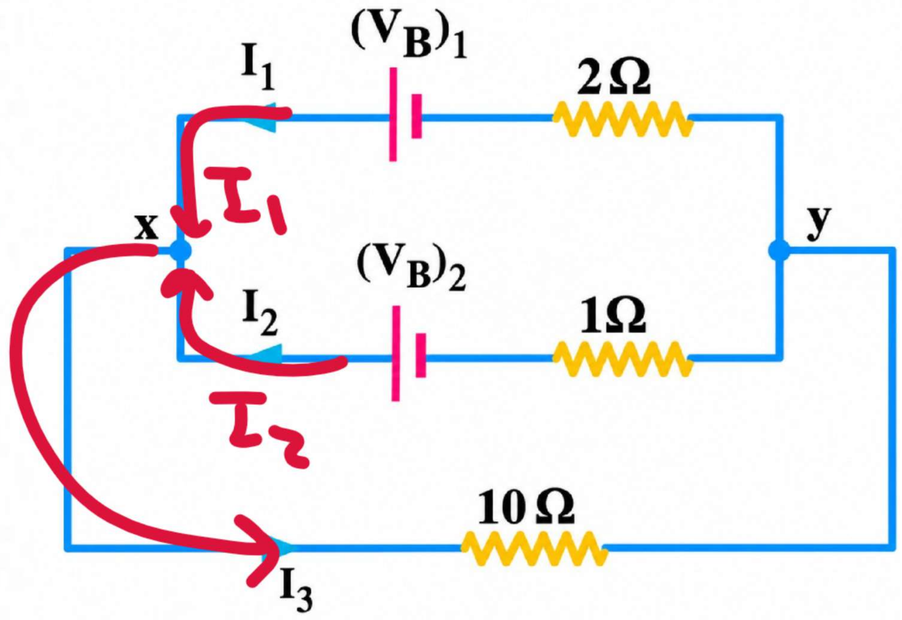
(2nd session 21)

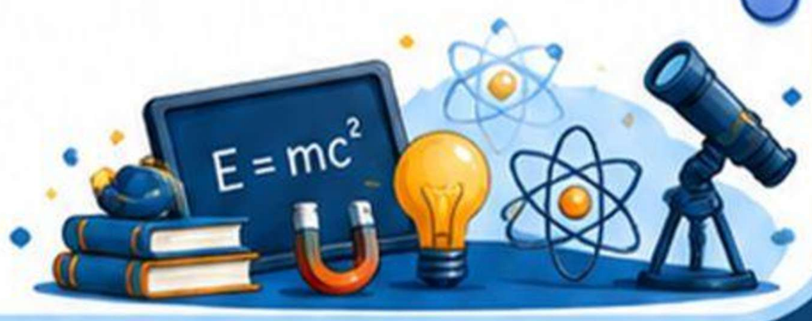
- $-I_1 - I_2 + I_3 = 0$
- $I_1 - I_2 - I_3 = 0$
- $-I_1 + I_2 + I_3 = 0$
- $I_1 + I_2 + I_3 = 0$

$I_1 + I_2 = I_3$

$I_1 + I_2 - I_3 = 0$

$-I_1 - I_2 + I_3 = 0$





Mr.Abdelwahab

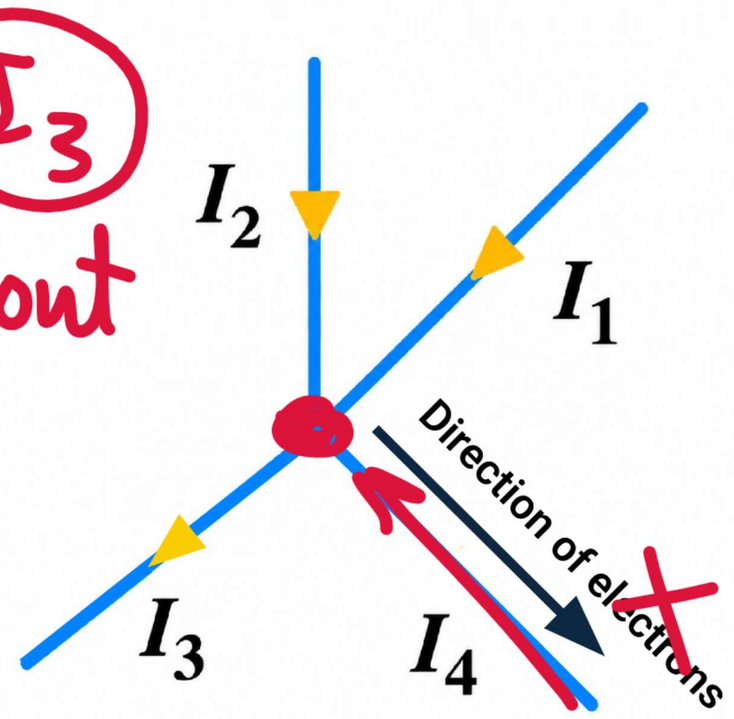
Questions

The figure shows a part of a closed electric circuit: If the directions of I_1, I_2 and I_3 represent the conventional currents while I4 represents the direction of the electron's motion. So, current (I_3) = ……. .

- $I_1 + I_2 - I_4$
- $I_1 + I_2 + I_4$
- $I_4 + I_1 - I_2$
- $I_4 + I_2 - I_1$

$I_1 + I_2 + I_4 = I_3$

$\Sigma I_{in} = \Sigma I_{out}$



Questions

The figure shows a part of a closed electric circuit, then the value of (I) = ……. .

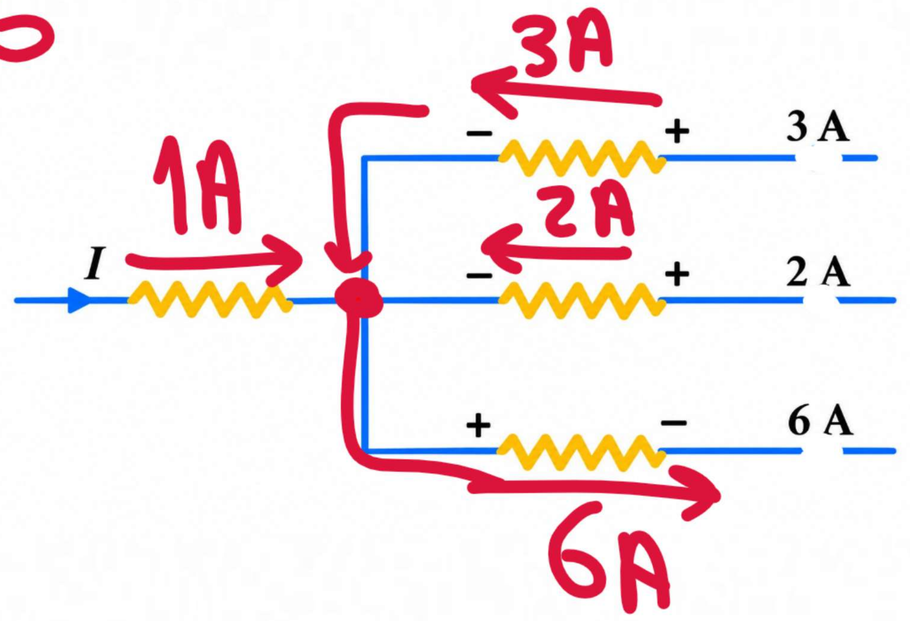
- 11 A
- 2 A
- 1 A
- 4 A

$\Sigma I_{in} = \Sigma I_{out}$

$3 + 2 + I = 6$

$I = 1A$

(2nd session 23)



Mr.Abdelwahab

Questions

The opposite figure represents electric currents through multiple branches. So, the value of current (I) and the direction of current (7 A) respectively are ..... .

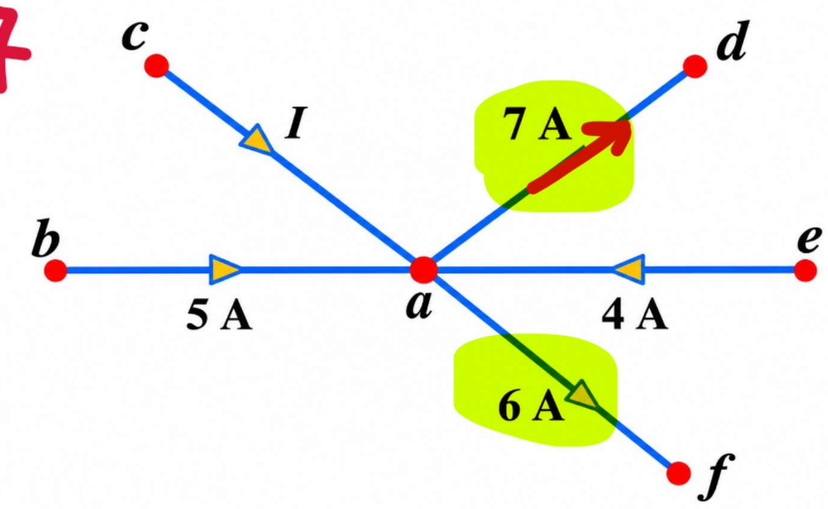
(Exp. 25)

- ✓ 4 A, from a to d
- ✗ 4 A, from d to a
- 13 A, from a to d
- ✗ 13 A, from d to a_1

$\Sigma I_{in} = \Sigma I_{out}$

$5 + I + 4 = 6 + 7$

$I = 4A$



Questions

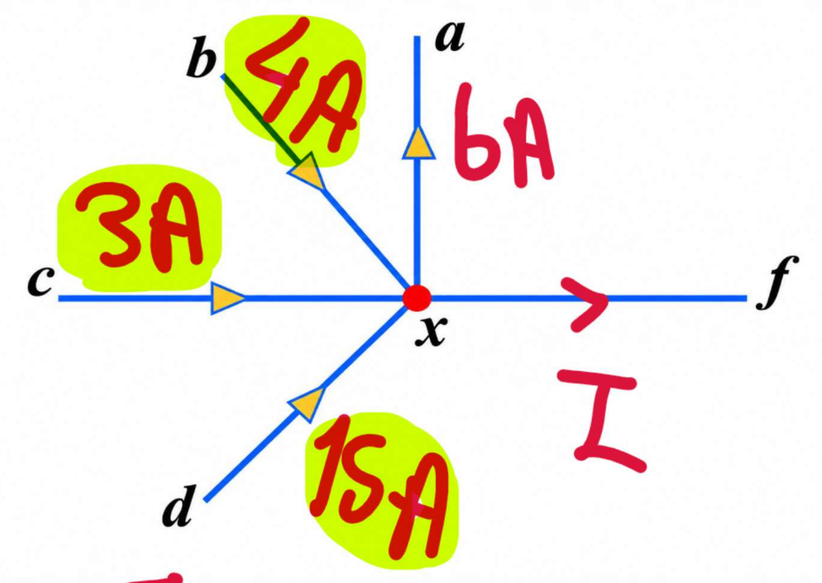
In the figure, if the directions shown represent the directions of electrons motion, and the intensities of the currents in branches xa, xc and xd are 6 A, 3 A, 15 A respectively, and 2.5×10^{19} electrons/second pass through branch xb, then the number of electrons passing through branch xf and their direction of motion are respectively ..... , ...

(Exp. 25)

$I = Rate \times e$

$= 2.5 \times 10^{19} \times 1.6 \times 10^{-19} = 4A$

- ✓ 1×10^{20} electrons/second, from x to f
- ✗ 1×10^{20} electrons/second, from f to x
- ✓ 1.6×10^{19} electrons/second, from x to f
- ✗ 1.6×10^{19} electrons/second, from f to x



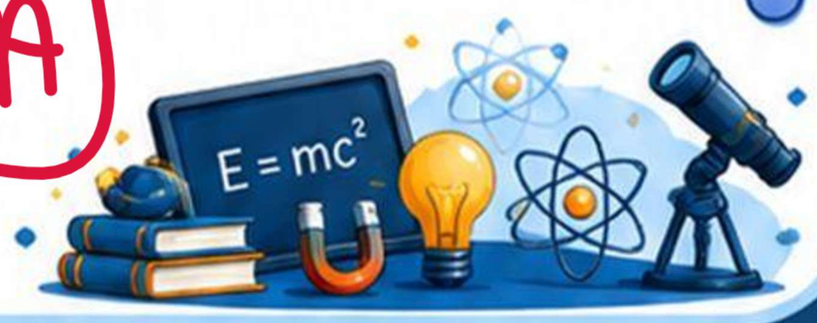
$Rate = \frac{I}{e}$

$= \frac{16}{1.6 \times 10^{-19}} = 1 \times 10^{20}$ e/s

$\Sigma I_{in} = \Sigma I_{out}$

$4 + 3 + 15 = 6 + I$

$I = 16A$



KVL Loop BEDCB

~~1+I2-I3-1=0~~

Using the data shown on the opposite circuit, the ratio (I_2 / I_1) = ... $\frac{1}{3} \div \frac{2}{3} = \frac{1}{2}$

let R=1Ω V_B=1V

$0I_1+I_2-I_3=0$

(Exp. May 21)

Kcl at node(B)

$I_1=\frac{2}{3}$ $I_2=\frac{1}{3}$

$I_1=I_2+I_3$

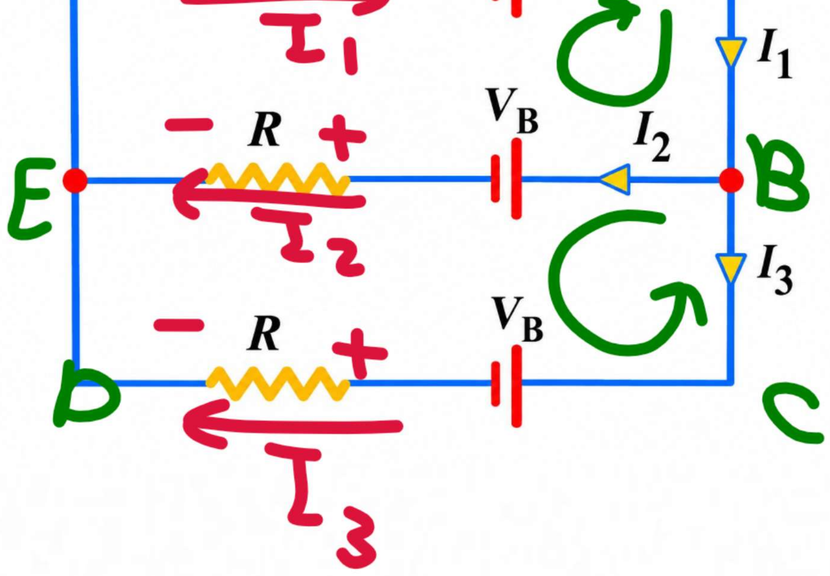
$I_1-I_2-I_3=0$

- ½
- ⅓
- ¾
- ⅔

KVL ABEFA

$1+I_2+I_1-2=0$

$I_1+I_2+0I_3=1$



In the opposite circuit, the value of electric current I_3 is ..... .

(Exp. June 21)

KVL Loop①

$-6-4+8I_2=0$

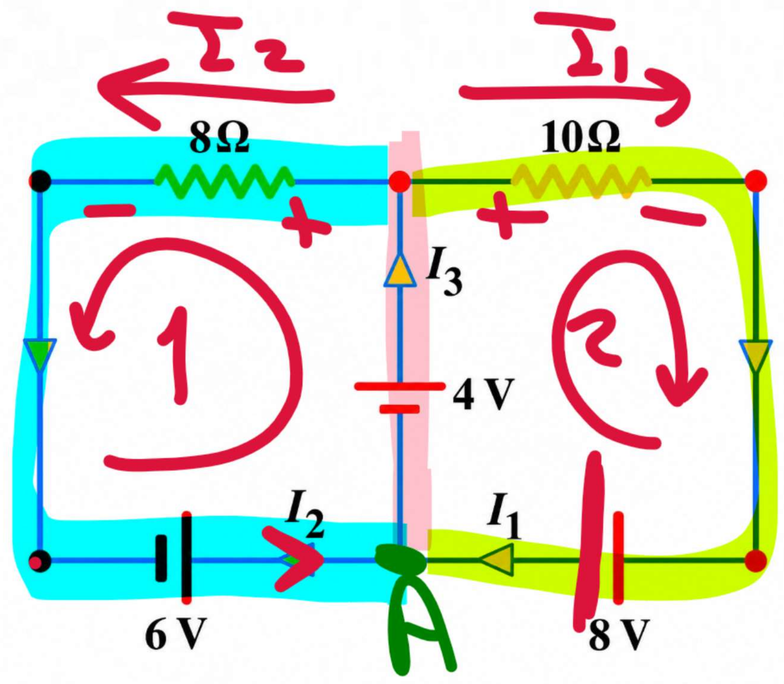
$I_2=1.25A$

KVL Loop②

$-8-4+10I_1=0$

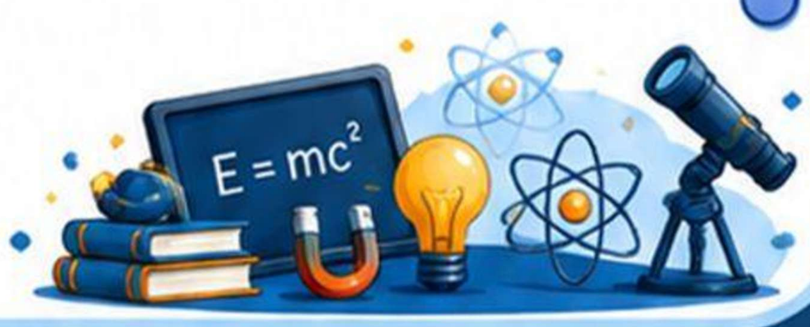
$I_1=1.2A$

- 1.2 A I_2
- 1.25 A I_1
- 2 A
- 2.45 A I_3



KCL at node A

$I_1+I_2=I_3=1.2+1.25=2.45A$



Mr.Abdelwahab

Questions

In the opposite electric circuit, we can apply Kirchhoff's second law to the closed path (adcba) as follows …….. .

(1st session 21)

- $3I_1 - I_3 - 4 = 0$
- ~~$2I_1 + I_2 + 4 = 0$~~
- ~~$2I_1 - I_2 - 4 = 0$~~
- ~~$2I_1 - I_2 + 4 = 0$~~

$8 - I_2 + 2I_1 - 12 = 0$

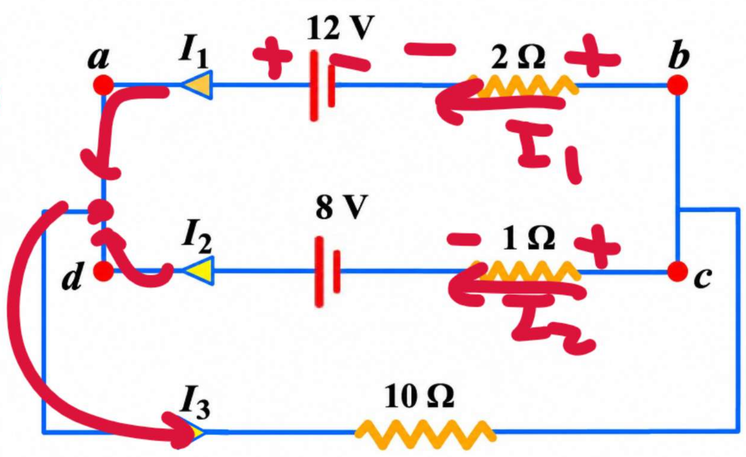
$2I_1 - I_2 - 4 = 0$

$I_1 + I_2 = I_3$

$I_2 = I_3 - I_1$

$2I_1 - [I_3 - I_1] - 4 = 0$

$2I_1 - I_3 + I_1 - 4 = 0 \rightarrow 3I_1 - I_3 - 4 = 0$



Questions

In the opposite circuit, if ($I_3 = -2\,I_1$), the value of the passing electric current in resistor R_3 equals …….. .

(1st session 21)

- $^2/_7$ A
- $^3/_7$ A
- $^4/_7$ A
- 1 A

KVL at loop ABCDA

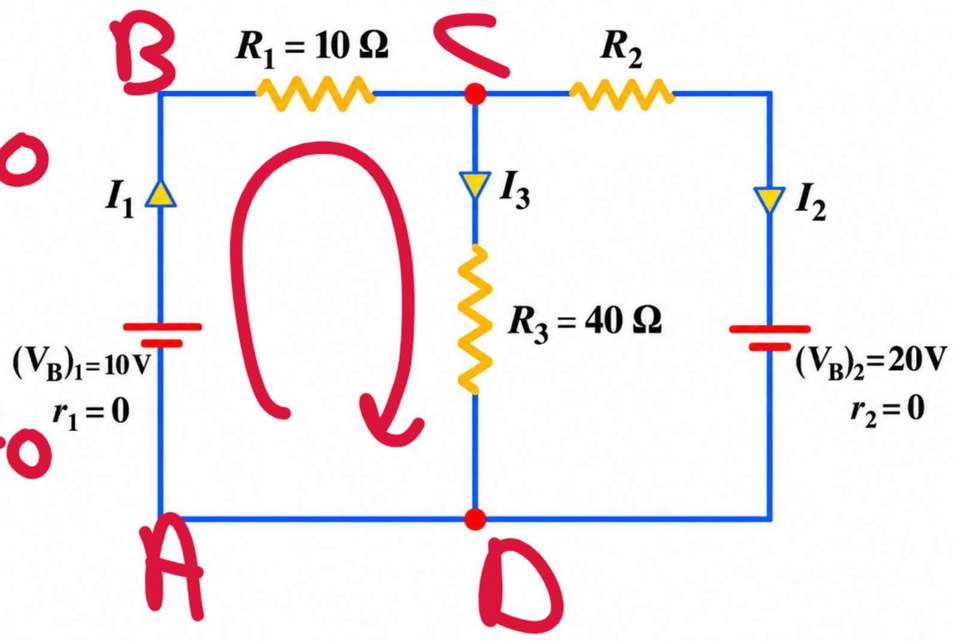
$-10 + 10I_1 + 40I_3 = 0$

$-10 + 10\left[-\frac{1}{2}I_3\right] + 40I_3 = 0$

$-10 - 5I_3 + 40I_3 = 0$

$-10 + 35I_3 = 0 \quad 35I_3 = 10 \quad I_3 = \frac{10}{35} = \frac{2}{7}A$

$I_1 = -\frac{1}{2}I_3$



Questions

In the shown electric circuit, if $I_3 = 2$ A, the value of I_2 is …….. .

(2nd session 21)

- 1 A
- 2 A
- 3 A ✓
- 4 A

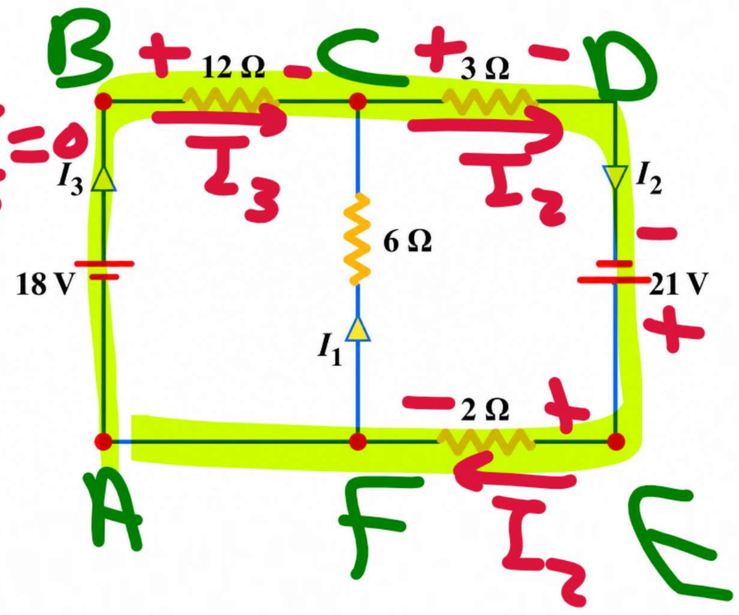
KVL ABCDEFA

$-18+12(2)+3I_2-21+2I_2=0$

$-18+24+5I_2-21=0$

$-15+5I_2=0$

$I_2=\frac{15}{5}=3A$



Questions

In the shown electric circuit: The value of $(V_B)_2$ that makes the reading of the ammeter vanishes =…….. .

(1st session 22)

- 6 V
- 4.5 V ✓
- 8 V
- 12 V

KVL DCBED

$-6+IR+3IR=0$

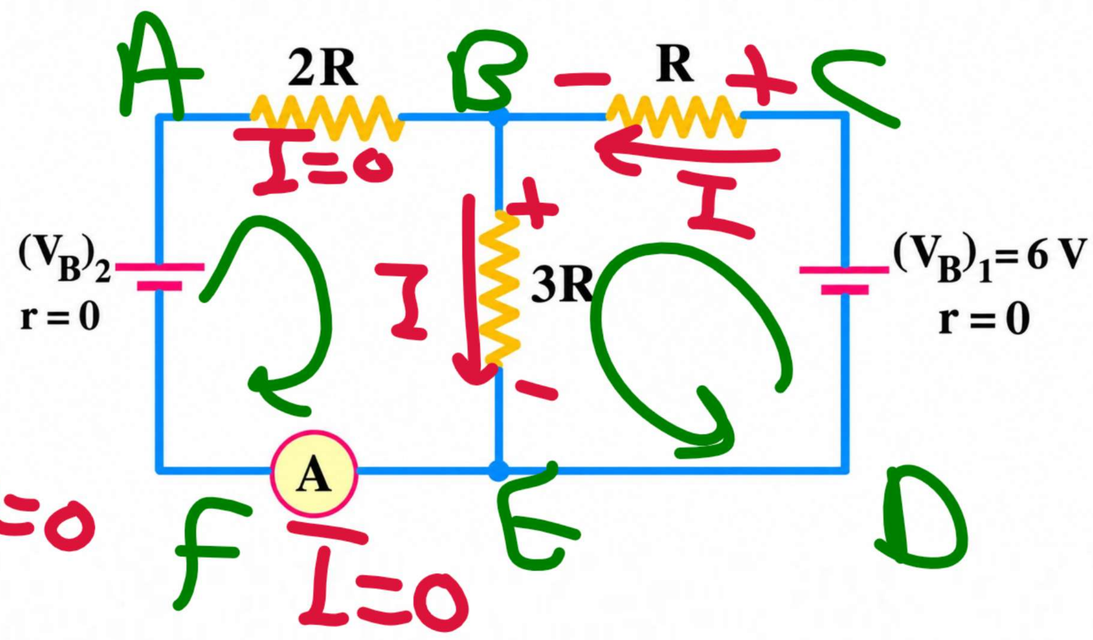
$4IR=6$

$IR=1.5$

KVL FABEF

$-V_{B_2}+0\times R+3IR=0$

$-V_{B_2}+3\times1.5=0$



Questions

In the shown electric circuit, the value of the intensity of electric current passing through the 8 Ω resistor equals …….. .

(1st session 22)

- 0.23 A I_2
- 0.846 A I_1
- 1.077 A I_3
- 1.306 A

KCL AT node (A)

$I_1+I_2-I_3=0$

KVL loop ABCDA

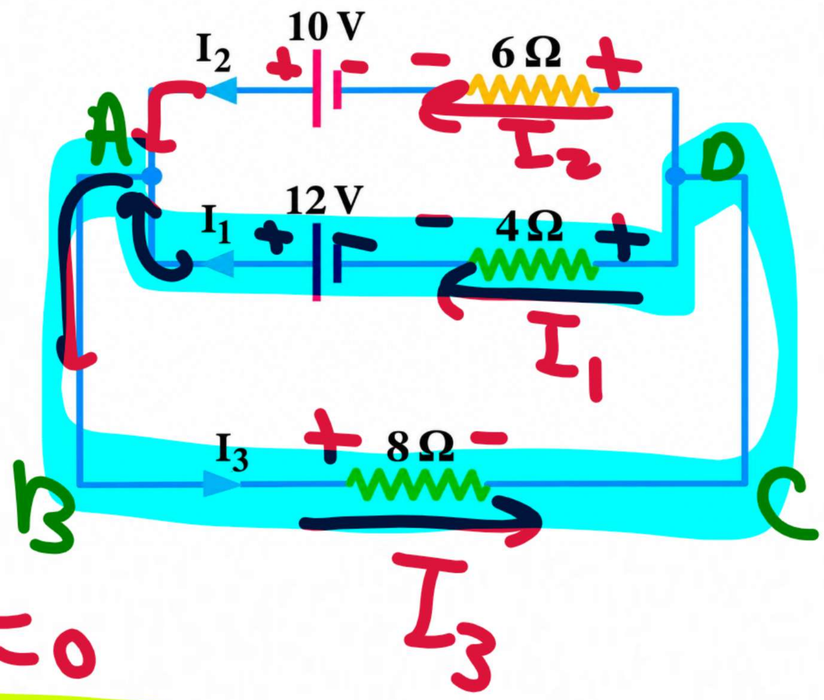
$8I_3+6I_2-10=0$

$0I_1+6I_2+8I_3=10$

KVL loop BCDAB

$8I_3+4I_1-12=0$

$4I_1+0I_2+8I_3=12$



Questions

In the shown closed electric circuit, the value of V_B =……. .

(2nd session 22)

- 10 V
- 4 V
- 15 V
- 6 V

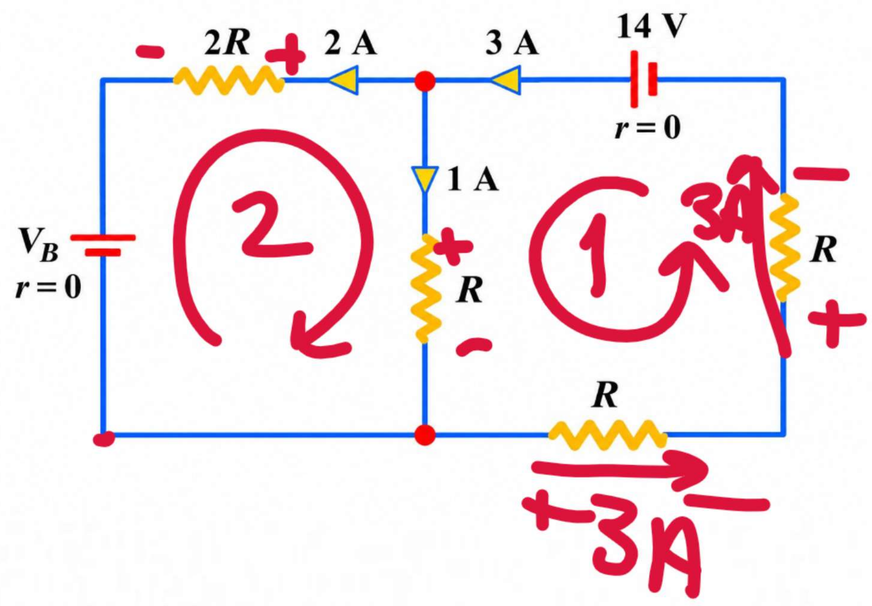
KVL Loop ①

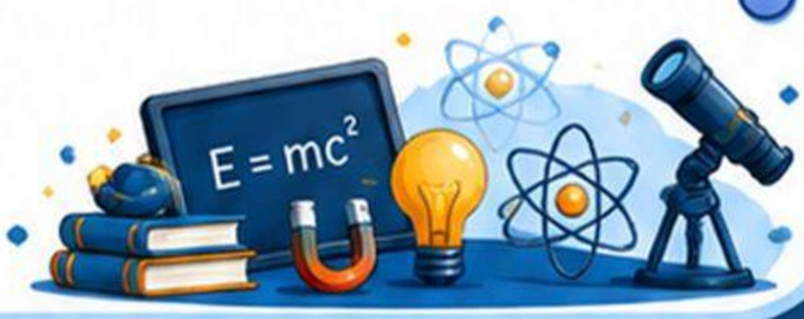
$-14+1\times R+3R+3R=0$

$R=2\Omega$

$-V_B-2R\times 2+R\times 1=0$

$V_B=6V$





Questions

In the illustrated electric circuit, the electromotive force of the battery V_B is …….. .

- $\frac{4}{3}$ V
- $\frac{40}{3}$ V
- $\frac{44}{3}$ V
- 12 V ✓

KVL ABbcdeA

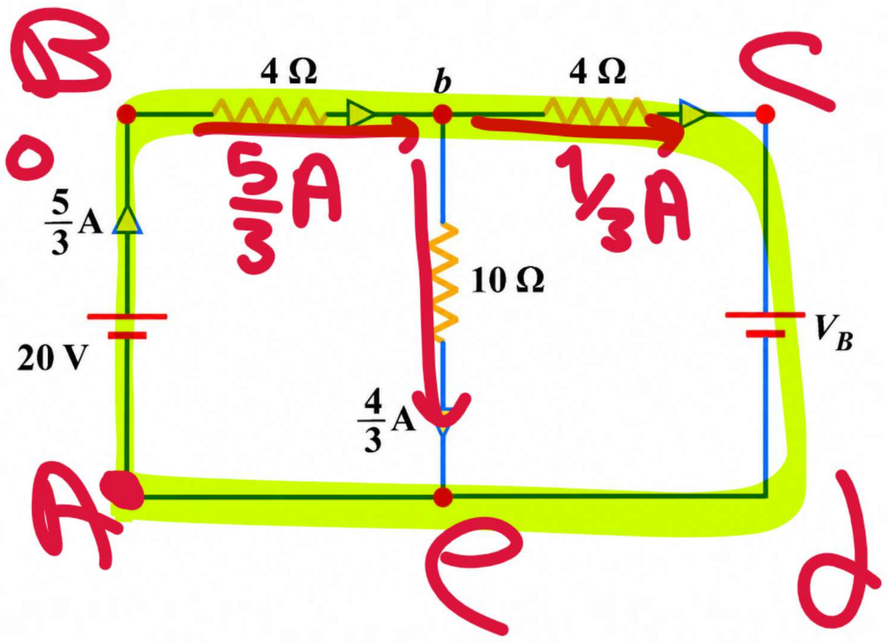
$-20+4\times\frac{5}{3}+4\times\frac{1}{3}+V_B=0$

$-20+\frac{20}{3}+\frac{4}{3}+V_B=0$

$-20+8+V_B=0$

$-12+V_B=0$

$\frac{5}{3}-\frac{4}{3}=\frac{1}{3}$ A



(Exp. 23)

Questions

In the shown closed electric circuit, the ratio between: (I_3 / I_2)= …….

- ½
- ⁴⁄₁ ✓
- ¼
- ²⁄₁

$+2I_3-2V_B=0$

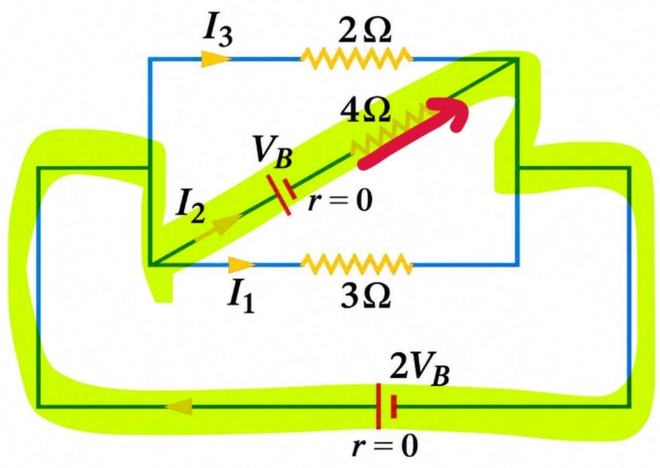
$I_3=\frac{2V_B}{2}=V_B$

$V_B+4I_2-2V_B=0$

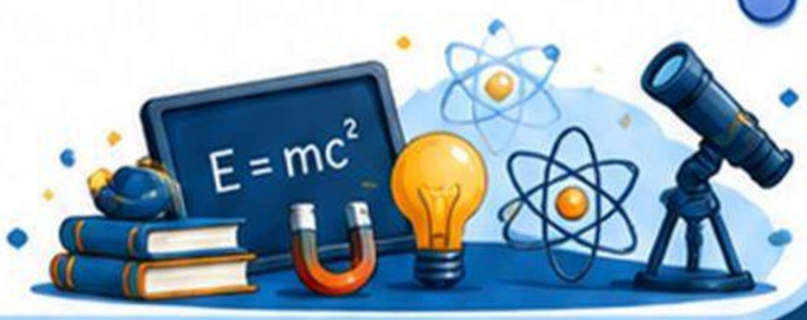
$4I_2-V_B=0$

$I_2=\frac{V_B}{4}$

$\frac{V_B}{V_B}=\frac{4}{1}$



(1st session 23)

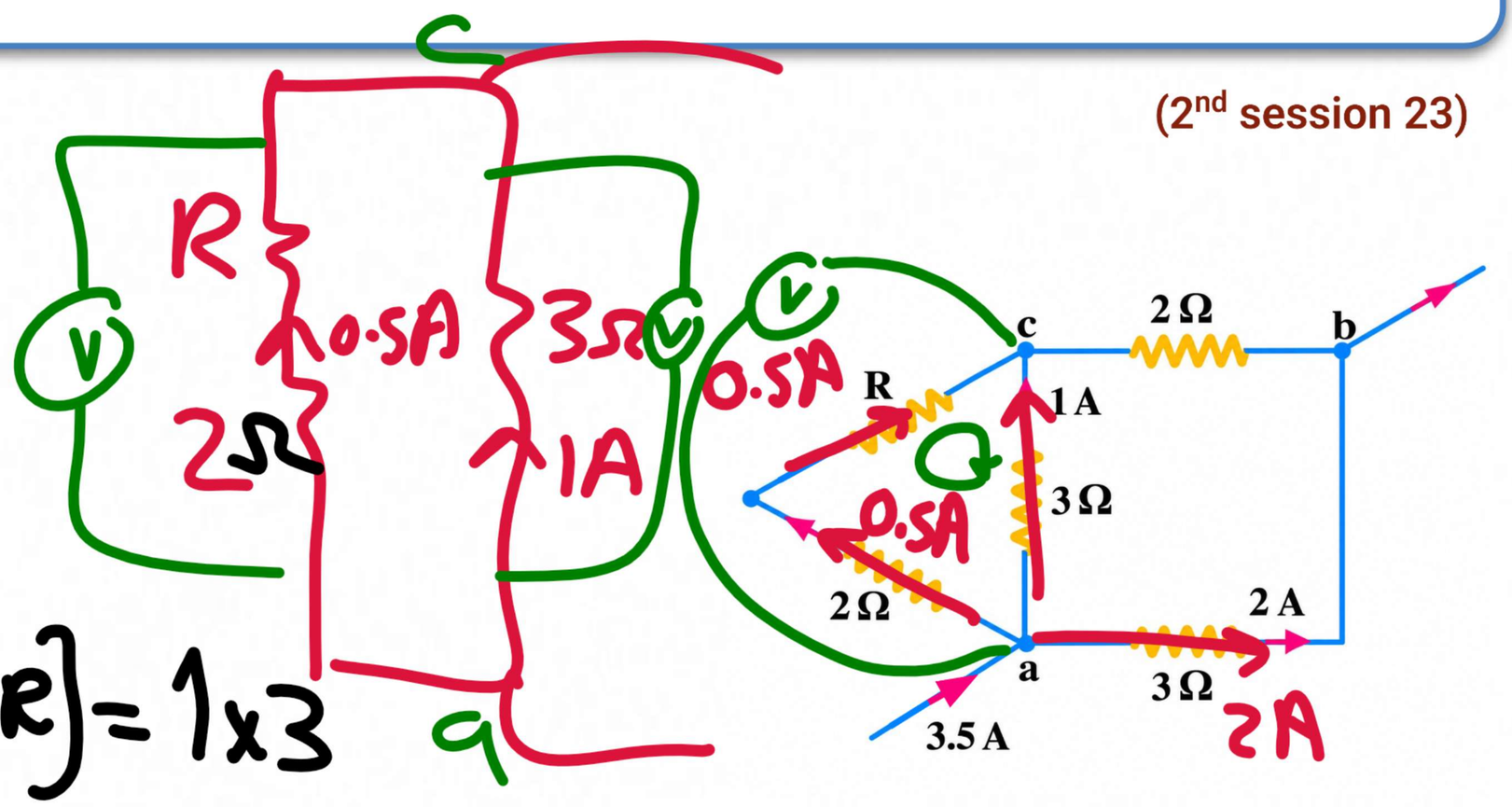


Questions

The figure illustrates a part of a closed electric circuit. Then the resistance (R) equals …….. .

(2nd session 23)

- 4 Ω ✓
- 2 Ω
- 6 Ω
- 3 Ω



$0.5[2+R]=1\times3$

$2+R=6 \quad R=4\Omega$

Questions

From the shown electric circuit, I_1 = …….. I_3 → $\frac{I_1}{I_3}$ = ……

- 2.25 ✓
- 1.25
- 0.8
- 1.8

KCL at node (B)

$I_1+I_2+I_3=0$

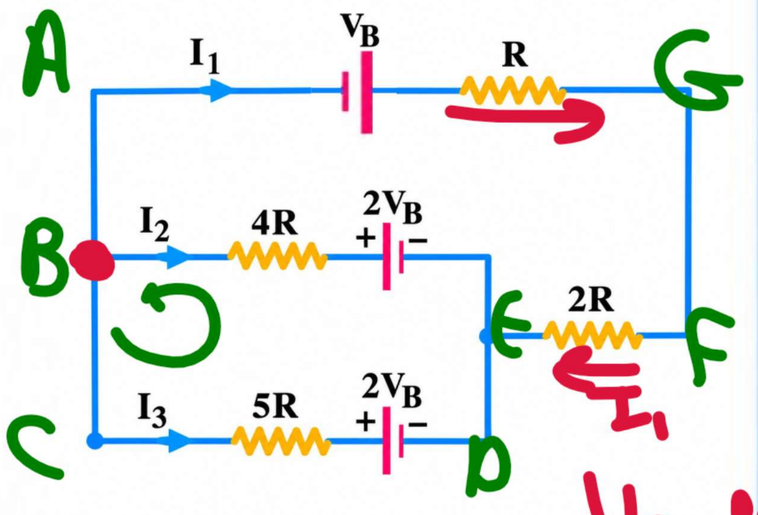
$\frac{0.574}{0.255}=2.25$

(1st session 24)

KVL at loop BCDEB

$5I_3+2-2-4I_2=0$

$0I_1-4I_2+5I_3=0$



$V_B=1V$

$R=1V$

$I_1=0.574A$

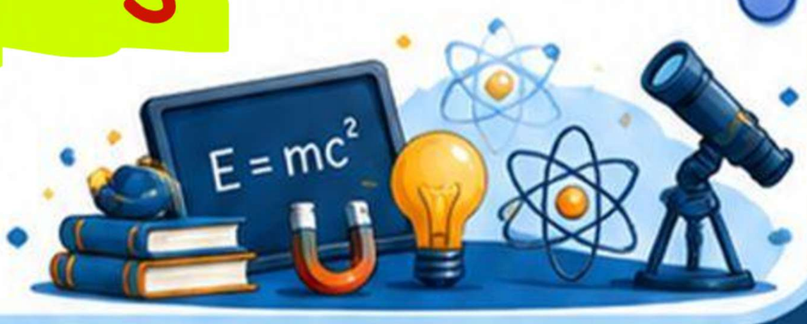
$I_2=-0.1319A$

$I_3=-0.255A$

KVL at loop ABCDEFGA

$5I_3+2-2I_1-I_1+1=0$

$-3I_1+0I_2+5I_3=-3$



Questions

The opposite figure shows a part from an electric circuit, by using Kirchhoff's laws, which equation is correct?

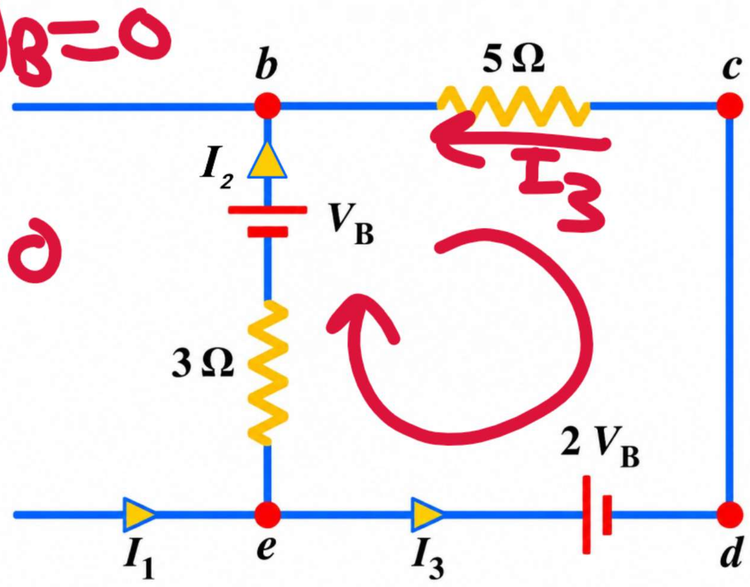
(2nd session 24)

KVL loop bcdeb

$-5I_3 - 2V_B + 3I_2 - V_B = 0$

$3I_2 - 5I_3 - 3V_B = 0$

- ~~$3I_1 + 7I_2 = -3V_B$~~
- ~~$3I_2 - 5I_3 = -3V_B$~~
- $3I_2 - 5I_3 = 3V_B$ ✓
- ~~$3I_1 - 8I_2 = 3V_B$~~



Questions

The opposite figure represents a section of electric circuit that carries an electric current, then the value of V_B is …… .

(Exp. 25)

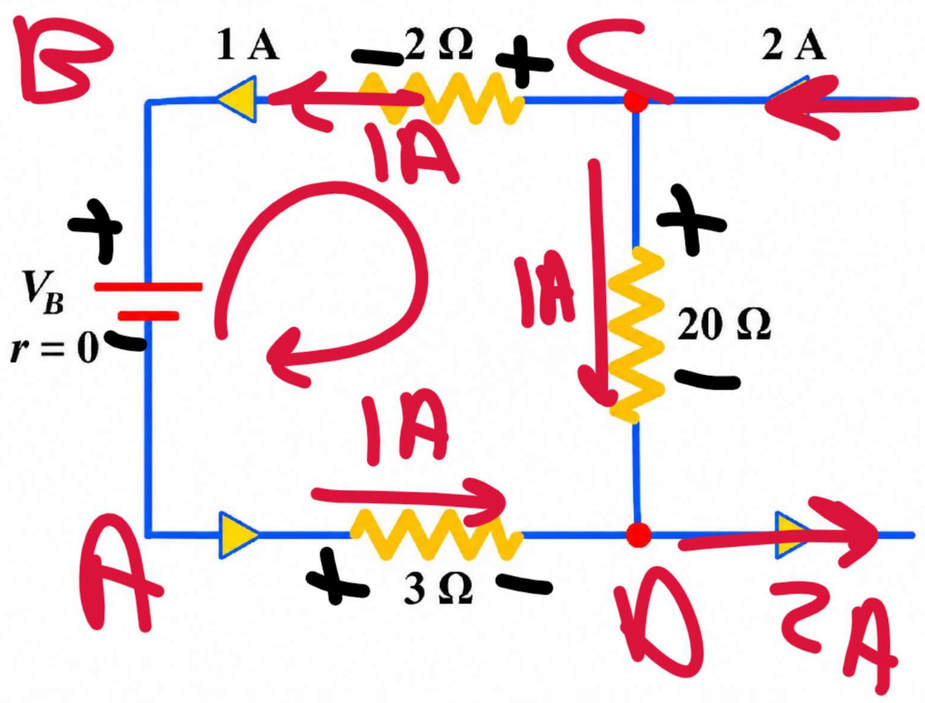
KVL ABCDA

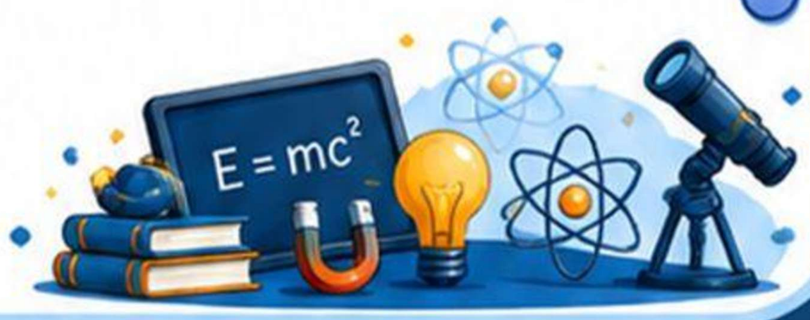
$-V_B - 1x2 + 1x20 - 3x1 = 0$

$-V_B - 2 + 20 - 3 = 0$

$-V_B + 15 = 0$

- 30 V
- 25 V
- 20 V
- 15 V ✓





Mr Abdelwahab

Etqan Way
MASTER TO RISE